Department of Mathematics and Applied Mathematics

University of Cape Town

A First Course in Metric Spaces

(with a generous sprinkling of applications,
and a touch of topology)

Notes written by Jurie Conradie
And used, with acknowledgement and thanks, by Anneliese Schauerte

On the mountains of truth you can never climb in vain: either you will reach a point higher up today, or you will be training your powers so that you will be able to climb higher tomorrow.

Friedrich Wilhelm Nietzsche

One does not discover new lands without consenting to lose sight of the shore for a very long time.

André Gide

I was at the mathematical school, where the master taught his pupils after a method scarce imaginable to us in Europe. The proposition and demonstration were fairly written on a thin wafer, with ink composed of a cephalic tincture. This the student was to swallow upon a fasting stomach, and for three days following eat nothing but bread and water. As the wafer digested, the tincture mounted to his brain, bearing the proposition along with it. Jonathan Swift, *Gulliver's Travels* (1726)

Preface

The notion of distance pervades a great deal of mathematics; sometimes it is there in the open, for all to see, sometimes it lurks beneath the surface. Measuring the distance between two points in the plane is something quite natural that we learn to do quite early in our mathematical career. Much more sophisticated is the idea of measuring the "distance" between two continuous functions, but this is essential to the understanding of the concept of uniform convergence of sequences of functions.

If the notion of distance is such a common and important one, it may make sense to study it on its own, stripped of all the baggage that it carries in specific examples. This is exactly what we are going to do in this course. A metric space is a set equipped with a "distance function", or "metric", which enables us to measure the distance between any two elements of the set. This sounds like an extremely meagre fare for a whole book. As you will soon discover, we are not only going to make a meal of it, but there will still be a great deal left untasted by the time we finish. Many ideas that you have come across in Real Analysis will surface here again in a more general setting. The greater generality will allow us to apply these ideas in many new contexts.

This course tries to address the interests of both those who want to find out about metric spaces so that they can use them elsewhere, and those who may get addicted to them because they are fascinating and challenging. You will therefore find in what follows both applications as well as the more esoteric delights of topology. Taste, eat and digest it. May you find enough sustenance to keep you going to the end!

Jurie Conradie
February 2002

Contents

Chapter 1

Starters

In which a large variety of problems with apparently little in common is introduced.

In a book with the title " A First course on Metric Spaces", it would be reasonable to expect a sentence fairly early on in the first chapter starting with " A metric space is ..." This book is an exception. In fact, you will probably not find the words "metric space" anywhere (else!) in this chapter.

What you will find is a collection of problems which at first seems to have very little in common. A careful reader will soon begin to notice a common thread, despite the apparent differences. Do read on, but do so carefully, and try to spot the theme.

Example 1.1 Suppose we are faced with the problem of finding the solutions (if there are any) of the equation

$$x = \cos x.$$

On the left in Figure 1.1 on the next page the graphs of the functions $f(x) = \cos x$ and $g(x) = x$ are shown; the solution of the equation above is given by the x-coordinate of the point of intersection of the two graphs. From the figure it seems to be clear that the equation has a solution. Unfortunately there is no formula for solving an equation like this. One possibility is to resort to a "trial and error" method. As a first step, guess a solution, say $x = 1$ (radian, of course). To check our guess, we calculate $\cos 1$. Since $\cos 1 \neq 1$, we do not have a solution yet. But the graph on the right in Figure 1.1 above suggests that $\cos 1$ may be a better approximation to the solution than 1. So we try $x = \cos 1$ as a second approximation. More formally, we can write $x_1 = 1$ (our first guess), $x_2 = \cos x_1 = \cos 1$ (our second

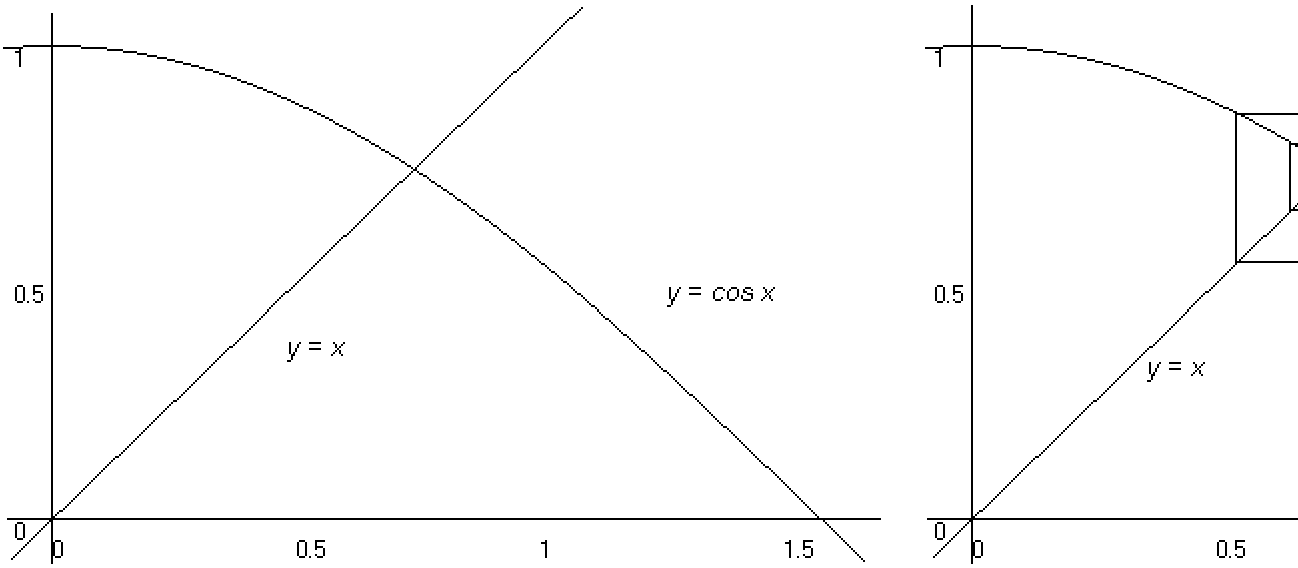


Figure 1.1: Solving the equation $x = \cos x$.

guess, or approximation). A check shows that x_2 is still not a solution, so we take $x_3 = \cos x_2$ as our third approximation. The graph suggests that if we continue in this way, we are going to get better and better approximations to the solution. It looks as if the sequence (x_n) defined inductively by

$$x_1 = 1, \quad x_{n+1} = \cos x_n, \quad n \geq 1$$

will converge to the solution of the equation $x = \cos x$.

A natural question to ask at this stage is: Can we always use this method to solve an equation of the form $x = f(x)$? To get a better "feel" for the method, we look at a second example. This time we try to solve the equation $x = e^{0.5x} - 1$. If we take $x_1 = 2$ as our initial guess, and put $x_{n+1} = e^{0.5x_n} - 1$, we see from Figure 1.2 that the sequence (x_n) obtained seems to converge to the solution $x = 0$. This is not all that useful, since it is easy to spot this solution immediately. Starting with $x_1 = 3$ leads to a sequence that clearly diverges to infinity. It is clear from the graph, however, that there is a solution other than $x = 0$ to this equation. A little bit of experimentation and a good look at the graph should be enough to convince you that we'll end up with a sequence converging to 0 or a sequence diverging to infinity, no matter which starting point (other than the solution itself) we choose. It is natural to ask why the method works for the first equation, but not for the second.

The method can go wrong in other ways as well. If it is used to try to solve the equation

$$x = 4x^2 e^{-x^2},$$

most starting points lead to a sequence with terms which eventually alternate between two different values and therefore does not converge to one of the solutions of the equation.

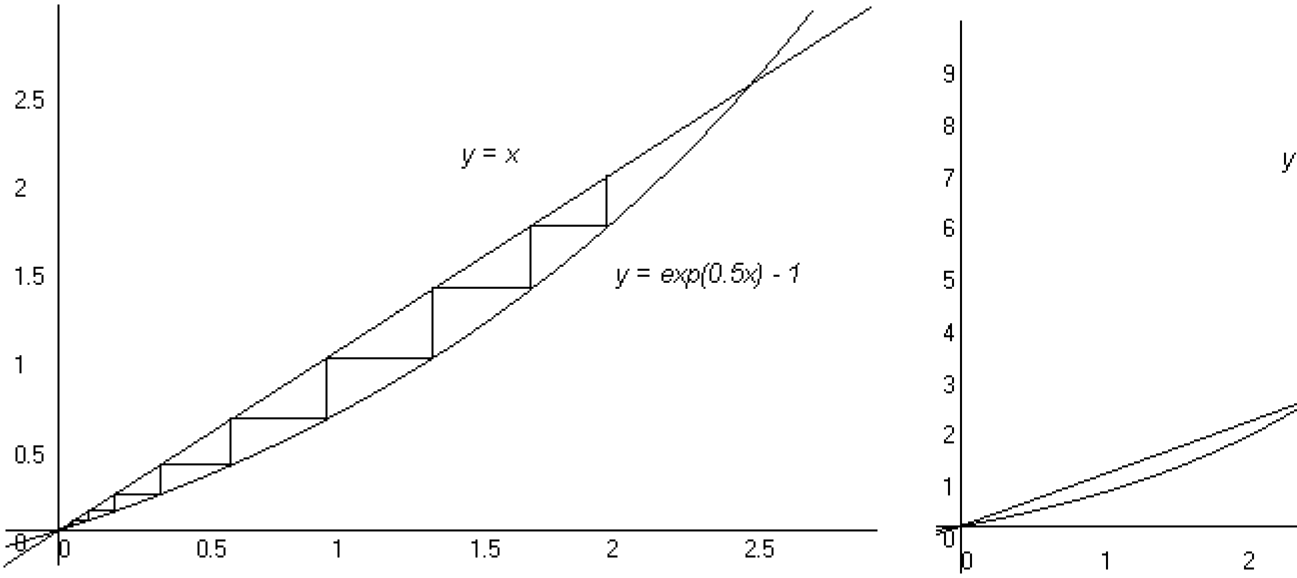


Figure 1.2: Attempting to solve the equation $x = e^{0.5x} - 1$.

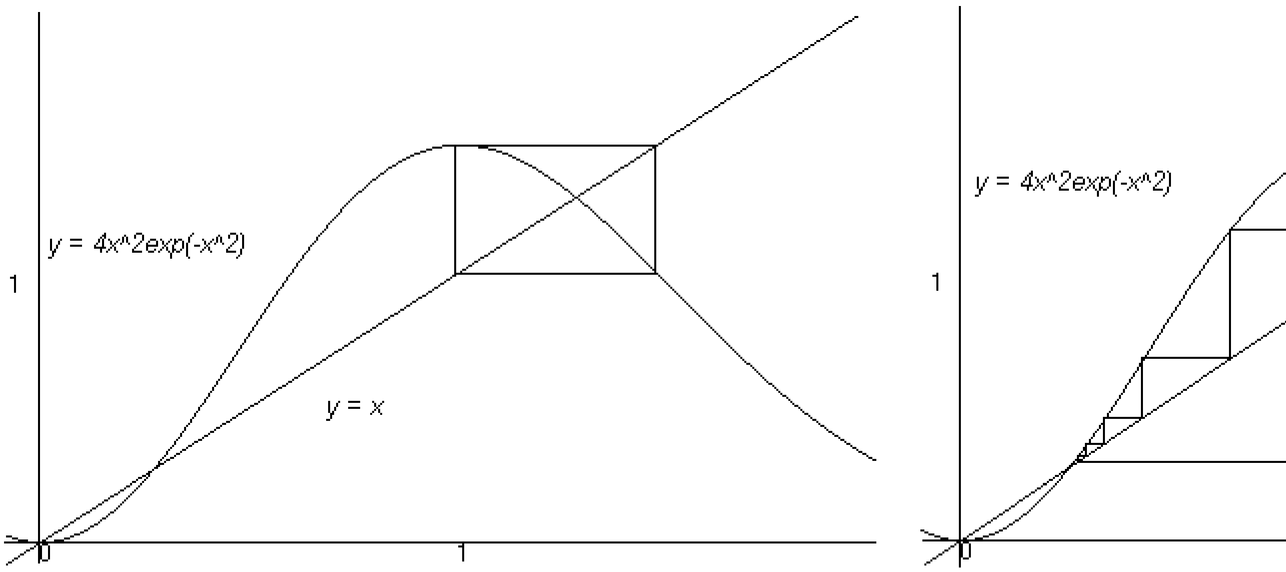


Figure 1.3: Attempting to solve the equation $x = 4x^2 e^{-x^2}$.

On the left in Figure 1.3 on the previous page we see what happens if we start with $x_1 = 1$, and on the right we see what happens if $x_1 = 2$ is used as a starting point. These examples suggest the following question: Suppose we try to solve the equation $x = f(x)$ by guessing a solution x_1 and then constructing a sequence (x_n) by putting $x_{n+1} = f(x_n)$ for every $n \geq 1$. Will this sequence converge, and if it does, will it converge to a solution of the equation? For practical applications, even just being able to find necessary conditions on x_1 and f to ensure convergence to a solution will be very useful.

This is a problem about real-valued functions of a real variable, and one would therefore suspect that a sound knowledge of Real Analysis would be enough to make some headway with this problem. This is indeed the case. But the following problem, though in some ways similar, moves to a higher level.

Example 1.2 Let's look at the differential equation

$$\frac{dy}{dx} = xy + 1$$

together with the initial condition $y(0) = 2$. You may recall that this is known as an *initial value problem.* If we integrate both sides of the differential equation from 0 to x and make use of the initial condition, we get

$$y(x) = 2 + \int_0^x (ty(t) + 1)\,dt.$$

An equation like this is called an *integral equation.* It can easily be checked that a function y of x satisfies the initial value problem given if and only if it satisfies this integral equation (see Exercise 2).

In what may appear to be a rather desperate attempt to solve this integral equation, let's guess a solution. A solution, we have to remember, must be a continuous function of x. A particularly simple function of this kind is the one given by $y(x) = 1$ for every $x \in \mathbb{R}$. A quick check shows that it is not a solution, but let's retain it as a first approximation to a solution. So we write $y_1(x) = 1$. Now we use y_1 to get a second approximation in a way that will look quite familiar: we put

$$y_2(x) = 2 + \int_0^x (ty_1(t) + 1)\,dt = 2 + \frac{1}{2}x^2 + x.$$

Another check shows that y_2 is still not a solution, so we put

$$y_3(x) = 2 + \int_0^x (ty_2(t) + 1)\,dt.$$

We can clearly continue in this way to find a sequence of *functions* that we may regard as "approximate solutions". It is now natural to ask whether there any sense in which this sequence of functions converges, and if so, whether it converges to a solution of the integral equation.

You can probably spot the similarities between these two examples. There is also a rather obvious difference: In the first case we considered a sequence of *real numbers*, in the second a sequence of *functions*. The third example looks at a sequence of yet another kind.

Example 1.3 Figure 1.4 below shows the first few steps in the construction of the so-called Sierpinski gasket S (an example of a *fractal*). Is there any sense in which

Figure 1.4: Constructing the Sierpinski gasket.

we could say that this sequence of subsets of $\mathbb{R}^2$ converges to a limit? If so, what is this limit?

Example 1.4 In the previous example we produced what looks like a very intricate geometrical figure by considering a sequence of simpler figures. There often is a need for doing something that goes in the opposite direction: given a complicated figure, try to find simpler figures which can be used to construct the original figure. This is a problem faced in the field of *image processing*, where efficient ways are sought to transmit complicated images. It is hardly ever necessary to transmit the exact image; a sufficiently good approximation will do. But what do we mean by "sufficiently good"? How do we measure how close the approximation is to the original image?

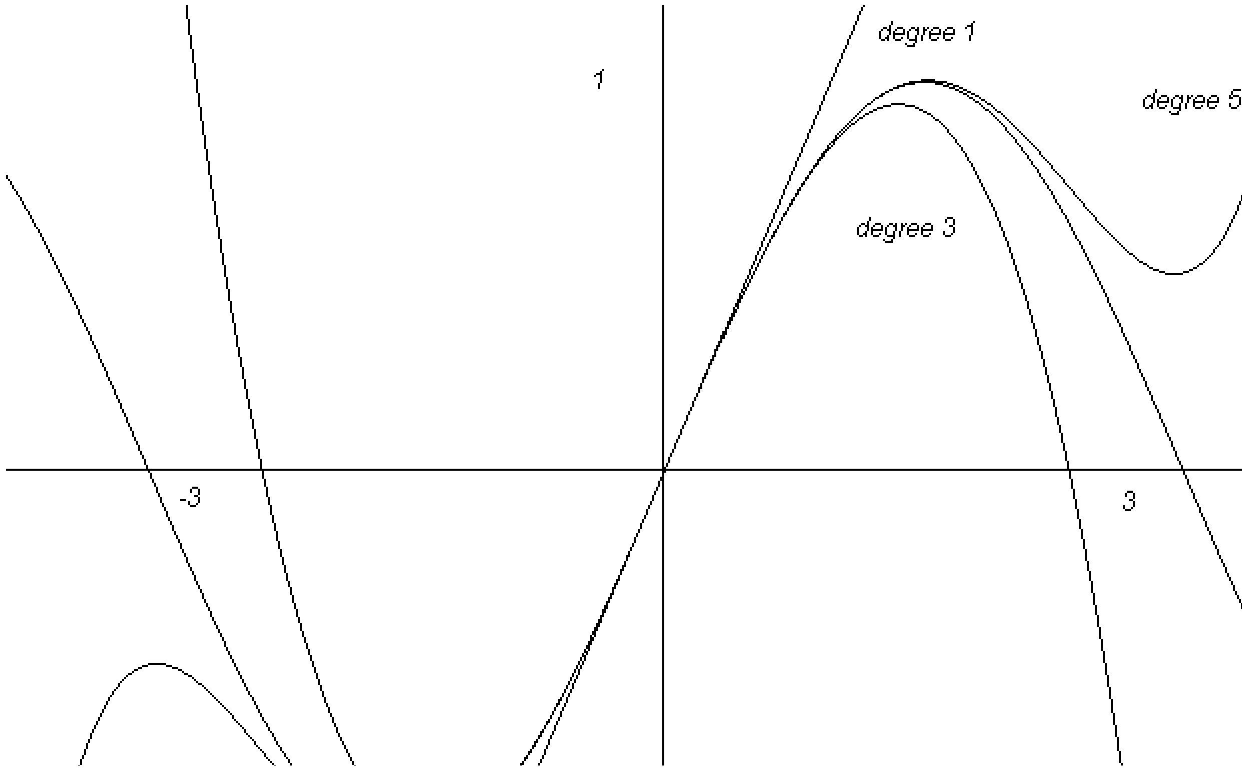
Figure 1.5: Approximating an image: the image on the left, an approximation on the right.

Example 1.5 You are familiar with the idea of approximating a (sufficiently often differentiable) function by one of its Taylor polynomials. Figure 1.6 on the next page shows the graphs of the function $f(x) = \sin x$ and its Taylor polynomial of degree 3 about $x = 0$ on the interval $[-\pi, \pi]$. It is natural to ask: "How good is this approximation?" (or, if you like, "How close is the polynomial to the function?"). We could go even further. Since f is infinitely differentiable, for each n, we can find its Taylor polynomial of degree n about $x = 0$, in this way producing a sequence of polynomials. Is there a sense in which this sequence converges to the function f?

Example 1.6 Non-differentiable functions cannot be approximated by Taylor polynomials. But such functions occur frequently in practice. An example of such a function is the so-called square wave function, which is defined on the interval $[-\pi, \pi)$ by

$$f(x) = \begin{cases} 0 & \text{for } -\pi \leq x \leq 0 \\ 1 & \text{for } 0 < x < \pi. \end{cases}$$

(Note that f is discontinuous, and therefore certainly not differentiable. We can extend the definition of f to the whole real line by requiring that $f(x + 2\pi) = f(x)$ for all $x \in \mathbb{R}$; f then becomes a *periodic function* with *period* 2π. Such functions can be approximated by linear combinations of functions of the form $\sin kx$ and $\cos kx$, where $k \in \mathbb{N}$, which are periodic functions themselves. These linear combinations are known as Fourier polynomials. Figure 1.7 shows the graph of the square wave function f and its Fourier polynomials of degree 1, 3 and 5. As before, we can ask how good this approximation is. Should we use the same "measure of closeness"



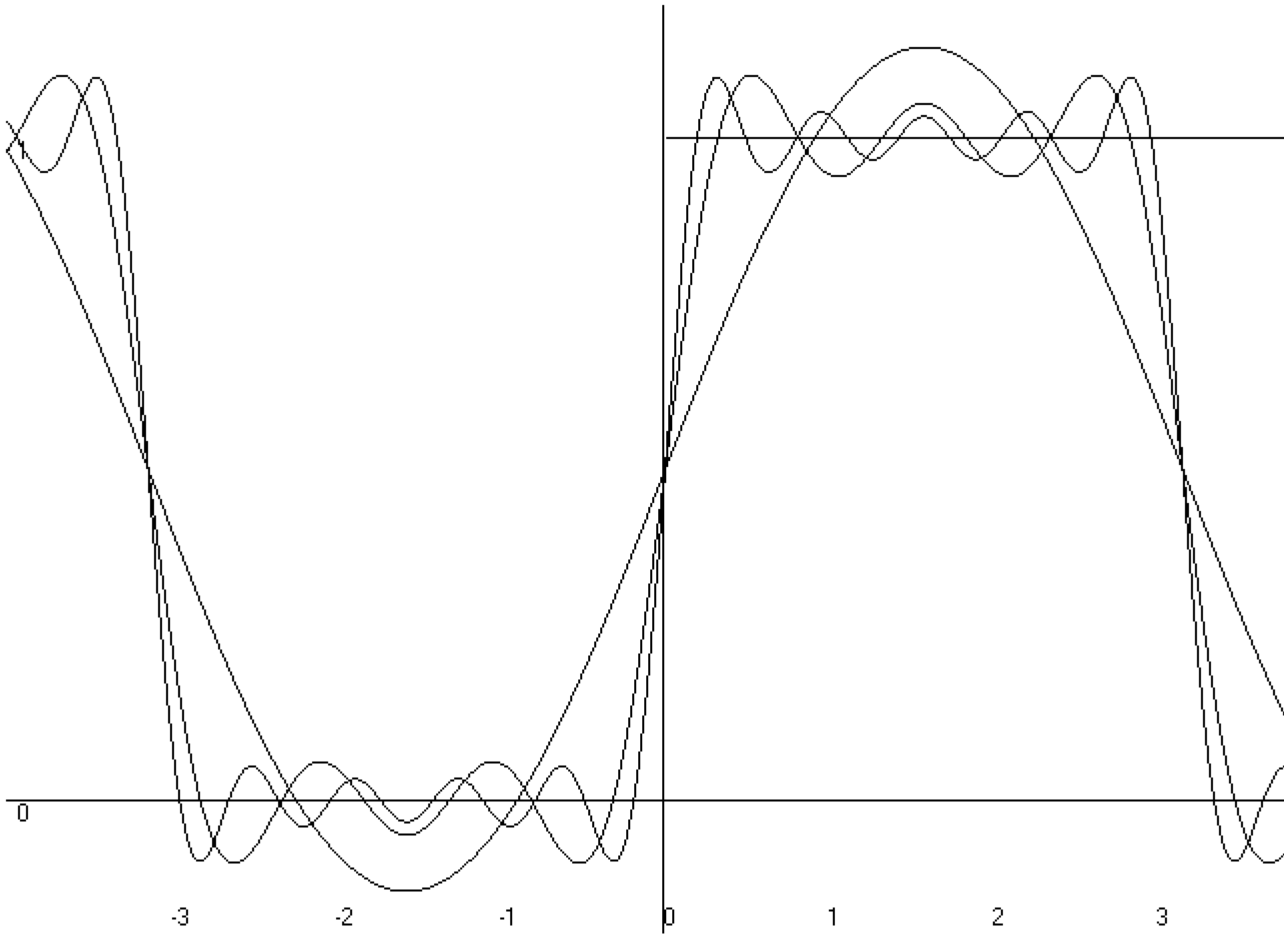


Figure 1.7: Fourier polynomials for the square wave function

here as in Example 1.5? And if we can produce a *sequence* of Fourier polynomials, is there a sense in which this sequence will converge to the square wave function f?

Example 1.7 In Example 1.1 we used a real number x_1 and a function f to generate a sequence (x_n) of real numbers inductively by defining $x_{n+1} = f(x_n)$ for $n \geq 1$. This sequence depends on both x_1 and f, and is usually known as the *orbit* of x_1 under f. From the examples we have considered it is clear that such an orbit can behave in different ways. The orbit can be a convergent sequence, or it can diverge to ∞ (or $-\infty$). But Figure 1.3 shows that the orbit could also be a sequence which eventually alternates between two values. But even "worse" things can happen: Figure 1.8 shows an orbit with what seems to be randomly distributed terms filling up an interval. This is known as *chaotic* behaviour. Sequences like

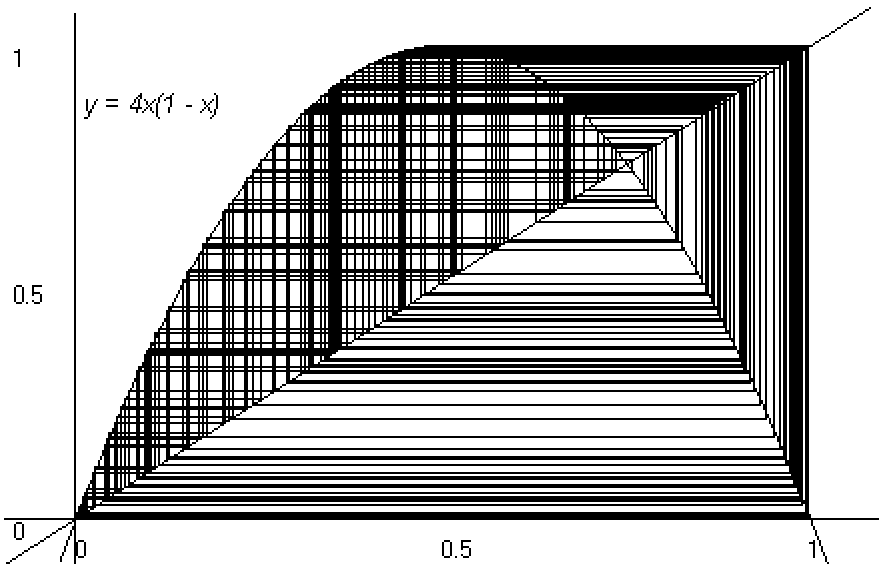


Figure 1.8: A chaotic orbit.

these arise in the study of systems that change with time if we only consider the time at discrete time intervals. We call such systems *discrete dynamical systems.* Is there any way in which we can predict the behaviour of such a system?

Example 1.8 Information is often stored and transmitted as binary strings, i.e. as finite sequences of zeros and ones. If we use strings of length 8 (say), two examples of such strings are 11000101 and 11100110. In transmitting such data it may get corrupted; so, for example, the first string above could change to the second. In trying to establish whether an error has occurred and when trying to correct it, it is important to "measure" by how much the second string differs from the first. The problem here is to decide on such a measure. How far is the first string from the second?

We've come to the end of our list of examples. Notions that you may have noticed occurring in some or all of these examples are those of iteration, sequences, convergence and approximation. But these notions occurred in rather different contexts.

With this in mind, we could ask: Is there a suitably general context in which we could look at all these problems at the same time? What is the least amount of structure we need on a set to be able to talk about approximation and convergence? A little thought should convince you that we cannot talk about approximation or convergence without being able to measure in some sense how far things are apart (or how close they are together!). If we try to home in on the essential idea, it seems that we need a situation in which are be able to measure the distance between elements of a set. But what properties should such a measure of distance have to be useful? It is to this question that we turn in the next chapter.

Historical Notes

Brook Taylor (1685 – 1731)

Taylor was an English mathematician best known for his discovery of the formula for what is now known as the Taylor expansion of a function. This first appeared in his book *Methodus incrementorum directa et inversa* of 1715, in which he also develops the branch of mathematics known as the "calculus of finite differences." He is credited with the invention of integration by parts, and gave the first systematic account of the basic principles of perspective. In 1712 he was elected a Fellow of the Royal Society and became a member of the committee who had to settle the priority dispute on the invention of the calculus between Newton and Leibniz.

Joseph Fourier (1768 – 1830)

Fourier was born in Auxerre in France, one of fifteen children. He showed great interest in mathematics at an early age, but initially decided to train for the priesthood. However, he never took religious vows, but took a job as a teacher in Auxerre, at the same time doing mathematical research. It was the time of the French Revolution and in 1793 Fourier became heavily involved in politics on the side of the revolutionaries. His support for one faction within the revolution led to his imprisonment on two occasions. In 1794 he became one of the first students at the newly established Ecole Normal in Paris and later taught mathematics at the College de France and the Ecole Polytechnique.

He joined Napoleon's expedition to Egypt as scientific advisor in 1798 and on his return to France was appointed by Napoleon as Prefect (an administrative position) in Grenoble. Although he was not very keen on this position, he acquitted himself well of this task and in addition managed to do some of his best research, into the mathematical theory of the conduction of heat, during this time. He established the partial differential equation governing heat diffusion and used infinite series of trigonometric functions (now called Fourier series in his honour) to solve the equation. This work at first proved to be very controversial and his essay *Theorie analytique de la chaleur* was only published in 1822. Fourier returned to Paris after Napoleon was finally defeated and worked there until his death in 1830.

Waclaw Sierpinski (1882 – 1969)

Sierpinski was born in Warsaw, Poland, at the time of the Russian occupation of Poland. He had to overcome many difficulties brought on by the attempt of the Russians at the time to impose their language and culture on Poland. Despite this he managed to graduate from the University of Warsaw, obtained a doctorate from the University of Cracow and was appointed at the University of Lvov in 1908.

It was at this time that he read the work of Cantor and started contributing to the field of set theory. During the First World War he was interned in Russia. Shortly after his return to Poland after the war he was appointed professor at the University of Warsaw. He remained in Warsaw until his death in 1969. He was an extremely prolific mathematician and made important contributions to set theory, topology and the theory of functions of a real variable. The Sierpinski space-filling curve (a continuous curve that goes through every point of the unit square) and the Sierpinski gasket were named after him.

Exercises

1. Find an approximate solution of the equation

$$x^3 + 2x^2 - 7x - 1 = 0$$

   by rewriting the equation in the form $x = f(x)$. [Hint: Take the term $-7x$ to the right hand side.] Try starting with $x = 0$ and then with $x = 2$. [Hint: If you have access to a graphing program like NEWT or are prepared to do some simple programming, you can avoid the tedious calculations.]

2. Show that a function $y(x)$ will be a solution of the initial value problem in Example 1.2 if and only if it is a solution of the integral equation in the same example.

3. Staying with Example 1.2, calculate the functions y_3, y_4 and y_5 (starting with $y_1(x) = 1$). Also calculate the first four terms of the sequence (y_n) if you start with $y_1(x) = x$.

4. List as many ways as you can think of in which a sequence of functions can converge to a function.

5. Do you think the sequence of Fourier polynomials of the square wave function f in Example 1.6 will converge to f at every point of $[-\pi, \pi]$? Give a reason for your answer.

6. What do you think is the best way of defining the "distance" between two functions?

7. Write down as many ways as you can think of measuring the distance between two subsets of $\mathbb{R}^2$.

Chapter 2

Distances, distances and nothing but distances

In which we have our first taste of metric spaces and solve some of the problems encountered in Chapter 1.

The examples we looked at in the previous chapter suggests that there are many different kinds of sets for which it may be quite useful to have available a way of measuring the "distance" between elements of the set. We also saw that this would really be essential if we want to talk about things such as approximation and convergence. In fact, a notion of distance appears to be *all* we need to do this. So let's be quite bold, and try a "minimalist" approach. To be more precise, let's see what we can say about sets that come equipped only with a notion of distance between any two elements of the set.

Our first problem is to get more clarity on what we mean by "a notion of distance". The situation we have in mind is the following: we are given a set X, and for any two elements x and y of X, we have something that we want to call "the distance between x and y"; let's write $d(x, y)$ for this. We can think of d as a *function* that assigns to the (ordered) pair (x, y) of elements of X a real number, the distance between x and y. The question now becomes: "What properties should the function d have to qualify as a *distance function*?" Qualification will obviously depend on the criteria we use. Ideally we would like to limit the number of properties to those that are essential to allow us to say something useful about things like approximation and convergence.

Let's start with a few properties that are so natural that one would not like to call anything which does not possess these properties a distance function.

- The distance between two distinct elements should be a *positive* real number.
- The distance between two elements should be 0 if the two elements are identical.
- The distance between two elements should be 0 only if the two elements are identical.
- The distance between two elements should not depend on the order in which we list them.

We'll soon discover that there is very little one can do without these properties. There is another one that is as important, but not quite so easy to spot. If we have three elements of a set X, say x, y and z, we can find $d(x,y), d(x,z)$ and $d(y,z)$. Are these distances related in any way? To try and answer this question, let's look at a notion of distance we're familiar with: distance between points in the plane. Figure 2 shows that we may be very fortunate and have $d(x,z) = d(x,y) + d(y,z)$, but that in general the best we can hope for is that

$$d(x,z) \leq d(x,y) + d(y,z). \tag{2.1}$$

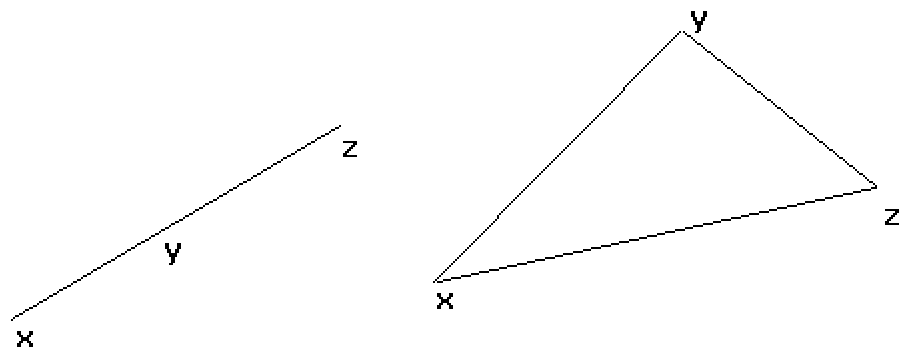


Figure 2.1: Distances between three points in the plane.

It should also be clear from the sketch why this property is usually referred to as the **triangle inequality**.

You have probably come across this name for the inequality

$$|a+b| \leq |a| + |b|, \tag{2.2}$$

where a and b are real numbers. You may remember that this inequality was quite useful in Real Analysis. The distance between two real numbers x and y on the number line is given by $|x-y|$, and if we put $a = x - y, b = y - z$ in (2.2), the inequality above becomes

$$|x-z| = |(x-y)+(y-z)| \le |x-y| + |y-x|.$$

This shows that the familiar triangle inequality for real numbers is just a special case of the inequality (2.1) if we put $d(x, y) = |x - y|$.

It is quite surprising that the properties we have listed so far are the only ones we need to develop a remarkably rich and useful theory. The mathematical structure we want to investigate will be nothing more than a set together with a distance function satisfying these properties. Such a structure is traditionally called a "metric space" and a "distance function" is known as a "metric". We are now ready to write down a definition at last.

Recall that if X is a non-empty set, then $X \times X$ stands for the set of all ordered pairs (x, y), with x and y in X.

Definition 2.1 A ***metric*** *on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all x and y in X:*

(M1) $d(x, y) = 0$ if and only if $x = y$ *(separation property)*

(M2) $d(y, x) = d(x, y)$ *(symmetry property)*

(M3) $d(x, z) \le d(x, y) + d(y, z)$ *(triangle inequality)*

A ***metric space*** *is a non-empty set X which has a metric d defined on it; we refer to it as "the metric space (X, d)".*

The notion of a metric space has two components: a *set* X and a *metric* d. If we want to talk about a particular metric space, we should therefore specify both, and use notation that reflects this. This is particularly important since it is quite possible to have more than one metric on a set (as we'll see shortly). However, there are many situations in which there is no danger of confusion, and we'll often lapse into simply talking about "the metric space X" instead of the more pedantic "the metric space (X, d)".

The definition of a metric space given above was first formulated by Maurice Fréchet in his doctoral thesis of 1906; the name "metric space" was first used by Felix Hausdorff in 1914.

There is an immediate objection you could raise to the definition above: It says nothing about a distance having to be non-negative. The reason for this is that one gets this for free:

Proposition 2.2 I*n any metric space* (X, d), $d(x, y) \geq 0$ *for all* x *and* y *in* X.

Proof: For all x and y in X,

$$0 = d(x, x) \leq d(x, y) + d(y, x) = 2d(x, y).$$

■

A definition without examples is no good. Fortunately we do not have look very far to find examples of metric spaces. In the process we'll begin to address some of the problems raised in Chapter 1.

Example 2.3 The *usual metric* or *Euclidean metric* on the set $\mathbb{R}$ of real numbers is given by

$$d(x, y) = |x - y|, \qquad x, y \in \mathbb{R}.$$

It is easy to check, using the properties of the absolute value (see Appendix A.1) that this is indeed a metric. Since this is our first example, we give the details:

(M1) If $x = y$, then $d(x, y) = |x - y| = |x - x| = |0| = 0$, and if $d(x, y) = 0$, then $0 = d(x, y) = |x - y|$, so $x - y = 0$, or $x = y$.

(M2) $d(y, x) = |y - x| = |-(x - y)| = |x - y| = d(x, y)$.

(M3) $d(x, z) = |x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z)$.

This is the natural way of measuring the distance between real numbers, and the metric that we'll use when looking at the problems raised in Example 1.1

Example 2.4 The *usual* or *Euclidean metric* on $\mathbb{R}^2$ is really just the familiar distance between points in the plane. If $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$, then

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

The properties M1 and M2 are easy to check (do this!); the triangle inequality needs more work:

$$\begin{aligned}
[d(\mathbf{x}, \mathbf{z})]^2 &= (x_1 - z_1)^2 + (x_2 - z_2)^2 \\
&= (x_1 - y_1 + y_1 - z_1)^2 + (x_2 - y_2 + y_2 - z_2)^2 \\
&= (x_1 - y_1)^2 + 2(x_1 - y_1)(y_1 - z_1) + (y_1 - z_1)^2 \\
&\qquad + (x_2 - y_2)^2 + 2(x_2 - y_2)(y_2 - z_2) + (y_2 - z_2)^2 \\
&= [d(\mathbf{x}, \mathbf{y})]^2 + 2(\mathbf{x} - \mathbf{y}).(\mathbf{y} - \mathbf{z}) + [d(\mathbf{y}, \mathbf{z})]^2 \\
&\leq [d(\mathbf{x}, \mathbf{y})]^2 + 2d(\mathbf{x}, \mathbf{y})\, d(\mathbf{y}, \mathbf{z}) + [d(\mathbf{y}, \mathbf{z})]^2 \\
&= [d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})]^2
\end{aligned}$$

The inequality in the second last line needs a bit of explanation. Recall that if $\mathbf{u} = (u_1, u_2), \mathbf{v} = (v_1, v_2)$ are two elements of $\mathbf{R}^2$, then the norm of $\mathbf{u}$ is defined by $||\mathbf{u}|| = \sqrt{u_1^2 + u_2^2}$ and the inner product of $\mathbf{u}$ and $\mathbf{v}$ by $\mathbf{u}.\mathbf{v} = u_1v_1 + u_2v_2$. The Euclidean metric and the norm are related by the equation $||\mathbf{x} - \mathbf{y}|| = d(\mathbf{x}, \mathbf{y})$. The Cauchy-Schwarz inequality (see Appendix A.3) says that $|\mathbf{u}.\mathbf{v}| \leq ||\mathbf{u}||\,||\mathbf{v}||$. Taking $\mathbf{u} = \mathbf{x} - \mathbf{y}$ and $\mathbf{v} = \mathbf{y} - \mathbf{z}$, the inequality in the second last line follows from the Cauchy-Schwarz inequality. The triangle inequality now follows from the inequality above on taking square roots on both sides of the inequality.

Example 2.5 The usual metric on the set $\mathbf{C}$ of complex numbers is defined in a similar way to the usual metric for real numbers. For $z, w \in \mathbf{C}$, we put

$$d(z, w) = |z - w|.$$

The check that this defines a metric is very similar to that for the usual metric on $\mathbf{R}$, this time using the properties of the modulus of a complex number (see Appendix A.2).

If $z = a+ib, w = c+id$, it follows from the definition of the modulus of a complex number that

$$d(z, w) = |(a + ib) - (c + id)| = |(a - c) + i(b - d)| = \sqrt{(a - c)^2 + (b - d)^2}.$$

If this makes you feel that in some sense this is really the same example as the previous one, you are not wrong. In Chapter 6 we'll make precise the phrase "in some sense the same."

Example 2.6 Example 2.4 can be generalized to $\mathbf{R}^n$. For any positive integer n, the *Euclidean metric* on $\mathbf{R}^n$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + ... + (x_n - y_n)^2},$$

where $\mathbf{x} = (x_1, x_2, ..., x_n), \mathbf{y} = (y_1, y_2, ..., y_n)$. The proof that this is a metric is essentially the same as in the special case $n = 2$ that we looked at above. (See also Appendix A.3.)

Example 2.7 We mentioned above that the same set may have more than one metric. To substantiate this claim, we give another metric on $\mathbf{R}^2$. For $\mathbf{x} = (x_1, x_2), \mathbf{y} = (y_1, y_2) \in \mathbf{R}^2$ put

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|.$$

It is easy to check that this defines a metric on $\mathbf{R}^2$; all you'll need are the properties of the absolute value. Do this check! Also try to explain why this metric is sometimes called the *taxi metric*. Figure 2.2 may be helpful.

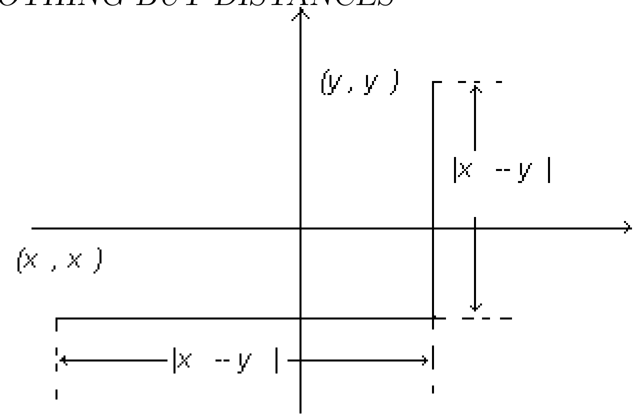


Figure 2.2: The taxi metric on $\mathbf{R}^2$.

In future, when we talk about "the usual metric on $\mathbf{R}^n$" we mean the Euclidean metric. If no metric is specified on $\mathbf{R}^n$, the Euclidean metric is assumed. In cases where we consider other metrics on $\mathbf{R}^n$, we shall denote the Euclidean metric by d_E.

Example 2.8 Let B_8 be the set of all binary strings of length 8; 11010011 and 11100101 are two examples of such strings. Let us define the distance between two such strings to be the number of positions in which there entries differ. The distance between the two strings given above, for example, is 4, since their entries differ in positions 3,4, 6 and 7. It is easy to check that this distance function satisfies properties M1 and M2. To verify the triangle inequality, use the fact that for three strings $\mathbf{r} = r_1r_2...r_8, \mathbf{s} = s_1s_2...s_8$ and $\mathbf{t} = t_1t_2...t_8$, the strings $\mathbf{r}$ and $\mathbf{t}$ can only differ in those position where either the strings $\mathbf{r}$ and $\mathbf{s}$, or the strings $\mathbf{s}$ and $\mathbf{t}$ differ. This metric is known as the *Hamming metric.* There is, of course, nothing special about the number 8; we can define the Hamming metric in the same way for strings of any (finite) length.

Example 2.9 There is a useful extension of the previous example to infinite binary sequences (i.e. infinite sequences with only 0's and 1's as terms). Let X be the set of all such sequences. For $\mathbf{x} = (x_n), \mathbf{y} = (y_n) \in X$, put

$$d(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^{\infty} 2^{-n}|x_n - y_n|.$$

Then d is a metric on X (we ask you to check this in the exercises; make sure that you understand why the factor 2^{-n} should be in each term). The metric space (X, d) is used extensively in the study of *chaotic dynamical systems.* There is a way of associating with a chaotic orbit an infinite binary sequence, and reducing the study of such orbits to the study of orbits of a relatively simple function (the so-called *shift map*) on the space X of binary sequences.

Example 2.10 Let a and b be real numbers with $a < b$, and write $C([a, b])$ for the set of all real-valued continuous functions on the interval $[a, b]$. How should one measure the distance between two such functions f and g? For each x in $[a, b]$ one can find the distance between the two real numbers $f(x)$ and $g(x)$, i.e. $|f(x)-g(x)|$, but this gives infinitely many distances, not one. If one has to choose one of these, it is quite natural to go for the largest one, i.e. the distance between the two functions at the point where they are furthest apart. But is there such a point? A little bit of Real Analysis comes to the rescue here: since f and g are continuous functions, so is $|f - g|$. And since a continuous function attains its maximum on a closed bounded interval, $\max\{|f(x) - g(x)| : x \in [a, b]\}$ exists (see Appendix A.6). So we can put

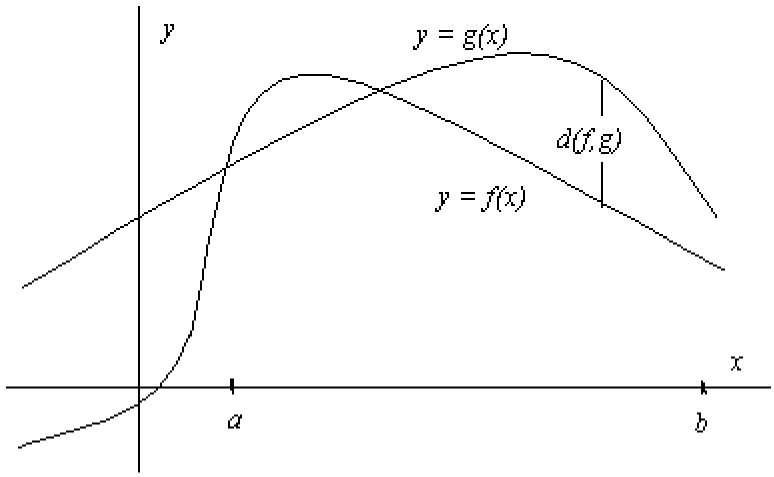


Figure 2.3: The maximum metric for $C([a, b])$.

We check that this is a metric. If $f = g$, then $f(x) = g(x)$ for all $x \in [a, b]$, so clearly $d(f, g) = 0$. If $0 = d(f, g) = \max\{|f(x) - g(x)| : x \in [a, b]\}$, it follows that $|f(x) - g(x)| = 0$ for all $x \in [a, b]$, or $f(x) = g(x)$ for all $x \in [a, b]$. Hence $f = g$. That $d(f, g) = d(g, f)$ follows at once from the fact that $|f(x)-g(x)| = |g(x)-f(x)|$ for all $x \in [a, b]$. To check the triangle inequality, consider the functions f, g and h in $C([a, b])$. For any $x \in [a, b]$,

$$\begin{aligned} |f(x) - h(x)| &= |f(x) - g(x) + g(x) - h(x)| \\ &\leq |f(x) - g(x)| + |g(x) - h(x)| \\ &\leq \max\{|f(x) - g(x)| : x \in [a, b]\} + \max\{|g(x) - h(x)| : x \in [a, b]\} \\ &= d(f, g) + d(g, h). \end{aligned}$$

Since this inequality holds for all x in $[a, b]$, it is true in particular for the point at which $f - h$ attains its maximum. Hence

$$d(f, h) = \max\{|f(x) - h(x)| : x \in [a, b]\} \leq d(f, g) + d(g, h).$$

So we have a metric. A small distance between two functions in terms of this metric means that at every point the distance between the values of the two functions is small, i.e. the graphs of the two functions stay close together. Therefore this metric is the appropriate one to use if one wants to look at the "local" behaviour of functions; it measures the distance between functions at the *point* where they are furthest apart. It is the metric used when measuring the "error" that occurs in approximating functions by their Taylor polynomials (Example 1.5). As we'll see much later, it is also the metric we use to solve the problems posed in Example 1.2. We shall refer to this metric as the *maximum metric*. Note that in this case we could also have written "sup" instead of "max", since the the maximum value is attained, and hence equals the supremum (see Appendix A.4 and A.6) . For this reason this metric is often called the *supremum metric* or *sup metric*. In the exercises (Exercise 8) we'll ask you to look at a set of functions on which the supremum metric makes sense, but not the maximum metric.

Example 2.11 We stay with the set $C([a, b])$, and define another metric on it. For f and g in $C([a, b])$, put

$$d(f, g) = \int_a^b |f(x) - g(x)|\, dx.$$

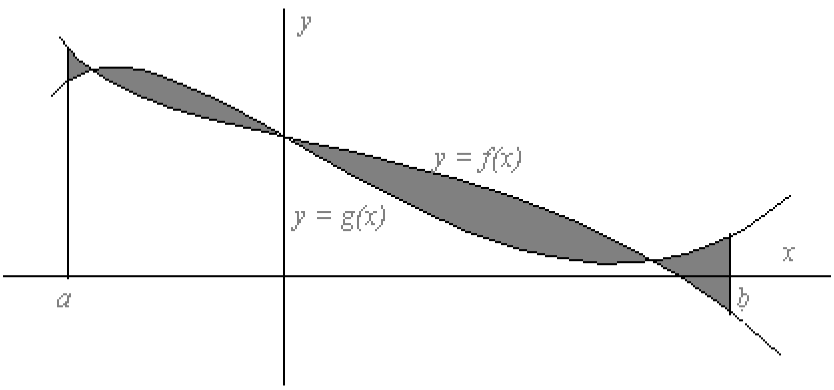


Figure 2.4: The integral metric on $C([a, b]) : d(f, g)$ is the shaded area

It is easy to check the properties M2 and M3 and the fact that $d(f, g) = 0$ if $f = g$ using the properties of the absolute value and the definite integral (see Appendix A.7). It remains to show that if $d(f, g) = 0$, then $f = g$. This equivalent to showing that if $f \neq g$, then $d(f, g) > 0$. Suppose $f \neq g$. Then there is a $u \in [a, b]$ such that

$f(u) \neq g(u)$, or $|f(u)-g(u)| = k > 0$. Since $|f-g|$ is a continuous function, there is an open interval (c,d) contained in $[a,b]$ such that $u \in (c,d)$ and $|f(x)-g(x)| \geq \frac{1}{2}k$ for all $x \in (c,d)$. But then

$$d(f,g) = \int_a^b |f(x)-g(x)|\,dx \geq \int_c^d |f(x)-g(x)|\,dx \geq \frac{1}{2}k(d-c) > 0.$$

We call this the *integral metric* on $C([a,b])$.

The metric in this example "adds up the differences" (in the sense of integration) between the two functions; it is therefore the appropriate one to use if one wants to look at the "global" behaviour of functions. A small distance between two functions when using this metric means that the average distance between the graphs of the the two functions are small (even though they may be quite far apart at certain points). Since Fourier polynomials in general give good global approximations, this metric (or a modification of it) is often used in the study of Fourier approximations.

In the definition of a metric space (X,d) the set X can be *any* non-empty set. A look at the examples we've considered so far shows that in many cases the set X has an algebraic structure as well. Quite often X turns out to be a *vector space*. A vector space has a rather special element, namely the zero element. If there is a metric defined on such a space, one would expect the distance between an element x and the zero element 0 to play a special role. The rather surprising thing is that in this context there is a sense in which it is enough to know the distance $d(0,x)$ between 0 and x. We make this rather vague statement precise in the next definition and proposition. What we do here in general you have come across before in the context of $\mathbf{R}^n$.

Definition 2.12 L*et X be a real or complex vector space. A function which assigns to every $x \in X$ a non-negative real number $||x||$ is called a* **norm** *on X if for every $x, y \in X$ and every scalar λ*

(N1) $||x|| = 0$ *if and only if* $x = 0$*;*

(N2) $||\lambda x|| = |\lambda| ||x||$*;*

(N3) $||x+y|| \leq ||x|| + ||y||$.

A vector space with a norm is called a **normed space***. The property N3 is (not surprisingly) referred to as the triangle inequality for norms.*

Example 2.13 For $\mathbf{x} = (x_1, x_2, ..., x_n) \in \mathbf{R}^n$, put

$$||\mathbf{x}|| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Then it can be checked (and you have probably seen it done in the Linear Algebra course) that this defines a norm on $\mathbf{R}^n$.

The next proposition shows that we can think of every normed space as a metric space. It helps to think of $||x||$ as the distance between 0 and x; the proposition then says that if you know the distance between 0 and x for every x in a vector space, you can find the distance between any two elements in the vector space. The way this can be done is suggested by the way the usual metric on $\mathbf{R}$ is defined in terms of the absolute value: $d(x, y) = |x - y|$.

Proposition 2.14 L*et X be a normed space. For $x, y \in X$ put*

$$d(x, y) = ||x - y||.$$

Then d is a metric on X (and hence (X, d) is a metric space).

Proof: We check that d satisfies the properties M1, M2, M3. For x, y and $z \in X$,

(M1) $d(x, y) = 0$ iff $||x - y|| = 0$ iff $x - y = 0$ iff $x = y$.

(M2) $d(y, x) = ||y - x|| = ||(-1)(x - y)|| = |-1|||x - y|| = ||x - y|| = d(x, y)$.

(M3) $d(x, z) = ||x-z|| = ||(x-y)+(y-z)|| \leq ||x-y||+||y-z|| = d(x, y)+d(y, z)$.

Note how the properties (N1), (N2) and (N3) are used in the proof. ∎

If a norm on a vector space is used in this way to define a metric, we say that the norm **induces** the metric on the space. Note that the norm on $\mathbf{R}^n$ defined in Example 2.13 induces the usual Euclidean metric on $\mathbf{R}^n$.

Example 2.15 Is it possible to define a metric on *any* non-empty set X? It is, and here is how it's done: For $x, y \in X$, put

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y. \end{cases}$$

Check that this is a metric. For the triangle inequality it is best to consider the different cases that can arise ($x = y = z; x = y, y \neq z; \ldots$) separately. This metric is known as the *discrete metric*.

The requirement that the distance between two points should be independent of the order in which they are listed seems a perfectly natural one. There are, however, situations in which this is not necessarily the case. Think, for example, of a situation in which the "distance" between two points is defined as the amount of energy needed to cycle from the first to the second. Now take as the two points the Baxter Theatre and Jameson Hall!

The need for such "distances" does arise more often than one would expect; they are often used in Computer Science, for example. This motivates the following definition. We use $\mathbf{R}^+ = \{a \in \mathbf{R} : a \geq 0\}$.

Definition 2.16 A ***quasi-metric*** *on a set* X *is a function* $d : X \times X \to \mathbf{R}^+$ *such that for all* x *and* y *in* X*:*

(QM1) $d(x, y) = 0$ *if* $x = y$ *and* $x = y$ *if* $d(x, y) = 0 = d(y, x)$ *(separation property)*

(QM2) $d(x, z) \leq d(x, y) + d(y, z)$ *(triangle inequality)*

A ***quasi-metric space*** *is a non-empty set* X *which has a quasi-metric* d *defined on it; we refer to it as "the quasi-metric space* (X, d)*."*

As an example of a quasi-metric we give a simple modification of the discrete metric. For this we need the underlying set X to be a partially ordered set (see Appendix A.8).

Example 2.17 Let X be a non-empty set and $\leq$ a partial order on X. For $x, y \in X$, put

$$d(x, y) = \begin{cases} 0 & \text{if } x \leq y \\ 1 & \text{otherwise.} \end{cases}$$

Then d is a quasi-metric on X (check this!).

If (X, d) is a metric space, then any non-empty subset Y of X can easily be turned into a metric space as well. The metric d is a function on $X \times X$ and $Y \times Y$ is a subset of $X \times X$. The restriction of d to $Y \times Y$ is clearly a metric on Y (check this!). Hence (Y, d) is a metric space as well. Note that we have used the same symbol d for the original function on $X \times X$ and its restriction to $Y \times Y$. Since this will not usually cause any confusion, we'll adopt this slightly imprecise notation. Should there be a need to distinguish between the function d on $X \times X$ and its restriction to $Y \times Y$, we could denote the latter by d_Y. We say that the metric d **induces** the metric d_Y on Y, and we call Y a **subspace** of X.

Example 2.18 The set of rational numbers $\mathbf{Q}$ is a subset of the set of real numbers $\mathbf{R}$. The usual metric on $\mathbf{R}$ induces a metric on $\mathbf{Q}$, so that we can regard $\mathbf{Q}$ as a metric space in its own right; it is a subspace of $\mathbf{R}$.

Example 2.19 The set $C^1([a,b])$ of all real-valued functions on $[a,b]$ with continuous first derivative is a subset of $C([a,b])$. It becomes a metric space in its own right when given the metric induced by the maximum metric of $C([a,b])$ (see Example 2.10).

You will soon notice that we often lapse into geometrical language when talking about metric spaces, even though many of these spaces are very different to the spaces of ordinary geometry. The word "space" is already a case in point. We'll also often talk about the *points*, rather than elements, of a metric space. The next definition gives another example of this kind of geometrical language.

Definition 2.20 L*et (X,d) be a metric space, $a \in X$ and $r > 0$. The* **open ball** *with centre a and radius r is the set*

$$B(a;r) = \{x \in X : d(a,x) < r\}.$$

The **closed ball** *with centre a and radius r is the set*

$$B[a;r] = \{x \in X : d(a,x) \leq r\}.$$

Example 2.21 In $\mathbf{R}$ with its usual metric, $B(3;2)$ is the interval $(1,5)$, and $B[3;2]$ is the interval $[1,5]$. If $\mathbf{R}$ has the discrete metric, $B(3;1)$ consists of the single number 3, and $B[3;1]$ is the whole of $\mathbf{R}$.

Example 2.22 In $\mathbf{R}^2$ with the Euclidean metric $B((0,0),1)$ and $B[(0,0),1]$ are respectively the open and closed disks with centre at the origin and radius 1.

Summary.

The notions of a metric (distance function) on a set and of a metric space were introduced, and a number of important examples given. A norm on a vector space was defined, and it was shown that it can be used to define a metric on the space. Quasi-metrics were introduced, and also the idea of a metric on a set inducing a metric on a subset. Open and closed balls in metric spaces were also defined.

Historical Notes.

Maurice Fréchet (1878 – 1973)

Fréchet was born in Maligny, France in 1878. In his thesis *Sur quelques points du calcul fonctionel*, written in 1906 under the supervision of Hadamard, he introduced the definition of a metric space still in use today. He was a professor of mechanics at the University of Poitiers, professor of mathematics at the University of Strasbourg and professor of mathematics and probability at the University of Paris. He made important contributions to the study of topology, functional analysis, statistics, probability theory and calculus. He died in Paris in 1973.

Felix Hausdorff (1869 – 1942)

Felix Hausdorff was born in Breslau in Germany (now Wroclaw in Poland) in 1869. He graduated from the University of Leipzig in 1891, taught there until 1910 and then accepted a position in Bonn. Hausdorff was a Jew, and despite the fact that he realized as early as 1932 that he was in danger from the Nazi regime, he continued to work in Bonn until he was forced to retire by the government in 1935. His position became more and more difficult; when it became clear in 1942 that he could no longer avoid being sent to an internment camp, he committed suicide together with his wife and sister-in-law.

Hausdorff's most important contributions to mathematics were in the fields of set theory and topology. He did important work on cardinality and formulated a generalized version of Cantor's continuum hypothesis. In his influential book *Grundzüge der Mengenlehre* (Principles of Set Theory) which appeared in 1914 he defined topological spaces and developed the theory of metric and topological spaces. In 1919 he introduced a notion of dimension now called *Hausdorff dimension* which is used in the study of *fractals*. He was the first one to use the name "metric space".

Richard Wesley Hamming (1915 – 1998)

Richard Hamming was born in Chicago in the USA. He received his first degree from the University of Chicago in 1937, followed by a Master's degree from the University of Nebraska and a PhD from the University of Illinois at Urbana-Cham-paign. In 1945 he joined the Manhattan Project which had as its aim the production of an atomic bomb. After the Second World War he worked for Bell telephone Laboratories until 1976, when he became professor of computer science at the Naval Post-graduate School in California.

Hamming did fundamental work on error-detecting and error-correcting codes, starting with a paper which appeared in 1950. The so-called Hamming codes use the Hamming metric introduced in this chapter. Hamming's work on an early computer (the IBM 650) led to the development of a programming language which forms the basis of many of today's high-level computer languages. He also did important work in numerical analysis and developed a technique for smoothing data in preparation for Fourier analysis.

Exercises

1. Check that

   (a) the *taxi metric* is a metric on $\mathbf{R}^2$ (Example 2.7);

   (b) the *discrete metric* is a metric on any non-empty set (Example 2.15).

2. Prove that any positive scalar multiple of a metric is again a metric (i.e. if d is a metric on X and c a positive real number, then d_c defined by $d_c(x, y) = cd(x, y)$ is also a metric on X).

3. For $x, y \in \mathbf{R}$, put
$$d(x, y) = |\arctan x - \arctan y|.$$
   (a) Prove that d is a metric on $\mathbf{R}$.
   (b) A metric d on a set X is a *bounded metric* if there is a $C > 0$ such that $d(x, y) \le C$ for all $x, y \in X$. Prove that the metric d on $\mathbf{R}$ defined above is a bounded metric.
   (c) More generally, let $f : \mathbf{R} \to \mathbf{R}$ be a function, and for $x, y \in \mathbf{R}$, put
$$d_f(x, y) = |f(x) - f(y)|.$$
   Is d_f necessarily a metric on $\mathbf{R}$? If not, what properties should f have to ensure that d_f is a metric? What properties should f have to ensure that d_f is a bounded metric?

4. (a) Let $n \in \mathbf{N}$, B_n be the set of all binary strings of length n and d_n the Hamming metric on B_n (Example 2.8). Show that for $\mathbf{r} = r_1 r_2 ... r_n$, $\mathbf{s} = s_1 s_2 ... s_n \in B_n$,
$$d_n(\mathbf{r}, \mathbf{s}) = \sum_{i=1}^{n} |r_i - s_i|.$$
   (b) Let X be the set of all infinite binary sequences (i.e. infinite sequences with only 0 and 1 as terms). For $\mathbf{x} = (x_n), \mathbf{y} = (y_n) \in X$, put
$$d(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^{\infty} 2^{-n} |x_n - y_n|$$
   (Example 2.9). Prove that d is a metric on X. [Hint: One of the things you'll have to check is that $d(\mathbf{x}, \mathbf{y})$ is always a real number; in other words, that the appropriate series converges.]

5. Show that the function d defined in Example 2.17 is indeed a quasi-metric.

6. Let X be the set of infinite binary sequences. For $\mathbf{x}, \mathbf{y} \in X$, write $\mathbf{x} \le \mathbf{y}$ if $x_n \le y_n$ for all $n \in \mathbf{N}$, and $\mathbf{x} \uparrow n$ for the binary sequence which coincides with $\mathbf{x}$ in the first n terms and has 0 for all the remaining terms. On $X \times X$ define the function d' by
$$d'(\mathbf{x}, \mathbf{y}) = \inf\{2^{-n} : \mathbf{x} \uparrow n \le \mathbf{y} \uparrow n\}.$$
Show that d' is a quasi-metric on X.

7. For $\mathbf{x} = (x_1, x_2, ..., x_n) \in \mathbf{R}^n$, put
$$||\mathbf{x}|| = \sum_{i=1}^{n} |x_i|.$$
Show that this defines a norm on $\mathbf{R}^n$, and write down the metric on $\mathbf{R}^n$ induced by this norm.

8. Let a and b be real numbers with $a < b$, and $B([a,b])$ be the set of all bounded real-valued functions on $[a,b]$. (A function $f : [a,b] \to \mathbf{R}$ is bounded if there is a $C > 0$ such that $|f(x)| \leq C$ for all $x \in [a,b]$.) For $f, g \in B([a,b])$, put
$$d(f,g) = \sup\{|f(x) - g(x)| : x \in [a,b]\}.$$
   (a) Prove that d is a metric on $B([a,b])$. One of the things you will have to check is that $d(f,g)$ is a non-negative real number for every $f, g \in B([a,b])$. For obvious reasons this metric is known as the **sup metric**. This is the set of functions on which the sup metric makes sense, but not the maximum metric. Explain why (see Example 2.10).
   (b) Show that this metric induces the metric of Example 2.10 on $C([a,b])$.
   (c) Let $f(x)$ be the greatest integer less than or equal to x, and $g(x) = 1 - x^2$. Consider f and g as elements of $B([-1,1])$ and calculate $d(f,g)$.

9. For $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ in $\mathbf{R}^2$, put
$$d_\ell(\mathbf{x}, \mathbf{y}) = \begin{cases} |x_1 - y_1| & \text{if } x_2 = y_2 \\ |x_1| + |x_2 - y_2| + |y_1| & \text{if } x_2 \neq y_2. \end{cases}$$
   (a) Show that we always have $|x_1 - y_1| \leq d_\ell(\mathbf{x}, \mathbf{y})$.
   (b) Check that d_ℓ satisfies the properties M1 and M2 of a metric.
   (c) Check that it also satisfies property M3. [Hint: Consider cases. First look at the case $x_2 = z_2$, then at the case $x_2 \neq z_2$.]
   (d) Explain why this metric could be called the "lift metric" . [Hint: Draw a sketch showing the distance between two points not on the y-axis with (i) the same y-coordinate (ii) different y-coordinates.]
   (e) Sketch the following sets:
      i. $B(\mathbf{0}; 1) = \{\mathbf{x} \in \mathbf{R}^2 : d_\ell(\mathbf{0}, \mathbf{x}) < 1\}$;
      ii. $B[(1,0); 2] = \{\mathbf{x} \in \mathbf{R}^2 : d_\ell((1,0), \mathbf{x}) \leq 2\}$.

10. For $z, w \in \mathbf{C}$ put
$$d(z,w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w \end{cases}$$
    (a) Prove that d defines a metric on $\mathbf{C}$.
    (b) Sketch the following balls in $(\mathbf{C}, d)$: (i) B(0,1) (ii) B(i,1) (iii) B[i,1].

11. Let (X, d_X) and (Y, d_Y) be metric spaces. Recall that the *Cartesian product* of the sets X and Y is the set
$$X \times Y = \{(x,y) : x \in X, y \in Y\}.$$
    (a) Prove that the function $d_{X\times Y} : (X \times Y) \times (X \times Y) \to \mathbf{R}$ defined by
$$d_{X\times Y}((x_1,y_1),(x_2,y_2)) = d_X(x_1,x_2) + d_Y(y_1,y_2)$$
    is a metric on $X \times Y$. (This is usually referred to as the *product metric*) on $X \times Y$.
    (b) Let $X = \mathbf{R}, Y = \mathbf{R}$ and the metric on both X and Y be the usual metric on $\mathbf{R}$. What is the product metric in this case?

12. Let d be a metric on a set X. Show that
$$d^*(x,y) = \frac{d(x,y)}{1 + d(x,y)}, \qquad x, y \in X$$
also defines a metric on X, and that d^* is a bounded metric (see Question 3(b) for the definition of a bounded metric).

Chapter 3

Nearness taken to the limit

In which we digest the notion of convergence of sequences in metric spaces.

In the examples in Chapter 1 we considered convergence of sequences of real numbers, but also of sequences of functions and sequences of sets. This suggests that it would be quite useful to have a notion of convergence of sequences available in *any* metric space. But how should we go about defining convergence in this general context?

Distance plays a crucial role in the way we think about convergence of sequences of real numbers. Saying that the sequence (x_n) of real numbers converges to the real number x means that the distance between x_n and x is getting smaller and smaller as n increases, or, more precisely, converging to 0. This is a bit of a cop-out, you may well say. What does it mean for a sequence of distances to converge to 0? The point is that a distance is a real number, and convergence of real numbers is something you have studied in detail before (in the Real Analysis course).There you saw that this is a notion one can define rigorously. So we can justifiably assume that we know what it means for a sequence of real numbers to converge to 0, and use the properties that such sequences have. This is exactly what we'll do in future.

So how does that help us to get to grips with convergent sequences in metric spaces? If (x_n) is a sequence in a metric space, and we want to test whether this sequence converges to the point x in the metric space, it makes sense to ask: "Does the distance between x_n and x tend to 0?" Since $(d(x_n, x))$ is a sequence of real numbers, we're on familiar ground. The definition of the convergence of a sequence in a metric space should come as no surprise now.

Definition 3.1 L*et (X, d) be a metric space, (x_n) a sequence in X and $x \in X$. We* *say the sequence* (x_n) *converges to* x *(with respect to the metric* d*) if* $d(x_n, x) \to 0$ *as* $n \to \infty$. *We write this*

$$x_n \to x \text{ as } n \to \infty, \text{ or } \lim_{n\to\infty} x_n = x.$$

The advantage of this definition is that to test whether a sequence in a metric space converges, we only have to check whether a sequence of real numbers converges to 0. We can therefore use everything we know about convergence of real sequences to prove things about convergence of sequences in metric spaces.

If you would prefer a definition that makes no reference to convergence of real sequences, we can of course translate the above definition into an "epsilon form" by using the definition of the convergence of a sequence of real numbers (see Appendix A.5). It then looks like this:

- *The sequence* (x_n) *in the metric space* (X, d) *converges to the element* $x \in X$ *if for every* $\epsilon > 0$ *there exists a natural number* N_ϵ *such that whenever* $n \geq N_\epsilon$, $d(x_n, x) < \epsilon$.

We'll occasionally (but not often) use the definition in this form.

The element x in the definition is called **a limit** of the sequence (x_n). Surely we should say **the** limit? Yes, but only after the next result.

Proposition 3.2 L*et* (x_n) *be a sequence in a metric space* (X, d) *and suppose* $x_n \to x$ *and* $x_n \to y$, *where* $x, y \in X$. *Then* $x = y$.
[In words: The limit of a convergent sequence in a metric space is unique.]

Proof: For any $n \in \mathbf{N}$,

$$d(x, y) \leq d(x, x_n) + d(x_n, y).$$

If we let $n \to \infty$ in this inequality and use the fact that $d(x, x_n) \to 0$ and $d(y, x_n) \to 0$, we get

$$0 \leq d(x, y) \leq 0 + 0 = 0,$$

so $d(x, y) = 0$. But then $x = y$. ■

We now look at convergence of sequences in a number of different metric spaces. The first one is not really new:

Example 3.3 In $\mathbf{R}$ a sequence converges with respect to the usual metric if and only if it converges in the usual sense.

Example 3.4 We'll call a sequence (x_n) (in any set) **eventually constant** if there is an $m \in \mathbf{N}$ such that $x_n = x_m$ for all $n > m$ (i.e. from a certain term on all the terms are equal). It is clear that in any metric space an eventually constant sequence converges (since, with m as above, $d(x_n, x_m) = 0$ for all $n \geq m$). In a discrete metric space these are the *only* convergent sequences, since in this case we can only have $d(x, x_n) < 1$ if $x = x_n$.

Example 3.5 If there is more than one metric defined on a particular set, a sequence in the set may well converge with respect to one metric, but not with respect to the other. As a simple example, consider the sequence $(\frac{1}{n})$ in $\mathbf{R}$. It converges to 0 with respect to the usual metric on $\mathbf{R}$, but not with respect to the discrete metric.

Example 3.6 Let (z_n) be a sequence in $\mathbf{C}$, with $z_n = x_n + iy_n$. Suppose z_n converges to $z = x + iy$ with respect to the usual metric of $\mathbf{C}$, i.e.

$$|z_n - z| = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0 \text{ as } \quad n \to \infty.$$

Then

$$0 \leq |x_n - x| \leq |z_n - z| \to 0, \quad \text{and } 0 \leq |y_n - y| \leq |z_n - z| \to 0.$$

Since $z_n \to z$, $|z_n - z| \to 0$, and therefore it follows from the inequality above that $|x_n - x| \to 0$. This means that $x_n \to x$ as $n \to \infty$; in the same way it follows that $y_n \to y$ as $n \to \infty$.
Conversely, suppose that $x_n \to x$ and $y_n \to y$ as $n \to \infty$. Then

$$\begin{aligned}
\lim_{n\to\infty} |z_n - z| &= \lim_{n\to\infty} \sqrt{(x_n - x)^2 + (y_n - y)^2} \\
&= \sqrt{\lim_{n\to\infty}((x_n - x)^2 + (y_n - y)^2)} \\
&= \sqrt{\lim_{n\to\infty}(x_n - x)^2 + \lim_{n\to\infty}(y_n - y)^2} \\
&= \sqrt{(\lim_{n\to\infty} x_n - x)^2 + (\lim_{n\to\infty} y_n - y)^2} \\
&= \sqrt{(x - x)^2 + (y - y)^2} \\
&= 0,
\end{aligned}$$

using properties of convergent real sequences (see Appendix A.5). This shows that $z_n \to z$ as $n \to \infty$ with respect to the usual metric in $\mathbf{C}$ if and only if $\text{Re}z_n \to \text{Re}z$ and $\text{Im}z_n \to \text{Im}z$ as $n \to \infty$. Hence we can check whether a sequence in $\mathbf{C}$ converges with respect to the usual metric on $\mathbf{C}$ by checking whether two real sequences converge with respect to the usual metric on $\mathbf{R}$.

We can formulate a very similar result in $\mathbf{R}^2$, and in fact more generally in $\mathbf{R}^m$, for any $m \geq 2$. To do this concisely, we introduce the following terminology:

Definition 3.7 *Let $m \in \mathbf{N}$ with $m \geq 2$. Let $(\mathbf{x}_n)$ be a sequence in $\mathbf{R}^m$ and $\mathbf{x} \in \mathbf{R}^m$, where $\mathbf{x} = (x_1, x_2, \ldots, x_m)$ and $\mathbf{x}_n = (x_{n1}, x_{n2}, \ldots, x_{nm})$. We say that $(\mathbf{x_n})$ converges* ***coordinatewise*** *to $\mathbf{x}$ if $x_{nk} \to x_k$ in $\mathbf{R}$ for $k = 1, 2, \ldots, m$.*

Example 3.8 A sequence in $\mathbf{R}^m$ (where $m \geq 2$) converges with respect to the usual Euclidean metric of $\mathbf{R}^m$ if and only if it converges coordinatewise. This can be proved in much the same way as the result for complex sequences in Example 3.6. Do this!

Example 3.9 A sequence (f_n) in the metric space $B([a, b])$ (see Exercise 8 of Chapter 2) converges to $f \in B([a, b])$ with respect to the supremum metric if and only if $\sup\{|f_n(x) - f(x)| : x \in [a, b]\} \to 0$ as $n \to \infty$. If this is the case we say that the sequence (f_n) of functions converges **uniformly** to f on $[a, b]$. This is a far stronger requirement than simply asking that $f_n(x) \to f(x)$ as $n \to \infty$ for every $x \in [a, b]$; the latter is referred to as **pointwise** convergence. You have seen this terminology in the Real Analysis course; what is new is that you can now think of uniform convergence as convergence with respect to a metric.

We conclude this chapter by restating the definition of the convergence of a sequence in terms of open balls. This form of the definition will turn out to be useful later.

Proposition 3.10 A *sequence (x_n) in a metric space (X, d) converges to an $x \in X$ if and only if for every $\epsilon > 0$ there is an $n_\epsilon \in \mathbf{N}$ such that for $n \geq n_\epsilon$ we have $x_n \in B(x, \epsilon)$.*

Proof: This is just a restatement of the "epsilon" definition of convergence. ■

Summary.

The notion of the convergence of a sequence in a metric space to a point (limit) in the space was introduced, and the limit shown to be unique. Coordinatewise convergence in $\mathbf{R}^m$ and uniform convergence of bounded functions were identified as convergence with respect to appropriate metrics.

Exercises

1. Check that in any metric space the eventually constant sequences always converge. Show that in a discrete metric space these are the *only* convergent sequences.

2. Let (f_n) be a sequence in $B([0,1])$ and $f \in B([0,1])$. Show that if $f_n \to f$ uniformly, $f_n \to f$ pointwise. Give an example to show that the converse is not true.

3. Let n be a positive integer and B_n the set of all binary strings of length n.

   (a) How many elements does B_n have?

   (b) Describe the sequences in B_n that converge with respect to the Hamming metric.

   (c) Let d be any metric on B_n. Describe the sequences in B_n that converge with respect to d. [Hint: Your answer to (a) may help you here.]

4. Let X be any non-empty set with finitely many elements, and d a metric on X. Describe the convergent sequences in (X, d).

5. The metrics d_m and d_1 on $C([0,1])$ are defined by

$$\begin{array}{ll} d_m(f,g) & = \max\{|f(x) - g(x)| : x \in [0,1]\} \\ d_1(f,g) & = \int_0^1 |f(x) - g(x)|\, dx. \end{array}$$

   (a) Show that if a sequence in $C([0,1])$ converges with respect to d_m, then it converges with respect to d_1, and the limits are the same.

   (b) Is the converse true? Give a proof or counter-example to substantiate your answer.

6. Let Σ_2 be the set of all infinite binary sequences (i.e. sequences with only 0 and 1 as terms), with the metric d defined in Example 2.9.

   (a) Let $\mathbf{x}, \mathbf{y} \in \Sigma_2$ and $n \in \mathbf{N}$. Show that if $x_i = y_i$ for $i = 1, 2, ..., n$, then $d(\mathbf{x}, \mathbf{y}) \leq 2^{-n}$, and that if $d(\mathbf{x}, \mathbf{y}) < 2^{-n}$, then $x_i = y_i$ for $i = 1, 2, ..., n$.

   (b) Let $\mathbf{x} \in \Sigma_2$. Give an example of a sequence in Σ_2 (other than an eventually constant sequence) that converges to $\mathbf{x}$ with respect to d.

7. Let d_ℓ be the lift metric on $\mathbf{R}^2$ (see Exercise 9 of Chapter 2). Which sequences in $\mathbf{R}^2$ converge with respect to this metric? Do not simply repeat the definition; try to find a description of the convergent sequences which does not mention the metric. You may have to consider more than one case.

8. Let d be the metric on $\mathbf{C}$ defined by

$$d(z,w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w \end{cases}$$

   Which sequences in $\mathbf{C}$ converge with respect to this metric?

Chapter 4

Sets which keep their limits inside.

In which we first meet closed sets and sample their properties.

One of the fairly elementary but still very useful results about convergent sequences of real numbers is the following:

If (x_n) is a convergent sequence of real numbers and $a \leq x_n \leq b$ for every $n \in \mathbf{N}$, then $a \leq \lim_{n\to\infty} x_n \leq b$.

We could express this by saying that the closed interval $[a, b]$ has the property that it contains the limit of every convergent sequence in it, i.e. if every term of a sequence is in this interval, and the sequence converges, then the limit of the sequence is also in the interval. Other kinds of intervals, like the open interval (a, b) or the half-open intervals $(a, b]$ and $[a, b)$, do not possess this property.

Although there is no way we can define the notion of an interval in a general metric space, there does seem to be some advantage in looking at sets in a metric space that are well-behaved in the sense that you can't go outside them by taking the limit of a sequence in them. One could say that such sets are "closed under taking limits", and therefore simply calling them *closed sets* does not seem to be such a bad idea. Here is the formal definition:

Definition 4.1 *Let (X, d) be a metric space and A a subset of X. We say A is* **closed** *if whenever $x_n \in A$ for every $n \in \mathbf{N}$ and $x_n \to x \in X$ as $n \to \infty$, then $x \in A$.*
In words: A is closed if it contains the limit of every convergent sequence in it.

It is important to note that to check whether a set is closed one only needs to consider sequences in the set which are *convergent*; this means that one can assume that the sequence has a limit, and need only check whether the limit is also in the set. For a set to be closed, *every* convergent sequence in the set must have its limit in the set as well. On the other hand, one convergent sequence in the set with a limit that is not in the set is enough to show that the set is not closed.

You will notice that the letter F is often used to denote closed sets; this can probably be traced back to the French word for closed ("fermé").

Example 4.2 Let $\mathbf{R}$ have its usual metric.

- Every closed bounded interval in $\mathbf{R}$ is a closed set. This follows from the result mentioned at the start of the chapter.
- Intervals of the form $(-\infty, b]$ and $[a, \infty)$ are also closed. This can be proved in the same way.
- $\mathbf{R}$ itself is closed.
- The set $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, ...\}$ is not closed, since the sequence $(\frac{1}{n})$ is in the set, but its limit, 0, is not in the set.
- The set $\{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, ...\}$ is closed. To see this, note that convergent sequences in this set are either eventually constant, and therefore have their limits in the set, or otherwise are, at least from a certain term onwards, a subsequence of the sequence $(\frac{1}{n})$, and so converges to 0, which is in the set.
- The set $\mathbf{N}$ is closed. Since every two distinct elements of this set are a distance of at least 1 apart, the only convergent sequences in the set are the eventually constant sequences, and all their limits are therefore in the set itself.
- An important example of a subset of $\mathbf{R}$ which is not closed is the set $\mathbf{Q}$ of rational numbers. To see this, all you have to do is to find an example of a sequence of rational numbers which converges to an irrational number. Find one!

Example 4.3 If $\mathbf{C}$ has its usual metric, the set

$$A = \{z \in \mathbf{C} : 0 \leq \mathrm{Re}z \leq 1, -1 \leq \mathrm{Im}z \leq 2\}$$

is closed. For suppose (z_n) is a convergent sequence in A with limit $z \in \mathbf{C}$. Then $0 \leq \mathrm{Re}z_n \leq 1$ and $\mathrm{Re}z_n \to \mathrm{Re}z$; similarly $-1 \leq \mathrm{Im}z_n \leq 2$ and $\mathrm{Im}z_n \to \mathrm{Im}z$. Both $(\mathrm{Re}z_n)$ and $(\mathrm{Im}z_n)$ are sequences of real numbers converging with respect to the usual metric on $\mathbf{R}$. Since closed intervals in $\mathbf{R}$ are closed, we have $0 \leq \mathrm{Re}z \leq 1$ and $-1 \leq \mathrm{Im}z \leq 2$. This shows that $z \in A$, and so A is closed.

Example 4.4 In any metric space, the whole space is closed; this follows immediately from the definition of convergence. Perhaps a little more surprisingly, the empty set is also closed (since it contains no sequences, there is nothing to check; we say the condition is satisfied *vacuously*). Sets consisting of only one element (we call such sets **singletons**) are closed since the only sequences they can contain are constant sequences. In much the same way it follows that any finite set is closed; the only convergent sequences such sets can contain are eventually constant sequences.

Example 4.5 Let $C([a, b])$ have the maximum metric. The set

$$F = \{f \in C([a, b]) : f(a) = 0\}$$

is closed. To see this, let (f_n) be a sequence in F, and suppose $f_n \to f \in C([a, b])$. Then (f_n) converges to f uniformly, and hence also pointwise. It follows that $f(a) = \lim_{n\to\infty} f_n(a) = 0$, since $f_n(a) = 0$ for every $n \in \mathbf{N}$. Hence $f \in F$, and so F is closed. The set $A = \{f \in C([a, b]) : f(a) > 0\}$ is not a closed subset of $C([a, b])$. To show this, it is enough to find a sequence in A which converges to a function not in A. The sequence (f_n), with $f_n(x) = n^{-1}$ for all $x \in [a, b]$ is an example of such a sequence.

In Chapter 2 we defined what we mean by a *closed ball* in a metric space. It would be a pity if such closed balls are not closed in the above sense. Fortunately, all is well:

Proposition 4.6 *In any metric space a closed ball is a closed set.*

Proof: Let (X, d) be a metric space, $a \in X$ and $r > 0$. We want to show that $B[a; r] = \{x \in X : d(a, x) \leq r\}$ is closed in X. Let (x_n) be a sequence in $B[a; r]$ such that $x_n \to x \in X$ as $n \to \infty$. Then $d(a, x_n) \leq r$ for every $n \in \mathbf{N}$, and so

$$d(a, x) \leq d(a, x_n) + d(x_n, x) \quad \text{for every } n \in \mathbf{N}.$$

Let $n \to \infty$ in the above inequality, then we get

$$d(a, x) \leq r + 0 = r.$$

This shows that $x \in B[a; r]$, and so $B[a; r]$ is closed. ■

Once we have some examples of closed sets, we can now ask whether there is any way in which we can combine them to make further closed sets. The next two results show that taking unions and intersections produce more closed sets. There is an important distinction between using unions and intersections, though. Watch out for it! To start, we need a simple but useful lemma.

Lemma 4.7 *If a sequence (x_n) in a metric space (X,d) converges to $x \in X$, then any subsequence of (x_n) also converges to x.*

Proof: Exercise. If you get stuck, look at the proof of the same result for real sequences in your Real Analysis course. ■

Theorem 4.8 *In any metric space, the union of two closed sets is again a closed set.*

Proof: Let F_1 and F_2 be two closed subsets of the metric space (X,d), and suppose (x_n) is a sequence in $F_1 \cup F_2$ such that $x_n \to x \in X$ as $n \to \infty$. Then at least one of the two sets F_1 and F_2 contains infinitely many terms of the sequence (x_n). Suppose, for the sake of the argument, it is F_1. Then F_1 contains a subsequence of (x_n), and this subsequence also converges to x, by Lemma 4.7. Since F_1 is closed, we must have $x \in F_1 \subset F_1 \cup F_2$. This shows that $F_1 \cup F_2$ is closed. ■

Corollary 4.9 *The union of any finite number of closed subsets of a metric space is closed.*

Proof: This follows easily by induction on the number of subsets in the union. Write out the proof for yourself! ■

What can we say about the union of an infinite number of closed subsets? The proof we have above will not work in this case. But this may just mean that we have to look for another proof, difficult as that may seem. If we can find a counter-example, though, that will settle the question once and for all. The question now becomes: Is it possible to find an example of an infinite family of closed subsets of a metric space such that the union of this family is not closed? We do not have to look very far: In $\mathbf{R}$, the infinite family $\{[\frac{1}{n}, 2] : n \in \mathbf{N}\}$ of closed intervals has as its union the interval $(0,2]$, which is not closed.

For intersections we need not make any exceptions:

Theorem 4.10 *The intersection of any family of closed sets in a metric space is again a closed set.*

Proof: Let $\{F_i : i \in I\}$ be a family of closed sets in the metric space (X,d), and put $F = \cap\{F_i : i \in I\}$. (The set I is an *index set*, used to index, or label, the members of the family of sets. It could be a finite, countable or uncountably infinite set.) Let (x_n) be a sequence in F, and suppose $x_n \to x \in X$ as $n \to \infty$. For each

$i \in I$, (x_n) is a sequence in F_i, and since F_i is closed, we must have $x \in F_i$. But this shows that $x \in F = \cap\{F_i : i \in I\}$, so F is closed. ■

We list below three very important properties of closed sets in metric spaces that we have derived so far. We shall see later that they play a crucial role when we want to generalise the concept of a metric space even further.

In any metric space:

1. The whole space is closed, and so is the empty set.
2. The union of a finite number of closed sets is again a closed set.
3. The intersection of an arbitrary family of closed sets is again a closed set.

A set in a metric space that is not closed fails to be closed because there is at least one convergent sequence with terms in the set, but with its limit not in the set. We can try to enlarge such a set to make it closed; to do this we'll have to include the limits of all convergent sequences which have their terms in the set. When we do this we get a set called the closure of the original set. Here is the precise definition:

Definition 4.11 *Let A be a subset of the metric space (X, d). The* **closure** *of A, denoted by $\overline{A}$, is the set of all limits of convergent sequences in A, i.e.*

$$\overline{A} = \{x \in X : \text{ there is a sequence } (x_n) \text{ such that } x_n \in A \text{ for every } n \text{ and } x_n \to x\}$$

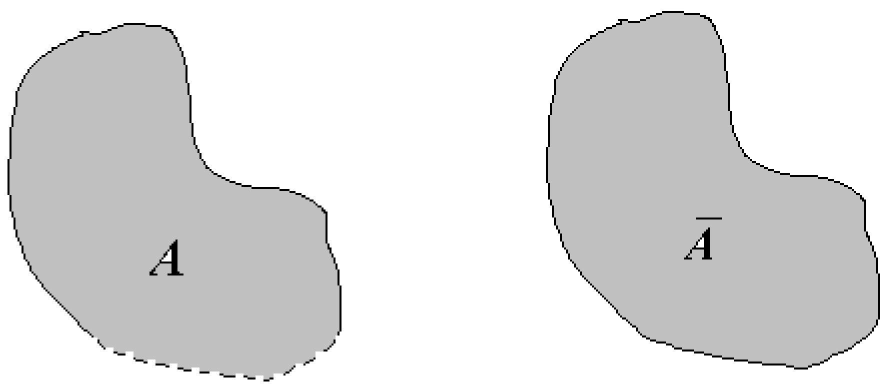


Figure 4.1: On the left is a set A in $\mathbf{R}^2$; on the right its closure $\overline{A}$ ($\mathbf{R}^2$ has its usual metric).

Note that it follows easily from this definition that every set is contained in its closure. For if $a \in A$, then we can put $a_n = a$ for every $n \in \mathbf{N}$. Then (a_n) is a sequence in A which converges to a, and therefore $a \in \overline{A}$.

Example 4.12 Let $\mathbf{R}$ have its usual metric. From the remark above it follows that $(0, 1)$ is a subset of the closure of $(0, 1)$. But we can write both 0 and 1 as limits of sequences in $(0, 1)$, and therefore both are in the closure as well. Therefore $[0, 1]$ is contained in the closure of $(0, 1)$. On the other hand, if (x_n) is a convergent sequence in $(0, 1)$ with limit x, then we must have $0 \leq x \leq 1$. This shows that the closure of $(0, 1)$ is contained in $[0, 1]$. Putting this all together, we find that the closure of $(0, 1)$ is exactly $[0, 1]$. In the same way we can also show that the interval $[0, 1]$ is the closure of both the sets $(0, 1]$ and $[0, 1)$.

Example 4.13 The previous examples were perhaps not unexpected; a little more interesting is the fact that the closure of the set $\mathbf{Q}$ of rational numbers in the metric space $\mathbf{R}$ (with its usual metric) is $\mathbf{R}$ itself. This follows from the fact that every real number is the limit of a sequence of rational numbers, and that $\mathbf{R}$ is therefore contained in the closure of $\mathbf{Q}$.

Example 4.14 Let $\mathbf{C}$ have its usual metric. Then the closure of the set $B(0; 1) = \{z \in \mathbf{C} : |z| < 1\}$ is the set $B[0; 1] = \{z \in \mathbf{C} : |z| \leq 1\}$. This seems entirely reasonable; as a challenge, see if you can prove it. [Hint: Show that every convergent sequence in $B(0; 1)$ has its limit in $B[0; 1]$, and that every z with $|z| = 1$ is the limit of a sequence in $B(0; 1)$.]

We would, of course, like the closure of A to be closed. This is not entirely obvious. The reason is that by adjoining limits of sequences to A, we may be creating new convergent sequences in the closure of A, and we have to check that their limits lie in the closure again. In the next proposition we prove this, and some other useful properties of the closure.

Proposition 4.15 *Let A be a subset of a metric space (X, d). Then*

1. *$A \subset \overline{A}$;*
2. *$\overline{A}$ is closed;*
3. *A is closed if and only if $A = \overline{A}$;*
4. *$\overline{\overline{A}} = \overline{A}$.*

Proof:

1. This was proved in the remark following the definition.

2. Let (x_n) be a sequence in $\overline{A}$ such that $x_n \to x \in X$. Since $x_n \in \overline{A}$, we can find, for every n, a sequence (x_{nm}) in A such that $x_{nm} \to x_n$ as $m \to \infty$. We can therefore choose an $m_n \in \mathbf{N}$ such that $d(x_{nm_n}, x_n) < \frac{1}{n}$. Then (x_{nm_n}) is a sequence in A, and

$$\begin{aligned} d(x, x_{nm_n}) &\leq d(x, x_n) + d(x_n, x_{nm_n}) \\ &< d(x, x_n) + \tfrac{1}{n} \\ &\to 0 \text{ as } n \to \infty. \end{aligned}$$

   Hence $x_{nm_n} \to x$ as $n \to \infty$; this shows that x is the limit of a sequence in A and therefore $x \in \overline{A}$. It follows that $\overline{A}$ is closed.

3. If $A = \overline{A}$, it follows from 2 that A is closed. Conversely, suppose A is closed. We have to show $A = \overline{A}$; from 1 we have $A \subset \overline{A}$, so it suffices to prove that $\overline{A} \subset A$. If $x \in \overline{A}$, there is a sequence (x_n) in A such that $x_n \to x$. Since A is closed, $x \in A$, so $\overline{A} \subset A$.

4. Follows from 2 and 3.

■

Example 4.16 Example 4.14 may suggest that in any metric space (X, d), the closure of an open ball $B(a; r)$ is the closed ball $B[a; r]$. This is in fact not the case, as the following example shows. Let X be any set with more than one element, and d the discrete metric on it. For $a \in X$ it is easy to check that $B(a, 1) = \{a\}$; since $\{a\}$ is closed, it follows from Proposition 4.15 that $\overline{B(a, 1)} = \{a\}$. However, $B[a; 1] = X$.

There is some terminology often used to describe points in the closure of a set A in a metric space. Points in the closure are referred to as **adherence points** or **limit points** of the set A; a point which is the limit of a sequence of points in A other than an eventually constant sequence is known as a **point of accumulation** or **cluster point** of the set A. If we use this terminology we can say that a set is closed if and only if it contains all its points of accumulation.

In the next chapter we'll need the following characterization of points in the closure of a set, in terms of open balls, rather than convergent sequences.

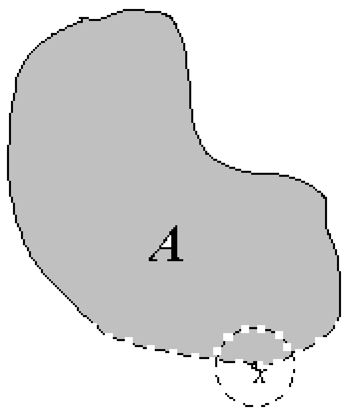


Figure 4.2: The open ball $B(x, r)$ contains a point of A.

Proposition 4.17 *Let A be a subset of a metric space (X, d) and $x \in X$. Then $x \in \overline{A}$ if and only if for every $r > 0$, the open ball $B(x, r)$ contains a point of A.*

Proof: If $x \in \overline{A}$, there is a sequence (x_n) in A such that $x_n \to x$. Then for every $r > 0$ there is an $n_r \in \mathbf{N}$ such that $x_n \in B(x, r)$ for $n \geq n_r$, by Proposition 3.10. Conversely, if for every $r > 0$ there is a point of A in $B(x, r)$, then, in particular for every $n \in \mathbf{N}$ there is an $x_n \in A$ such that $d(x, x_n) < \frac{1}{n}$. But then (x_n) is a sequence in A such that $x_n \to x$, and so $x \in \overline{A}$. ■

Up to now we have only worked with distances between *points* in a metric space. It is also possible to introduce the idea of the distance between a *a set and a point.* By this we mean the "least distance" between the point and a point in the set. Since there may not always be a point in the set which is closest to the given point, we have to be careful with what we mean by "least distance". Here is the definition:

Definition 4.18 *Let (X, d) be a metric space, $A \subset X$ and $x \in X$. The distance between x and A is defined by*

$$d(x, A) = \inf\{d(x, y) : y \in A\}.$$

Example 4.19 Let $\mathbf{R}$ have its usual metric and $A = (0, 1]$. Then
$d(2, A) = d(2, 1) = 1$, $\quad d(-3, A) = 3$, $\quad d(0, A) = 0$ and $d(\frac{1}{2}, A) = d(\frac{1}{2}, \frac{1}{2}) = 0$.
Note that the first and last distances are attained, the other two not.

The example above shows that the distance between a point and a set may be 0 even if the point is not in the set. The next proposition tells us exactly when this distance is 0.

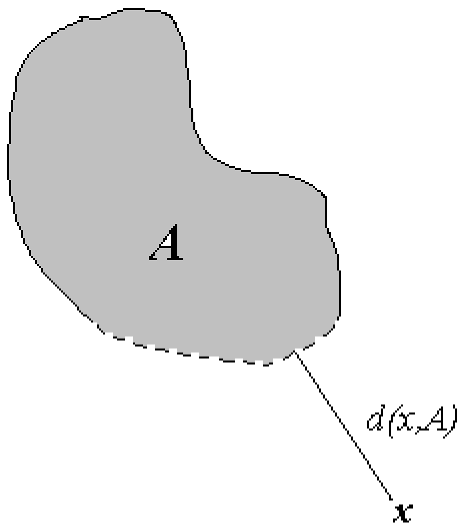


Figure 4.3: The distance between x and the set A in $\mathbf{R}^2$ with its usual metric.

Proposition 4.20 *If A is a subset of a metric space (X, d) and $x \in X$, then $d(x, A) = 0$ if and only if $x \in \overline{A}$.*

Proof: If $x \in \overline{A}$, there is a sequence (x_n) in A such that $x_n \to x$. Then $d(x, x_n) \to 0$, and so, since $(d(x, x_n))$ is also a non-negative sequence, $\inf\{d(x, x_n) : n \in \mathbf{N}\} = 0$. But then

$$d(x, A) = \inf\{d(x, y) : y \in A\} \leq \inf\{d(x, x_n) : n \in \mathbf{N}\} = 0.$$

Conversely, if $d(x, A) = 0$, then for each $n \in \mathbf{N}$, there is an $x_n \in A$ such that $d(x, x_n) < \frac{1}{n}$. But then $d(x, x_n) \to 0$, so $x_n \to x$, and so $x \in \overline{A}$. ■

The examples we've looked at in $\mathbf{R}, \mathbf{R}^2$ and $\mathbf{C}$ suggest that whether a set is closed or not depends on whether the "boundary" of the set is part of the set or not. This observation depends on some intuitive idea of what the boundary of a set should be. Before we can confirm our suspicion we'll have to make precise the notion of a boundary. One possibility is to say that a point is in the boundary of a set A if it is the limit of a sequence in A, and also the limit of another sequence in the complement A' of A. This leads to the following definition:

Definition 4.21 *Let A be a subset of a metric space (X, d). The* **boundary** *of A is the set $\mathrm{bd}(A)$ defined by*

$$\mathrm{bd}(A) = \overline{A} \cap \overline{A'},$$

where $A' = \{x \in X : x \notin A\}$ is the complement of A in X.

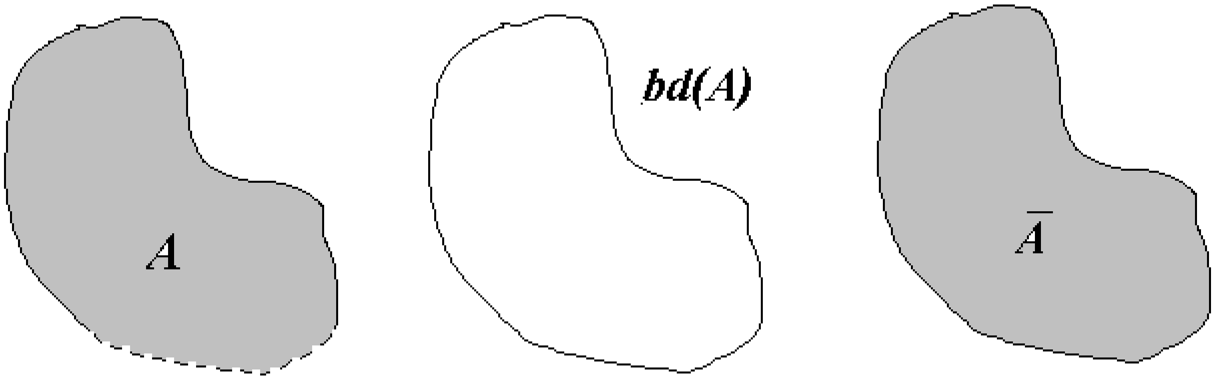


Figure 4.4: A set A in $\mathbf{R}^2$ (with its usual metric), its boundary and its closure.

Example 4.22 Let $X = \mathbf{R}$, with its usual metric, and $A = (-1, 2]$. Then $\overline{A} = [-1, 2], A' = (-\infty - 1] \cup (2, \infty), \overline{A'} = (-\infty, -1] \cup [2, \infty)$ and so $\mathrm{bd}(A) = \overline{A} \cap \overline{A'} = \{-1, 2\}$.

Example 4.23 Again, let $X = \mathbf{R}$, with its usual metric, and $A = \mathbf{N}$. Since $\mathbf{N}$ is closed (Example 4.2), $\overline{\mathbf{N}} = \mathbf{N}$. Also, for each $m \in \mathbf{N}, m$ is the limit of the sequence $(m + 2^{-n})$, which lies in the complement of $\mathbf{N}$), and so $\overline{\mathbf{N}'} = \mathbf{R}$. Hence $\mathrm{bd}(A) = \mathbf{N}$.

If $A = \mathbf{Q}$, $\overline{A} = \mathbf{R}$, and $\overline{A'} = \mathbf{R}$ (since each real number is the limit of a sequence of irrational numbers), and therefore

$$\mathrm{bd}(\mathbf{Q}) = \mathrm{bd}(A) = \overline{A} \cap \overline{A'} = \mathbf{R} \cap \mathbf{R} = \mathbf{R}.$$

Example 4.24 Let $X = \mathbf{C}$ and $A = \{z \in \mathbf{C} : |z| < 1\}$. Then $\mathrm{bd}(A) = \{z \in \mathbf{C} : |z| = 1\}$. If $A = \{z \in \mathbf{C} : \mathrm{Re}z = 1\}, \mathrm{bd}(A) = A$. Try to write out proofs of these facts.

Example 4.25 In a discrete metric space the boundary of every set is empty. Make sure you know why!

The examples show that one has to be careful not to take too much for granted.

We show now that our hunch about closed sets and boundaries was correct. To do this, we'll need some properties of the boundary.

Proposition 4.26 *Let A be a subset of a metric space (X, d). Then*

1. bd(A) *is closed;*

2. bd$(A') = $ bd(A)*;*

3. $\overline{A} = A \cup \text{bd}(A)$.

Proof: Exercise. (Use the definition of the boundary and the properties of the closure of a set listed in Proposition 4.15.) ∎

Proposition 4.27 *A subset of a metric space is closed if and only if it contains its boundary.*

Proof: If A is closed, $A = \overline{A}$, so $\text{bd}(A) = A \cap \overline{A'} \subset A$. Conversely, if $\text{bd}(A) \subset A$, then it follows from Proposition 4.26 that $\overline{A} = A$, and so A is closed (by Proposition 4.15). ∎

Summary.

In this chapter we introduced the notion of a closed set in a metric space. The empty set and the whole metric space are closed; so is any finite union and arbitrary intersection of closed sets. The closure and boundary of a set in a metric space were introduced, as well as the distance between a point and a set. Closedness was characterized in terms of the closure and the boundary.

Historical notes

Georg Ferdinand Ludwig Philipp Cantor (1845–1918)

Cantor introduced the concept of a limit point of a set of real numbers, and the set of all limit points of such a set, in 1872 and used it to define closed sets. He also defined the notion of an open set, something we'll look at in the next chapter.

Georg Cantor was born in St Petersburg in Russia in 1845, the son of a Danish merchant and a Russian mother. He inherited considerable artistic and musical talents from his parents and was himself an accomplished violinist. The family moved to Germany when he was eleven; Cantor fondly remembered the time he spent in Russia and never felt quite at ease in Germany. He did exceptionally well at school, especially in mathematics, and his father sent him to the Polytechnic in Zurich to study engineering. Georg preferred mathematics, and his father eventually allowed him to change course.

After the death of his father in 1863, he moved to Berlin, where he studied under Weierstrass, Kummer and Kronecker. Here he obtained his doctorate in number theory, after which he taught at a school and a college for teachers. In 1869 Cantor was appointed at the University of Halle, and under the influence of his colleague Heine he started doing research in analysis, and in particular working on problems connected with Fourier series. In 1872 he was promoted to professor, met Dedekind and published a paper in which he defined irrational numbers in terms of convergent sequences of rational numbers.

In 1873 Cantor started making a number of remarkable discoveries. He first showed that the set of rational numbers is countable, then that the same is true for the set of algebraic numbers (real numbers that are roots of polynomial equations with integer coefficients). This was followed by a proof that the set of real numbers is not countable. Even more surprising, he showed that the numbers in the unit interval could be put in a one-to-one correspondence with the points in the unit square in $\mathbf{R}^2$. In 1877 he proved more: the unit interval could be put into one-to-one correspondence with the points in $\mathbf{R}^n$, for any positive integer n. Cantor was so surprised by this discovery that he wrote: "I see it, but I don't believe it!" When he submitted his results for publication, it was treated with suspicion by Kronecker, and Dedekind had to intervene to ensure that it was published.

Cantor published six papers between 1879 and 1884 which was intended to be an introduction to set theory. There was much opposition to Cantor's ideas, and his cooperation with some of the leading mathematicians of the time ended as a result of this. He suffered an attack of depression in 1884, and this severely hampered his research. He was also worried by the fact that he could not prove the continuum hypothesis. He was asked to withdraw one of his papers submitted to *Acta Mathematica* because the editor thought it was " about one hundred years ahead of its time". All of this resulted in Cantor discontinuing his work on set theory. He turned to other matters and helped to found the German Mathematical Society, and became its first president. He did return to set theory and published two important papers on transfinite arithmetic in 1895 and 1897. By this time he had also discovered the first of a number of paradoxes in set theory.

At this time Cantor began to suffer increasingly from periods of depression, worsened by the deaths of his mother, younger brother and his youngest son. He had to be hospitalised a number of times, and turned his attention to literary matters and his theory that the plays of Shakespeare were written by Francis Bacon. After being forced to take leave frequently he retired in 1913. His illness was worsened by severe food shortages in Germany during the First World War. He died in a sanatorium, of a heart attack, in 1918.

Hilbert described his contribution to mathematics as “the finest product of mathematical genius and one of the supreme achievements of purely intellectual activity.”

Exercises

1. Which of the following subsets of $\mathbf{C}$ with its usual metric are closed? Give reasons for each of your answers and also give the closure of each of the sets.
   (a) $\{z \in \mathbf{C} : 1 < |z| \leq 2\}$;
   (b) $\{z \in \mathbf{C} : 0 < \arg z < \frac{\pi}{4}, 0 < |z| < 1\}$;
   (c) $\{z \in \mathbf{C} : |z| = n, n \in \mathbf{N}\}$;
   (d) $\{z \in \mathbf{C} : |z| = \frac{1}{n}, n \in \mathbf{N}\}$.
2. Find the boundary of each of the sets in Question 1.
3. Repeat Question 1, but this time let $\mathbf{C}$ have the metric defined by
$$d(z, w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$
   Also find the boundary of each of the sets in Question 1 when $\mathbf{C}$ has this metric.
4. Let $A = \{z \in \mathbf{C} : d(1 - i, z) < 1\}$. Find the distance between $z = i$ and A, first when $\mathbf{C}$ has its usual metric, and then when it has the metric of Question 3.
5. Let $B([0, 1])$ have its usual supremum metric (see Exercise 8 of Chapter 2).
   (a) Is the set $\{f \in B([0, 1]) : f(x) = 0 \text{ for every } x \in (0, \frac{1}{2})\}$ closed? Give a reason for your answer.
   (b) Prove that $C([0, 1])$ is a closed subset of $B([0, 1])$.
6. Give proofs for the three statements in Proposition 4.26.

7. Let (X, d) be any metric space and $A \subset X$. Prove that $x \in X$ is a point of accumulation of A if and only if every open ball with x as centre contains a point of A distinct from x. Also show that we may in fact replace "a point" by "infinitely many distinct points" in this statement.

Chapter 5

Sets without boundaries.

In which we encounter open sets, and our first taste of topology.

We think of closed intervals on the real line as intervals which include their endpoints. In the same way, open intervals may be regarded as intervals which do not include their endpoints. In the previous chapter we saw that the boundary of an interval is the set of endpoints of the interval. One possible way of generalising the idea of an open interval would therefore be to look at those sets in a metric space which do not contain any of their boundary points, and call such sets *open*. We adopt this as a definition.

Definition 5.1 A *subset A of a metric space (X, d) is* **open** *if it contains none of its boundary points, i.e, if*

$$A \cap \mathrm{bd}(A) = \emptyset.$$

We have already seen that a set in a metric space is closed if it contains *all* of its boundary points, and now we have defined an open set as one that contains *none* of its boundary points. It is, of course, possible for a set to contain some, but not all of its boundary points. According to our definitions this means that such a set will be *neither open nor closed*. We are so used to thinking of open and closed as opposites (e.g. if a door is not closed, it is open) that it takes a while to get used to the idea that in a metric space there are sets that are neither open nor closed. We look at some examples.

Example 5.2 If $\mathbf{R}$ has its usual metric, any interval of the form $(a, b), (-\infty, a)$ or (b, ∞) is open, since the endpoints are not included in the intervals. The set $\mathbf{R}$ itself is open (its boundary is empty). Intervals of the form $(a, b]$ and $[a, b)$ are neither open nor closed, since they contain part but not all of their boundaries.

Example 5.3 In $\mathbf{C}$ with its usual metric, the set $\{z \in \mathbf{C} : |z| < 1\}$ is open, since it is disjoint from its boundary, the set $\{z \in \mathbf{C} : |z| = 1\}$. The set $\{z \in \mathbf{C} : |z| < 1, \mathrm{Re} z \leq 0\}$ is neither open nor closed, since it contains part of its boundary (i.e. the set $\{z \in \mathbf{C} : \mathrm{Re} z = 0, -1 < \mathrm{Im} z < 1\}$), but not all of it (the part $\{z \in \mathbf{C} : |z| = 1, \mathrm{Re} z \leq 0\}$ is not included).

Example 5.4 In a discrete metric space every subset is open. This follows from the fact that the boundary of every subset of a discrete metric space is empty.

How are open and closed sets related? It is clear from the above examples and remarks that a set which is not closed need not be open, and a set which is not open need not be closed. The answer is given in the next theorem.

Theorem 5.5 A *subset A of a metric space (X, d) is open if and only if its complement $A' = \{x \in X : x \notin A\}$ is closed.*

Proof:

$$\begin{aligned} A \text{ is open} \quad &\Leftrightarrow A \cap \mathrm{bd}(A) = \emptyset \\ &\Leftrightarrow A \cap \mathrm{bd}(A') = \emptyset \\ &\Leftrightarrow \mathrm{bd}(A') \subset A' \\ &\Leftrightarrow A' \text{ is closed,} \end{aligned}$$

using Proposition 4.26. ∎

At the risk of sounding pedantic, we insert a warning:

WARNING: If a set is not open, it does NOT follow that it is closed; if it is not closed, it does NOT follow that it is open.

The definition we have given for open sets is not the one that you will come across in most textbooks. If closed sets are defined first, open sets are usually defined as sets whose complements are closed. The theorem we've just proved shows that our definition is equivalent to this definition. In other books open sets are defined before closed sets. We'll make some remarks about this approach at the end of this chapter.

We have already seen examples of sets that are neither open nor closed; the complements of such sets are neither open nor closed as well. It is even possible for a set to be *both* open and closed. This is true for the whole space X and the empty set $\emptyset$ in any metric space (X, d); in a discrete metric space *every* set is both open and closed. (Sets which are both open and closed are sometimes called **clopen**.)

We have seen that in every metric space closed balls are closed; we would likewise expect open balls to be open. The next proposition reassures us on this score.

Proposition 5.6 I*n any metric space open balls are open.*

Proof: Let (X, d) be a metric space, $a \in X$ and $r > 0$. We want to show that the open ball $B(a; r) = \{x \in X : d(a, x) < r\}$ is open. Put

$$A = [B(a; r)]' = \{x \in X : d(a, x) \geq r\};$$

it is enough to show that A is closed. Let (x_n) be a sequence in A with $x_n \to x \in X$. Then $d(a, x_n) \geq r$ for every n and $d(x, x_n) \to 0$ as $n \to \infty$. We also have

$$r \leq d(a, x_n) \leq d(a, x) + d(x, x_n) \text{ for every } n \in \mathbf{N}.$$

Letting $n \to \infty$ in this inequality gives $r \leq d(a, x)$. It follows that $x \in A$, and so A is closed. ■

It follows from what we've done in the previous chapter that a set in a metric space which is not closed can be enlarged to obtain a closed set, called the closure of the set. We can think of the closure as the set obtained by adjoining all the boundary points of the set which were not in the set already. Is it possible in the same way to obtain an open set from a set which is not open by removing all the boundary points of the set which are actually *in* the set? There seems to be no reason why not; we formalise this idea in the next definition.

Definition 5.7 L*et A be a subset of the metric space (X, d). A point of A is an* **interior point** *of A if it is not a boundary point of A. The set of all interior points of A is called the* **interior** *of A and is denoted by* $\text{int}(A)$*. In symbols,*

$$\text{int}(A) = A \cap [\text{bd}(A)]' = A \setminus \text{bd}(A).$$

Example 5.8 The interiors of the sets $(0, 2], [0, 2]$ and $(0, 2)$ in $\mathbf{R}$ with its usual metric are all the same, namely the set $(0, 2)$. (For example, the boundary of $[0, 2]$ is the set $\{0, 2\}$, and so every point in $[0, 2]$ other than 0 and 2 is an interior point.)

Example 5.9 We stay in $\mathbf{R}$ with its usual metric for some unusual examples. The interior of the set $\mathbf{N}$ is empty, since the boundary of $\mathbf{N}$ is $\mathbf{N}$ itself. The interior of $\mathbf{Q}$ is the empty set, since the boundary of $\mathbf{Q}$ is $\mathbf{R}$.

Example 5.10 Let $\mathbf{C}$ have its usual metric and $A = \{z \in \mathbf{C} : |z| \leq 1, \text{Re}z > 0\}$. Then $\text{bd}(A) = \{z \in \mathbf{C} : |z| = 1, \text{Re}z > 0\} \cup \{z \in \mathbf{C} : \text{Re}z = 0, -1 \leq \text{Im}z \leq 1\}$ and therefore $\text{int}(A) = A \setminus \text{bd}(A) = \{z \in \mathbf{C} : |z| < 1, \text{Re}z > 0\}$.

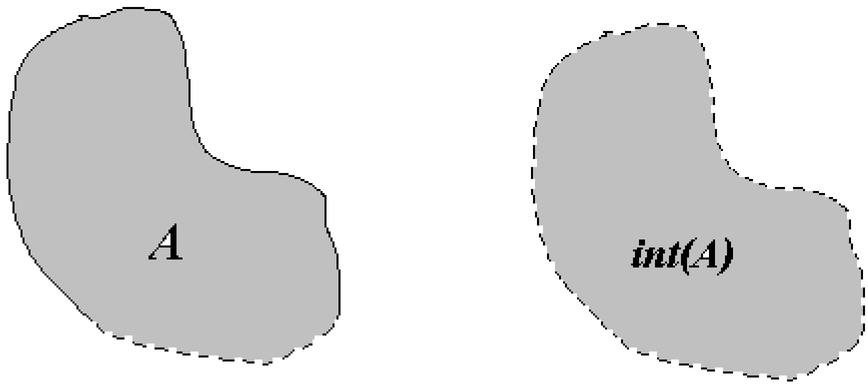


Figure 5.1: A set A in $\mathbf{R}^2$ (with its usual metric) and its interior.

The interior of a set is obtained by removing boundary points from the set; the closure is obtained by adjoining boundary points. It looks as if the two concepts are closely related. Can we express this relationship concisely? The next proposition does exactly that.

Proposition 5.11 *Let A be a subset of a metric space (X, d). Then* $\mathrm{int}(A) = (\overline{A'})'$.

Proof:

$$\begin{aligned}
[\mathrm{int}(A)]' &= [A \cap (\mathrm{bd}(A))']' \\
&= A' \cup \mathrm{bd}(A) \\
&= A' \cup (\overline{A} \cap \overline{A'}) \\
&= (A' \cup \overline{A}) \cap (A' \cup \overline{A'}) \\
&= X \cap \overline{A'} \\
&= \overline{A'},
\end{aligned}$$

using de Morgan's Law. Taking complements on both sides of this equation we get $\mathrm{int}(A) = (\overline{A'})'$. ∎

Proposition 5.12 *If A is a subset of a metric space, then*

1. $\mathrm{int}(A) \subset A$;
2. $\mathrm{int}(A)$ *is open;*
3. *A is open if and only if $A = \mathrm{int}(A)$.*

Proof:

1. By definition.
2. $\overline{A'}$ is closed, so $\mathrm{int}(A) = (\overline{A'})'$ is open, by Theorem 5.5.
3. If $A = \mathrm{int}(A)$, then A is open, by 2. Conversely, if A is open, $A \cap \mathrm{bd}(A) = \emptyset$ by definition, so $A \subset [\mathrm{bd}(A)]'$ and therefore $\mathrm{int}(A) = A \cap [\mathrm{bd}(A)]' = A$.

■

If one looks at examples of subsets of $\mathbf{R}^2$ with its Euclidean metric then it looks as if the closure of a set should be the union of its interior and its boundary. The next proposition shows that this is in fact the case in any metric space; this is an improvement on Proposition 4.26 in the sense that we can now write the closure as the union of two *disjoint* sets.

Proposition 5.13 *Let A be a subset of the metric space X. Then*

$$\overline{A} = \mathrm{int}(A) \cup \mathrm{bd}(A) \quad \textit{and} \quad \mathrm{int}(A) \cap \mathrm{bd}(A) = \emptyset.$$

Proof: It is clear from the definition of $\mathrm{int}(A)$ that $\mathrm{int}(A) \cap \mathrm{bd}(A) = \emptyset$. Since $\mathrm{int}(A) \subset A \subset \overline{A}$ and $\mathrm{bd}(A) = \overline{A} \cap \overline{A'} \subset \overline{A}$, it follows that $\mathrm{int}(A) \cup \mathrm{bd}(A) \subset \overline{A}$. Conversely, if $x \in \overline{A}$, then either

$x \in \overline{A'}$, in which case $x \in \overline{A} \cap \overline{A'} = \mathrm{bd}(A)$;

$x \notin \overline{A'}$, in which case $x \in (\overline{A'})' = \mathrm{int}(A)$.

Hence $\overline{A} \subset \mathrm{int}(A) \cup \mathrm{bd}(A)$. ■

We saw at the end of the previous chapter that the points in the closure of a set could be described using open balls. It is not surprising, in the light of the definition of the interior of a set, that we can also describe interior points in this way. This leads to a characterisation of open sets in terms of open balls.

Proposition 5.14 *Let (X, d) be a metric space and A a subset of X.*

1. *A point $x \in X$ is an interior point of A if and only if there is an $r > 0$ such that $B(x, r) \subset A$.*
2. *A is open if and only if for every $x \in A$ there is an $r > 0$ such that $B(x, r) \subset A$.*

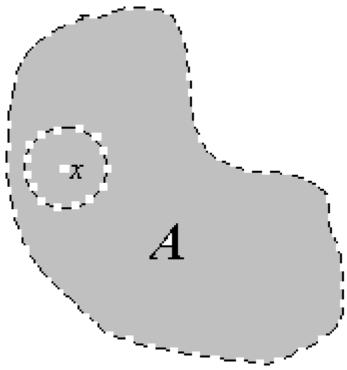


Figure 5.2: An interior point x of a set A in $\mathbf{R}^2$ (with its usual metric).

Proof:

1. If $x \in \mathrm{int}(A)$, then $x \notin \overline{A'}$ by Proposition 5.11. It follows from Proposition 4.17 that there is an $r > 0$ such that the open ball $B(x, r)$ does not contain any points of A', i.e. $B(x, r) \subset (A')' = A$. Conversely, if there is a ball $B(x, r)$ such that $B(x, r) \subset A$, then $B(x, r)$ does nor contain any point of A', and it follows from Proposition 4.17 that $x \notin \overline{A'}$, and therefore x is an interior point of A.

2. This follows at once from 1 and Proposition 5.12.

■

The property (M1) in the definition of a metric (Definition 2.1) is known as the "separation property". The next proposition shows why.

Proposition 5.15 *Let (X, d) be a metric space and $x, y \in X$, with $x \neq y$. Then there are disjoint open sets G_1 and G_2 in X with $x \in G_1, y \in G_2$.*

Proof: It follows from (M1) that $r = d(x, y) > 0$, since $x \neq y$. Put $G_1 = B(x; \frac{1}{2}r), G_2 = B(y; \frac{1}{2}r)$. Certainly G_1 and G_2 are open and $x \in G_1, y \in G_2$. Suppose $z \in G_1 \cap G_2$. Then $d(x, z) < \frac{1}{2}r$ and $d(y, z) < \frac{1}{2}r$ and hence

$$r = d(x, y) \leq d(x, z) + d(z, y) = d(x, z) + d(y, z) < \frac{1}{2}r + \frac{1}{2}r = r,$$

a contradiction. Therefore there is no $z \in G_1 \cap G_2$, i.e. G_1 and G_2 are disjoint. ■

This property of metric spaces is often referred to as the **Hausdorff Property** , after Felix Hausdorff (see page 25).

We now have three different descriptions of open sets:

1. A set is open if it is disjoint from its boundary (our *definition* of an open set).
2. A set is open if its complement is closed (Theorem 5.5).
3. A set is open if every point in it is the centre of an open ball contained in the set (Proposition 5.14).

The first description uses the notion of the boundary of a set. The boundary was defined in terms of closures, and the closure in terms of convergent sequences. The second description uses the notion of a closed set, which was also defined in terms of convergent sequences. The last description does not use convergent sequences at all, but gives us a way of defining open sets in terms of open balls (which are defined in terms of the metric, of course). We could in fact have used this as our starting point, and you will find that some books on metric spaces do exactly this. In this approach, once metric spaces have been introduced and open balls defined, one defines an open set as a set with the property that every point in it is the centre of an open ball contained in the set. Note that Proposition 5.14 guarantees that this definition agrees with our own definition of open sets. Closed sets can then be defined as the complements of open sets. The closure of a set can be defined in terms of open sets, as suggested by Proposition 4.17. In this approach the notion of an open set is the "primitive" notion in terms of which all other concepts are defined.

The advantage of this way of doing things is that it lends itself to even further generalisation. Before we say more about this, we use the properties of closed sets, the fact that open sets are exactly the complements of closed sets and de Morgan's Laws to see what we can say about unions and intersections of open sets.

Theorem 5.16 *Let (X, d) be a metric space.*

1. *X and $\emptyset$ are open sets.*
2. *The intersection of a finite number of open subsets of X is open.*
3. *The union of an arbitrary family of open subsets of X is open.*

Proof:

1. X and $\emptyset$ are closed, and so their complements $\emptyset$ and X are open.
2. Let $G_1, G_2, \ldots, G_n$ be open sets in X. Then $G'_1, G'_2, \ldots, G'_n$ are closed sets. Hence $F = G'_1 \cup G'_2 \cup \ldots \cup G'_n$ is a closed set in X (Corollary 4.9), and so $G_1 \cap G_2 \cap \ldots \cap G_n = F'$ is an open set.
3. Let $\{G_i : i \in I\}$ be an arbitrary family (finite or infinite) of open subsets of X. Then $\{G'_i : i \in I\}$ is a family of closed sets, and so by Theorem 4.10 its intersection $F = \cap\{G'_i : i \in I\}$ is a closed set. But then $\cup\{G_i : i \in I\} = F'$ is an open set.

■

Corollary 5.17 *A subset of a metric space is open if and only if it is a union of open balls. In particular, a subset of* **R** *with its usual metric is open if and only if it is a union of open intervals.*

Proof: This follows from Proposition 5.14 and Theorem 5.16. ■

In Chapter 2 we saw that if (X, d) is a metric space and A is a subset of X, then the metric d induces a metric d_A on A. In this way we can think of (A, d_A) as a metric space in its own right. What do the open sets in (A, d_A) look like? The next proposition, which we'll need again in Chapter 10, shows that there is a very satisfactory way of describing these open sets in terms of the open sets in (X, d).

Proposition 5.18 L*et the subset A of the metric space (X, d) have the metric d_A induced by d on A. Then a subset B of A is open in (A, d_A) if and only if there is an open set G in X such that $B = A \cap G$.*

Proof: Note that for $a \in A, r > 0$,

$$\{x \in A : d_A(x, a) < r\} = A \cap \{x \in X : d(x, a) < r\} = A \cap B(a; r).$$

We can interpret this equation as saying that a subset of A is an open ball in (A, d_A) if and only if it is the intersection of A with an open ball in (X, d). Now if B is an open subset of A, it is a union of open balls in A (Corollary 5.17), so there is some subset A_0 of A and for each $a \in A_0$ and $r_a > 0$ such that

$$B = \cup\{\{x \in A : d_A(x, a) < r_a\} : a \in A_0\}.$$

From the above remark it follows that B can be written in the form

$$B = \cup\{A \cap B(a; r) : a \in A_0\} = A \cap \cup\{B(a; r) : a \in A_0\}.$$

Since a union of open balls in X is an open set G in X, we have $B = A \cap G$, and one half of the result follows. Conversely, if $B = A \cap G$ for some open subset G of X, and $x \in B$, then $x \in G$ and so there is an open ball $B(x; r)$ in X such that $B(x; r) \subset G$. But then, again by the remark we started with, $A \cap B(x; r)$ is an open ball in (A, d_A) which is contained in B, so B is open in A. ■

The importance of the theorem we have just proved is that it can be used as a starting point for a generalisation of metric spaces. In the case of a metric space we start with a set X and a distance function (or metric) d, and all concepts (such as closed sets, open sets, convergence) are defined using the metric. The basic idea underlying the generalisation is to try to define as many of the notions we've had so far without using the metric. Of course we cannot start with nothing at all; as indicated above, it does seem likely that we could define most of the concepts we've come across so far in terms of *open sets*. This supplies the motivation for starting with a non-empty set X and a special family of subsets of X. We are going to think of this family of subsets as the family of *open subsets* of X and define all other concepts (such as convergence, closed sets, closure) in terms of these "open sets".

Can we, given a non-empty set, just take any family of subsets of it and call these the *open sets*? There is nothing stopping us, of course, but we'll find that if we want to prove similar results to the ones we've obtained in the setting of metric spaces, we'll have to be a bit more selective. The question becomes: What are the minimum conditions we have to impose on the family of subsets (the ones we are going to call "open") to be able to prove something useful?

It makes sense to look at the situation in metric spaces for motivation. The family of open sets in a metric space has the properties listed in Theorem 5.16. The surprising thing is that these properties are enough for a meaningful generalisation. This may all sound a bit esoteric; let's make it more precise.

Definition 5.19 L*et X be a non-empty set and $\mathcal{T}$ a family of subsets of X such that*

(T1) $X \in \mathcal{T}, \emptyset \in \mathcal{T}$;

(T2) the intersection of any finite number of sets in $\mathcal{T}$ is again in $\mathcal{T}$, i.e. if $G_1, G_2, \ldots, G_n \in \mathcal{T}$, then $G_1 \cap G_2 \cap \ldots \cap G_n \in \mathcal{T}$;

(T3) the union of an arbitrary family of sets in $\mathcal{T}$ is again in $\mathcal{T}$, i.e. if $G_i \in \mathcal{T}$ for each $i \in I$ (where I is an index set for the family), then $\cup\{G_i : i \in I\} \in \mathcal{T}$.

Then $\mathcal{T}$ is called a **topology** *on X. In addition, $(X, \mathcal{T})$ is called a* **topological space** *and the sets in $\mathcal{T}$ are called the* **open sets** *of the topological space.*

A technical remark: $\cup\{A_i : i \in \emptyset\} = \{x : (\exists i \in \emptyset)(x \in A_i)\} = \emptyset$ and $\cap\{A_i : i \in \emptyset\} = \{x : (\forall i \in \emptyset)(x \in A_i)\} = X$. (See Appendix A.8.) So condition (T1) could be subsumed under (T2) and (T3); but it seems much easier to write (T1) out as we have it here.

We have hinted that the concept we've just defined is going to be a generalisation of the concept of a metric space. But then there must be a way of regarding every metric space as a topological space. This means that given a metric space we must be able to find a family of subsets of the metric space which satisfies the conditions (T1), (T2) and (T3). The first example reassures us on this score.

Example 5.20 Let (X, d) be a metric space. Let $\mathcal{T}_d$ consist of all subsets A of X with the property that for every $x \in A$, there is an open ball centred at x which is contained in A. Then $\mathcal{T}_d$ is just the family of subsets of X that we've called "open" in the metric spaces context, and it follows from Theorem 5.16 that $\mathcal{T}_d$ is a topology on X. We call $\mathcal{T}_d$ the **topology induced by the metric** d.

Example 5.21 Let X be any non-empty set and let $\mathcal{T} = \{\emptyset, X\}$. Then it is easy to check that $\mathcal{T}$ is a topology on X, called the **indiscrete topology**. Since metric spaces have the Hausdorff property (Proposition 5.15) and indiscrete topological spaces cannot have this property (why not?), it follows that this topology is not induced by any metric.

Example 5.22 Let X be any non-empty set and $\mathcal{T}$ be the family of *all* subsets of X. Then it is easy to check that $\mathcal{T}$ is a topology for X, called the **discrete topology**. It also follows easily that this topology is induced by the discrete metric on X. (Make sure that you can prove both these facts.)

Example 5.23 Let X be a nonempty set and put

$$\mathcal{T} = \{A \subset X : A' \text{ is finite}\} \cup \{\emptyset\}.$$

Then it can be checked that $\mathcal{T}$ is a topology on X (do this!), called the **cofinite topology**.

The above examples show that a metric space may be regarded as a topological space (the metric induces a topology), but that there are topologies that are not induced by metrics. The focus of this course is on metric spaces, and we'll therefore not develop the theory of topological spaces much further. As we go along and introduce new concepts in metric spaces, we will, however, show that many of these concepts can be defined in terms of open sets only. This will serve (at least) two purposes. In the first place such a characterisation in terms of open sets will often add useful geometrical insights, and may in some cases lead to shorter or clearer proofs. In the second place it will be a great help to those of you who may want to learn more about topology, since it will give a way of defining these concepts in the setting of topological spaces.

We have seen that a metric on a set induces a topology on that set. We have also seen that it is possible to define more than one metric on a set. Is it possible for two different metrics on a set to induce the same topology? The following example gives an answer to this question.

Example 5.24 Let $X = \mathbf{R}^2$, and let d be the usual Euclidean metric on $\mathbf{R}^2$, and d_m the metric defined by

$$d_m(\mathbf{x}, \mathbf{y}) = \max\{|x_1 - y_1|, |x_2 - y_2|\}.$$

Let $\mathcal{T}$ be the topology induced by d and $\mathcal{T}_m$ the topology induced by d_m. The open

Figure 5.3: Open balls for the Euclidean (left) and maximum (right) metrics

balls of d are open discs, and those of d_m are open squares (see Figure 5.3). Let $A \in \mathcal{T}$, and $\mathbf{x} \in A$. Then there is an open disc $B(\mathbf{x}, r) \subset A$. There is an open square centred at $\mathbf{x}$ contained in this disc (see Figure 5.4). Since $\mathbf{x} \in A$ was arbitrary, it follows that $A \in \mathcal{T}_m$. Hence $\mathcal{T} \subset \mathcal{T}_m$.

Conversely, let $A \in \mathcal{T}_m$, and $\mathbf{x} \in A$. Then there is an open square S (say) centred at $\mathbf{x}$ contained in A, and there is an open disc $B(\mathbf{x}, r) \subset S \subset A$ (see Figure 5.4). It follows as before that $A \in \mathcal{T}$, and hence $\mathcal{T}_m \subset \mathcal{T}$. But then we have $\mathcal{T} = \mathcal{T}_m$, and so the two different metrics induce the same topology.

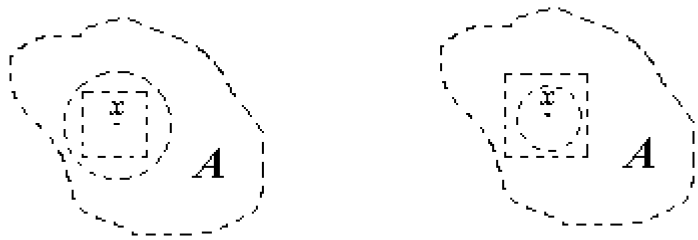


Figure 5.4: Squares in discs and discs in squares

Definition 5.25 *Two metrics d_1 and d_2 on a set X are called* **equivalent** *if they induce the same topology on X (i.e. if the two metric spaces (X, d_1) and (X, d_2) have the same open sets).*

The last example shows that d and d_m are equivalent metrics on $\mathbf{R}^2$.

This last definition may well feel a little out of keeping with the rest of the course at this stage. So far we have defined almost every concept in terms of convergent sequences, and you may rightly feel that we should call two metrics *equivalent* if the give rise to the same convergent sequences. The next result should reassure you. Not surprisingly, we first have to establish a link between open sets and convergent sequences.

Proposition 5.26 *Let X be a non-empty set.*

1. *Let d be a metric on X. A sequence (x_n) in X converges to $x \in X$ with respect to d if and only if for every open set G in (X, d) with $x \in G$ there is an $n_G \in \mathbf{N}$ such that $x_n \in G$ for every $n \geq n_G$.*
2. *The following two statements are equivalent:*
   (a) *The metrics d_1 and d_2 on X are equivalent;*
   (b) *A sequence (x_n) in X converges to $x \in X$ with respect to d_1 if and only if it converges to x with respect to d_2.*

Proof:

1. Suppose $x_n \to x \in X$ and let G be an open set in X with $x \in G$. Then there is an $r > 0$ such that $B(x, r) \subset G$; since $d(x, x_n) \to 0$, there is an $n_r \in \mathbf{N}$ such that $d(x, x_n) < r$ for $n \geq n_r$. Put $n_G = n_r$; then $x_n \in B(x, r) \subset G$ for $n \geq n_G$.

Conversely, if $\epsilon > 0$, then $G = B(x, \epsilon)$ is an open set in X and $x \in B(x, \epsilon)$. By assumption there is an $n_G \in \mathbf{N}$ such that $x_n \in G = B(x, \epsilon)$ for every $n \geq n_G$, i.e. $d(x, x_n) < \epsilon$ for $n \geq n_G$. This shows that $x_n \to x$ as $n \to \infty$, as required.

2. If the two metrics are equivalent, the same sets are open with respect to both metrics and hence, by 1, the same sequences are convergent, to the same limit, with respect to both metrics. Conversely, if the same sequences converge with respect to both metrics, then the same sets are closed with respect to both metrics. But then the same sets are open with respect to both metrics, i.e. the metrics are equivalent.

■

Example 5.27 The usual metric and the discrete metric on $\mathbf{R}$ are not equivalent, since the sequence $(\frac{1}{n})$ converges with respect to the usual metric, but not with respect to the discrete metric.

Example 5.28 Any two metrics on a finite set are equivalent. (Can you see why?)

Summary.

Open sets in metric spaces were introduced as sets that do not contain any boundary points, and characterised as complements of closed sets. The interior of a set was defined and the link between interiors and closures established. Interior points and open sets were characterised in terms of open balls. Topological spaces were introduced as generalisations of metric spaces, and equivalent metrics defined.

Historical notes

Augustus De Morgan (1806 – 1871)

Augustus De Morgan was born Madura, India where his father was a Lieutenant-Colonel in the British Army. He lost the sight of his right eye shortly after birth, and this disability no doubt contributed to his irascibility in later life. The De Morgan family returned to England soon after Augustus's birth. Even though he did not do particularly well at school, he went to the University of Cambridge at the age of 16. He graduated, but as a matter of principle refused to do a theological test that the university required for postgraduate study. In 1826 he started studying law in London, but in 1827, at the age of 21, he was appointed professor of Mathematics at the newly founded University College London.

He resigned from this position on a matter of principle in 1831, was reappointed in 1836 and stayed on until 1866, when he resigned on a matter of principle for the second time.
De Morgan's most important contributions to mathematics was in the fields of algebra and logic. He introduced the laws of complementation that are nowdays named after him, and was the first one to introduce the term "mathematical induction" and give a rigorous description of it. He also wrote on the geometrical representation of complex numbers. He was a co-founder of the London Mathematical Society and its first president.

Exercises

1. For each of the following subsets of $\mathbf{R}^2$ with its usual Euclidean metric say whether it is open, give a reason for your answer, and give the interior of the set:
   (a) $\{\mathbf{x} \in \mathbf{R}^2 : 0 \leq x_1 < 2, -1 \leq x_2 < 1\}$;
   (b) $\{\mathbf{x} \in \mathbf{R}^2 : -1 \leq x_2 < 1\}$;
   (c) $\{\mathbf{x} \in \mathbf{R}^2 : 0 < x_1 < 2, \cos x_2 > 0\}$.
2. Let $\mathbf{C}$ have the metric defined by
$$d(z, w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$
   For each of the following sets, say whether it is open, and give a reason for your answer. Also give the interior of the set.
   (a) $\{z \in \mathbf{C} : \mathrm{Re} z \geq 1, \mathrm{Im} z \geq 0\}$;
   (b) $\{z \in \mathbf{C} : \mathrm{Re} z \geq 0, \mathrm{Im} z \geq 0\}$.
3. Let (X, d) be a metric space and A and B be subsets of X. Prove or disprove the following statements:

(a) $\text{int}(\text{int}(A)) = \text{int}(A)$;

(b) $\text{int}(A \cup B) = \text{int}(A) \cup \text{int}(B)$;

(c) $\text{int}(A') = (\text{int}(A))'$.

4. Let (X, d) be a metric space and $A \subset X$.

   (a) Prove that $x \in X$ is an interior point of A if and only if there is an open ball containing x which is contained in A. [Note: The open ball need not be centred at x.]

   (b) Prove that $x \in X$ is an interior point of A if and only if $d(x, A') > 0$.

5. Let d be a metric on the non-empty set X, and $k > 0$. Prove that the metric kd is equivalent to d.

6. Let (X, d) be a metric space. Show that if X is a finite set, then d is equivalent to the discrete metric on X.

7. Let (X, d_X) and (Y, d_Y) be two metric spaces. For $(x_1, y_1), (x_2, y_2) \in X \times Y$, put

$$\begin{aligned} d_1((x_1, y_1), (x_2, y_2)) &= d_X(x_1, x_2) + d_Y(y_1, y_2) \\ d_m((x_1, y_1), (x_2, y_2)) &= \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\} \\ d_E((x_1, y_1), (x_2, y_2)) &= \sqrt{[d_X(x_1, x_2)]^2 + [d_Y(y_1, y_2)]^2} \end{aligned}$$

   (a) In Exercise 11b of Chapter 2 you were asked to check that d_1 is a metric on $X \times Y$. Now do the same for d_m and d_E.

   (b) Prove that the metrics d_1, d_m and d_E are equivalent metrics on $X \times Y$.

8. Let $X = \mathbf{R}$ and $\mathcal{T} = \{[a, b) : a, b \in \mathbf{R}, a < b\}$. Is $\mathcal{T}$ a topology on X? Answer the same question if $\mathcal{T} = \{A \subset \mathbf{R} : A \text{ is a union of intervals of the form } [a, b)\}$.

9. Let (X, d) be a quasi-metric space and let $\mathcal{T}$ be the collection of all subsets A of X with the property that if $x \in A$, then there is an $r > 0$ such that $\{y \in X : d(x, y) < r\} \subset A$. Show that $\mathcal{T}$ is a topology on X. This is known as the topology induced by the quasi-metric d.

10. Let X be a partially ordered set. For $x \in X$, let $\uparrow x = \{y \in X : y \geq x\}$ and

$$\mathcal{T} = \{A \subset X : x \in A \Rightarrow \uparrow x \subset A\}.$$

   (a) Show that $\mathcal{T}$ is a topology on X. (This topology is known as the *Alexandrov topology* on X.)

   (b) Show that the quasi-metric introduced in Example 2.17 induces the Alexandrov topology on X.

Chapter 6

Functions preserving convergence

In which we extend the notion of a continuous function to the setting of metric spaces.

Continuous functions play an important role in calculus, and by now you probably have a reasonably good intuitive feeling for such functions. One tends to think of them in terms of their graphs: continuous functions (from the real line to the real line) are functions with no "gaps" in their graphs. In the first year you also saw how this graphical understanding of continuity could be translated into a more rigorous definition:

CR A function $f : \mathbf{R} \to \mathbf{R}$ is continuous at $a \in \mathbf{R}$ if $lim_{x\to a}f(x) = f(a)$.

At the time we were not too concerned about saying exactly what we mean by the limit $lim_{x\to a}f(x)$. We were content with rather vague explanations like "$f(x)$ gets close to $f(a)$ when x gets close to a". But even imprecise explanations like these suggest that we should be able to describe the notion of continuity in terms of distances. This suspicion is confirmed when one looks at one of the ways the notion of a limit of a function (and hence also of continuity) is made precise in Real Analysis. You may remember this involves "epsilons and deltas"; here is the formal definition of continuity:

CR1 A function $f : \mathbf{R} \to \mathbf{R}$ is continuous at $a \in \mathbf{R}$ if for every $\epsilon > 0$ there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|f(x) - f(a)| < \epsilon$.

It is clear that this definition relies heavily on the notion of the distance between two real numbers. There is an obvious question to be asked now: Can we extend the notion of a continuous function to functions from one metric space to another? The definition above suggests that this should not be too difficult. All we need to do is to replace the distance between real numbers (expressed in terms of the absolute value) by distances as measured by metrics. Here is what we get:

CMS1 Let (X, d_X) and (Y, d_Y) be two metric spaces. The function $f : X \to Y$ is continuous at $a \in X$ if for every $\epsilon > 0$ there is a $\delta > 0$ such that if $d_X(x, a) < \delta$, then $d_Y(f(x), f(a)) < \epsilon$.

But you may feel, and quite justifiably so, that we have managed so far to do everything in terms of sequences and in that way to avoid using things like epsilons and deltas unnecessarily. Is it not possible to do this here as well? Fortunately there is a characterisation of continuity that you saw in Real Analysis that turns out to be exactly what we need:

CR2 A function $f : \mathbf{R} \to \mathbf{R}$ is continuous at $a \in \mathbf{R}$ if for every sequence $x_n \to a$ as $n \to \infty$, we have that $f(x_n) \to f(a)$ as $n \to \infty$.

We therefore have two equivalent ways of defining continuity for functions from $\mathbf{R}$ to $\mathbf{R}$; the "epsilon-delta", definition (CR1), and the definition in terms of convergent sequences (CR2). We have already seen above that we can translate the first definition to the more general context of metric spaces; this gave us (CMS1). Since we know what it means for a sequence to converge in a metric space, the second definition can also be used just as it stands in a metric space context. We are going to opt for this as our definition of continuity in metric spaces.

Definition 6.1 L*et (X, d_X) and (Y, d_Y) be metric spaces and f be a function from X to Y (i.e. $f : X \to Y$). Then f is* **continuous** *at $x \in X$ if for every sequence (x_n) in X such that $x_n \to x$ as $n \to \infty$ in X, we have that $f(x_n) \to f(x)$ as $n \to \infty$ in Y.*
We say f is continuous on X (or simply "f is continuous") if f is continuous at every point $x \in X$.

Note that this means that f is continuous at x if $d_X(x_n, x) \to 0$ implies that $d_Y(f(x_n), f(x)) \to 0$. We can also think of a continuous function as one for which we can write
$$\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n);$$
i.e. we can interchange the processes of taking limits and applying the function. We can express this by saying that a continuous function is a function that "preserves limits". Notice that this really implies two things:

- if the sequence (x_n) is convergent, then the sequence $(f(x_n))$ is convergent as well, **and**
- the limit of $(f(x_n))$ is just $f(\lim x_n)$.

It follows from the remark above that Definition 6.1 agrees with the usual definition of continuity in the case where $X = Y = \mathbf{R}$, both equipped with the usual metric on $\mathbf{R}$. We shall see later that Definition 6.1 is also equivalent to the other possible definition we mentioned earlier (CMS1).

Example 6.2 If $X = \mathbf{R}^2$ and $Y = \mathbf{R}$, both with their usual Euclidean metrics, then a function $f : \mathbf{R}^2 \to \mathbf{R}$ is continuous at the point (x, y) in $\mathbf{R}^2$ if and only if whenever $x_n \to x, y_n \to y$ in $\mathbf{R}$, then $f(x_n, y_n) \to f(x, y)$ in $\mathbf{R}$. To see this, first suppose f is continuous at $(x, y) \in \mathbf{R}^2$ and suppose that $x_n \to x$ and $y_n \to y$ as $n \to \infty$. Then it follows from the fact that convergence in $\mathbf{R}^2$ is equivalent to coordinatewise convergence (Example 3.8) that $(x_n, y_n) \to (x, y)$ as $n \to \infty$ in the Euclidean metric of $\mathbf{R}^2$. Since f is continuous at (x, y), $f(x_n, y_n) \to f(x, y)$ in the usual metric of $\mathbf{R}$. Conversely, suppose that $f(x_n, y_n) \to f(x, y)$ in $\mathbf{R}$ whenever $x_n \to x, y_n \to y$ in $\mathbf{R}$. Let $(x, y) \in \mathbf{R}^2$ and suppose $(x_n, y_n) \to (x, y)$ in the Euclidean metric of $\mathbf{R}^2$. Then it follows from Example 3.8 again that $x_n \to x$ and $y_n \to y$ in $\mathbf{R}$, and so by assumption $f(x_n, y_n) \to f(x, y)$ in $\mathbf{R}$. This shows that f is continuous at (x, y).

Example 6.3 If $X = Y = \mathbf{C}$, with its usual metric, a function $f : \mathbf{C} \to \mathbf{C}$ is continuous if whenever $\mathrm{Re} z_n \to \mathrm{Re} z, \mathrm{Im} z_n \to \mathrm{Im} z$, then $\mathrm{Re} f(z_n) \to \mathrm{Re} f(z)$ and $\mathrm{Im} f(z_n) \to \mathrm{Im} f(z)$. This follows from the fact that a sequence (z_n) in $\mathbf{C}$ converges to $z \in \mathbf{C}$ if and only if $\mathrm{Re} z_n \to \mathrm{Re} z$ and $\mathrm{Im} z_n \to \mathrm{Im} z$ (Example 3.6). The proof is similar to that in the previous example; make sure that you know how to do it.

The two examples so far considered familiar metric spaces. We now turn to something that you would probably not immediately recognise as a function: the definite integral. The definite integral of a continuous real-valued function over the interval $[a, b]$ is a real number, so we can think of the integral as a function from the metric space $C([a, b])$ to the metric space $\mathbf{R}$.

Example 6.4 Let $X = C([a, b])$ have the sup metric and $\mathbf{R}$ its usual metric. Let the function $F : C([a, b]) \to \mathbf{R}$ be defined by

$$F(f) = \int_a^b f(x)\,dx \qquad \text{for } f \in C([a, b]).$$

We check to see whether F is continuous. Let (f_n) be a sequence in $C([a,b])$ such that $f_n \to f \in C([a,b])$ as $n \to \infty$ with respect to the sup metric. This means

$$d(f, f_n) = \sup\{|f(x) - f_n(x)| : x \in [a,b]\} \to 0 \text{ as } n \to \infty.$$

Then

$$\begin{aligned} |F(f) - F(f_n)| &= |\textstyle\int_a^b f(x)\,dx - \int_a^b f_n(x)\,dx| \\ &= |\textstyle\int_a^b (f(x) - f_n(x))\,dx| \\ &\le \textstyle\int_a^b |f(x) - f_n(x)|\,dx \\ &\le \sup\{|f(x) - f_n(x)| : x \in [a,b]\} \textstyle\int_a^b 1\,dx \\ &= d(f, f_n)(b-a) \\ &\to 0 \text{ as } n \to \infty, \end{aligned}$$

using the properties of integrals and the inequalities listed in Appendix A.7. Hence $F(f_n) \to F(f)$ as $n \to \infty$. It follows that F is continuous on $C([a,b])$.

We can also formulate the result in the last example as follows:

- If (f_n) is a sequence of continuous functions on $[a,b]$ converging uniformly to the continuous function f on $[a,b]$, then

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx.$$

This result is no longer true when "converging uniformly" is replaced by "converging pointwise". We ask you to check this in the exercises (Exercise 3). It is this deficiency which is one of the motivations for generalising the theory of Riemann integration (integration as you know it) to Lebesque integration.

Functions like F above occur frequently in the *calculus of variations* and in the study of differential equations. At the end of the nineteenth century they were first called "functionals" (functions with functions as variables) to distinguish them from functions of a real or complex variable. There is still an important branch of mathematics called *Functional Analysis*, even though today it deals with much more than the analysis of such functionals.

Example 6.5 Let d_D be the discrete metric on a non-empty set X, and (Y,d) be any metric space. Then every function $f : (X, d_D) \to (Y, d)$ is continuous. Make sure that you know why!

The next proposition gives a way of using two continuous functions to produce a third one.

Proposition 6.6 *Let $(X, d_X), (Y, d_Y)$ and (Z, d_Z) be metric spaces and let $g : X \to Y$ and $f : Y \to Z$ be continuous functions. Then the composition $f \circ g$ is also continuous.*

Proof: Let $x \in X$ and (x_n) be a sequence in X such that $x_n \to x$. Since g is continuous, $g(x_n) \to g(x)$, and since f is continuous, $f(g(x_n)) \to f(g(x))$. This shows that $f \circ g(x_n) \to f \circ g(x)$, and so $f \circ g$ is continuous at x. Since $x \in X$ was arbitrary, it follows that $f \circ g$ is continuous. ∎

The metric on any metric space X is a function from $X \times X$ to $\mathbf{R}$. It therefore makes sense to ask whether it is continuous. To answer this question, we first prove a useful inequality.

Lemma 6.7 *Let (X, d) be a metric space and $a, b, x, y \in X$. Then*

$$|d(a, b) - d(x, y)| \leq d(a, x) + d(b, y).$$

Proof: It follows from the triangle inequality that

$$d(a, b) \leq d(a, x) + d(x, y) + d(y, b)$$

and so

$$d(a, b) - d(x, y) \leq d(a, x) + d(y, b) = d(a, x) + d(b, y).$$

Starting with $d(x, y)$ we obtain, in the same way,

$$d(x, y) - d(a, b) \leq d(a, x) + d(b, y).$$

Combining these two inequalities, we get

$$|d(a, b) - d(x, y)| \leq d(a, x) + d(b, y).$$

∎

Before we investigate the continuity of the metric, we first look at a slightly simpler question. If we fix a point in X, say the point $a \in X$, then $d(a, x)$ is a function of x. More precisely, for fixed $a \in X$, we can define the function $f_a : X \to \mathbf{R}$ by $f_a(x) = d(a, x)$ and ask whether it is a continuous function.

Proposition 6.8 *Let (X, d) be any metric space, let $\mathbf{R}$ have its usual metric, let $a \in X$ be fixed and $f_a : X \to \mathbf{R}$ be defined by $f_a(x) = d(a, x)$. Then f_a is a continuous function.*

Proof: If $x_n \to x \in X$, then $d(x, x_n) \to 0$, and so, using Lemma 6.7,

$$|f_a(x) - f_a(x_n)| = |d(a, x) - d(a, x_n)| \leq d(x, x_n) \to 0.$$

It follows that $f_a(x_n) \to f(x)$, and so f_a is continuous on X. ■

Note that this means that we can write

$$\lim_{n\to\infty} d(a, x_n) = d(a, lim_{n\to\infty} x_n).$$

One can think of this as saying that the distance function is continuous in one variable at a time (i.e. if one variable is fixed and the other allowed to vary).

You will recall that if we have a norm on a vector space, then we can think of $||x||$ as the distance from 0 to x, and that the norm induces a metric d on the vector space such that $||x|| = d(0, x)$. It follows from what we have just said that a norm on a vector space X is a continuous function from X to $\mathbf{R}$, since $||x|| = f_0(x)$ in the notation of Proposition 6.8.

To decide whether the metric is continuous in both variables "simultaneously" (or jointly), we have to decide how we can turn $X \times X$ into a metric space (since the metric d is a function from $X \times X$ to $\mathbf{R}$). We look at a slightly more general question: If (X, d_X) and (Y, d_Y) are two metric spaces, how can we define a metric on $X \times Y$? In Exercise 11b of Chapter 2 we saw that $d_{X\times Y}$ defined by

$$d_{X\times Y}((x, y), (u, v)) = d_X(x, u) + d_Y(y, v) \qquad \text{for } (x, y), (u, v) \in X \times Y$$

is a metric on $X \times Y$.

Proposition 6.9 L*et* (X, d) *be a metric space and let* $X \times X$ *have the metric*

$$d_{X\times X}((x, y), (u, v)) = d(x, u) + d(y, v),$$

and $\mathbf{R}$ *have its usual metric. Then the metric* d *is a continuous function from* $X \times X$ *to* $\mathbf{R}$*.*

Proof: Let $((x_n, y_n))$ be a sequence in $X \times X$ such that $(x_n, y_n) \to (x, y)$, i.e. $d(x_n, x) + d(y_n, y) = d_{X\times X}((x_n, y_n), (x, y)) \to 0$. It follows that $d(x_n, x) \to 0$ and $d(y_n, y) \to 0$. Using Lemma 6.7 we get

$$|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y) \to 0 + 0 = 0.$$

This shows that d is continuous. ■

Continuous functions are functions which preserve convergence. In order to do this, they have to preserve "closeness": if x_n gets close to x, $f(x_n)$ gets close to $f(x)$. Functions which actually preserve distances are more special; they will of course also preserve closeness. The same should be true for functions which scale distances by a specified factor. In what follows we give names to a number of different types of functions with properties like these, and then show that they are all continuous.

Definition 6.10 *Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ be a function between them. Then*

1. *f is an* **isometry** *if*

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2) \qquad \text{for all } x_1, x_2 \in X;$$

2. *f is* **Lipschitz-continuous** *if there is a $k > 0$ such that*

$$d_Y(f(x_1), f(x_2)) \leq k d_X(x_1, x_2) \qquad \text{for all } x_1, x_2 \in X;$$

   in this case k is called a **Lipschitz constant** *for f;*

3. *f is a* **contraction** *if $X = Y$ and $d_X = d_Y$ and f is Lipschitz-continuous with Lipschitz constant $k < 1$. Explicitly, if (X, d) is a metric space then a function $f : X \to X$ is a contraction if there is a constant $0 < k < 1$ such that*

$$d(f(x), f(y)) \leq k d(x, y) \qquad \text{for all } x, y \in X.$$

Note that we may think of a contraction as a function from a metric space into itself which reduces the distance between points by at least a factor k, where k is strictly less than 1.

Example 6.11 Let $\mathbf{R}^2$ have the Euclidean metric d_E and $\mathbf{C}$ its usual metric. Then the function $f : \mathbf{R}^2 \to \mathbf{C}$ defined by $f((x_1, x_2)) = x_1 + ix_2$ is an isometry, since

$$\begin{aligned} |f((x_1, x_2)) - f((y_1, y_2))| &= |x_1 - y_1 + i(x_2 - y_2)| \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \\ &= d_E((x_1, x_2), (y_1, y_2)). \end{aligned}$$

Example 6.12 The function F in Example 6.4 is Lipschitz-continuous with Lipschitz constant $k = b - a$. This follows immediately from the inequality derived in that example.

Example 6.13 Let $X = Y = [1, \infty)$, and let X and Y both have the metric induced by the Euclidean metric on $\mathbf{R}$. Fix a $k > \frac{1}{2}$ and define the function $f_k : X \to Y$ by $f_k(x) = k(\frac{1}{x} + x)$. For $x, y \in X$,

$$\begin{aligned}
|f_k(x) - f_k(y)| &= |k(\tfrac{1}{x} + x) - k(\tfrac{1}{y} + y)| \\
&= k|\tfrac{1}{x} - \tfrac{1}{y} + x - y| \\
&= k|\tfrac{y-x}{xy} + x - y| \\
&= k|x - y||1 - \tfrac{1}{xy}| \\
&\leq k|x - y|, \text{ since } 0 < \tfrac{1}{xy} \leq 1 \text{ for } x, y \geq 1.
\end{aligned}$$

It follows from this inequality that f_k is Lipschitz-continuous, and that k is a Lipschitz constant for f_k. This holds for any $k > 0$. In the case where $\frac{1}{2} < k < 1$, f_k is a contraction on X.

WARNING: A function $f : X \to X$ for which $d(f(x), f(y)) < d(x, y)$ for all $x, y \in X, x \neq y$ is not necessarily a contraction; for it to be a contraction we must be able to find a constant k with $0 < k < 1$ such that $d(f(x), f(y)) \leq kd(x, y)$ for all $x, y \in X$. The next example shows that a function may satisfy the first condition, but not the second.

Example 6.14 Let f_k be the function defined in Example 6.13 and consider the case where $k = 1$. Then it follows from the inequality derived in that example that $|f_1(x) - f_1(y)| \leq |x - y|$ for all $x, y \geq 1$. If $x, y \geq 1$ and $x \neq y$, then either $x > 1$ or $y > 1$ and therefore $xy > 1$. It follows from this that $|f_1(x) - f_1(y)| < |x - y|$ if $x, y \geq 1$ and $x \neq y$. However, this function is not a contraction. To see this, suppose we could find a $0 < c < 1$ such that $|f_1(x) - f_1(y)| = |x - y||1 - \frac{1}{xy}| \leq c|x - y|$ for all $x, y \geq 1, x \neq y$. Then we have $|1 - \frac{1}{xy}| \leq c$ for all such x and y. You can now check that choosing say $x = \frac{1}{1-c}, y = \frac{2}{(1-c)}$ leads to a contradiction. It follows that f_1 is not a contraction.

Proposition 6.15 *Let (X, d_X) and (Y, d_Y) be metric spaces. If $f : X \to Y$ is an isometry, or is Lipschitz-continuous, it is continuous.*

Proof: That an isometry is continuous follows at once from the definitions. If f is Lipschitz-continuous and $x_n \to x$ in X, then $d_X(x_n, x) \to 0$ and

$$0 \leq d_Y(f(x), f(x_n)) \leq kd_X(x, x_n) \to 0,$$

and so $f(x_n) \to f(x)$ and therefore f is continuous. ■

In line with our policy of trying to find characterisations of concepts in terms of open sets only, we now look at other ways of describing continuity. Since our definition at the moment is in terms of convergent sequences, and closed sets and closures are defined in terms of such sequences as well, we start of by exploring the link between continuity and closed sets. We need a few set-theoretic definitions and results which we collect below.

Definition 6.16 *Let X and Y be sets, $A \subset X, B \subset Y$ and $f : X \to Y$ be a function. Then we put*

1. $f(A) = \{f(x) : x \in A\}$ *($f(A)$ is called the **direct image** of A under f);*
2. $f^{-1}(B) = \{x \in X : f(x) \in B\}$ *($f^{-1}(B)$ is called the **inverse image** of B under f).*

The notation $f^{-1}(B)$ is somewhat misleading. It **does not** imply that the function f has an inverse f^{-1}; $f^{-1}(B)$ is defined even if f does not have an inverse, and simply denotes the set of all $x \in X$ which f maps into B. You may also have seen the notation $f^{\to}(A)$ for $f(A)$ and $f^{\leftarrow}(B)$ for $f^{-1}(B)$.

Proposition 6.17 *Let X and Y be sets, $A \subset X, B \subset Y$ and $f : X \to Y$ be a function. Then*

1. $A \subset f^{-1}(f(A))$;
2. $f(f^{-1}(B)) \subset B$;
3. $f^{-1}(B') = [f^{-1}(B)]'$.

Proof: (1) and (2) follow at once from the definitions, and (3) follows from

$$x \in f^{-1}(B') \Leftrightarrow f(x) \in B' \Leftrightarrow f(x) \notin B \Leftrightarrow x \notin f^{-1}(B) \Leftrightarrow x \in [f^{-1}(B)]'.$$

■

Theorem 6.18 *Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ be a function. Then the following statements are equivalent:*

1. *f is continuous;*
2. *for every $A \subset X, f(\overline{A}) \subset \overline{f(A)}$;*

3. *$f^{-1}(F)$ is closed in X for every closed subset F of Y.*

Proof: (1) $\Rightarrow$ (2): Let $y \in f(\overline{A})$. Then $y = f(x)$, with $x \in \overline{A}$, and so there is a sequence (x_n) in A such that $x_n \to x$. Since f is continuous, $f(x_n) \to f(x) = y$, so $y \in \overline{f(A)}$. It follows that $f(\overline{A}) \subset \overline{f(A)}$.
(2) $\Rightarrow$ (3): Let F be a closed set in Y. Then $f^{-1}(F) \subset X$, so it follows from (2) and Proposition 6.17.2 that

$$f(\overline{f^{-1}(F)}) \subset \overline{f(f^{-1}(F))} \subset \overline{F} = F.$$

It follows from this that $\overline{f^{-1}(F)} \subset f^{-1}(F)$. Since we always have $\overline{f^{-1}(F)} \supset f^{-1}(F)$, we get $\overline{f^{-1}(F)} = f^{-1}(F)$, and so $f^{-1}(F)$ is closed.
(3)$\Rightarrow$ (1): We show that if f is not continuous, then $f^{-1}(F)$ is not closed for some closed subset F of Y. If f is not continuous, it is not continuous at some $x \in X$, and then there is a sequence (x_n) in X such that $x_n \to x$ but $f(x_n)$ does not converge to $f(x)$. From this we can deduce that there is an $r > 0$ such that for every $k \in \mathbf{N}$, we can choose an $n_k \in \mathbf{N}$ such that $d_Y(f(x), f(x_{n_k})) \geq r$ and such that $n_k > n_m$ if $k > m$. Now put

$$F = \{y \in Y : d_Y(y, f(x)) \geq r\} = [B(f(x), r)]'.$$

Then F is closed in Y (since it is the complement of an open ball), and $(f(x_{n_k}))$ is a sequence in F. Hence (x_{n_k}) is a sequence in $f^{-1}(F)$, and $x_{n_k} \to x$. But since $d_Y(f(x), f(x)) = 0$, $f(x) \notin F$, i.e. $x \notin f^{-1}(F)$. It follows that $f^{-1}(F)$ is not closed.

■

Example 6.19 Let (X, d) be a metric space and $a \in X$. Since the function $f_a(x) = d(a, x)$ is a continuous function from X to $\mathbf{R}$ (Proposition 6.8), the set

$$\{x \in X : d(a, x) = 1\} = \{x \in X : f_a(x) = 1\} = \{x \in X : f_a(x) \in \{1\}\} = f_a^{-1}(\{1\})$$

is closed. In particular, the set $\{z \in \mathbf{C} : |z| = 1\}$ is closed.

We have established a link between closed sets and continuity; it is now easy to do the same for open sets.

Theorem 6.20 Let *(X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ be a function. Then f is continuous if and only if $f^{-1}(G)$ is open in X for every open set G in Y.*

Proof:

$$\begin{aligned} f \text{ is continuous } &\Leftrightarrow f^{-1}(F) \text{ is closed for each closed } F \subset Y \\ &\Leftrightarrow f^{-1}(G') \text{ is closed for each open } G \subset Y \\ &\Leftrightarrow [f^{-1}(G)]' \text{ is closed for each open } G \subset Y \\ &\Leftrightarrow f^{-1}(G) \text{ is open for each open } G \subset Y, \end{aligned}$$

using Proposition 6.17.3. ∎

Example 6.21 Let $\mathbf{R}$ have its usual metric. The set $\{x \in \mathbf{R} : e^{\sin x} < 1\}$ is an open set, since $(0, 1)$ is an open subset of $\mathbf{R}$, the function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = e^{\sin x}$ is continuous and

$$\{x \in \mathbf{R} : e^{\sin x} < 1\} = f^{-1}((0, 1)).$$

We stated earlier that our definition for continuity in terms of convergence of sequences (Definition 6.1) is equivalent to a “sequence-free” definition in terms of distances. One way to prove this equivalence is to make use of Theorem 6.20 and Proposition 5.14. We leave the details of the proof to you.

Proposition 6.22 L*et* (X, d_X) *and* (Y, d_Y) *be metric spaces and* $f : X \to Y$ *be a function. Then* f *is continuous at* $a \in X$ *if and only if for every* $\epsilon > 0$*, there is a* $\delta > 0$ *such that if* $d_X(x, a) < \delta$*, then* $d_Y(f(x), f(a)) < \epsilon$*.*

Proof: See Exercise 12 at the end of this chapter for hints. ∎

The above proposition states the conditions under which a function is continuous *at a point*. In general, the δ that goes with a particular ϵ will depend on the point a in the domain of the function. If the function is also continuous at another point, a different δ may well be needed for the same ϵ at the second point. A continuous function for which, given an $\epsilon > 0$, a δ can be found that will do *at every point* is rather more special, and deserves a special name:

Definition 6.23 L*et* (X, d_X) *and* (Y, d_Y) *be metric spaces,* $f : X \to Y$ *be a function and* $A \subset X$*. Then* f *is* **uniformly continuous** *on* A *if for every* $\epsilon > 0$*, there is a* $\delta > 0$ *such that if* $x_1, x_2 \in A$ *and* $d_X(x_1, x_2) < \delta$*, then* $d_Y(f(x_1), f(x_2)) < \epsilon$*.*

Example 6.24 The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = 3x - 7$ is uniformly continuous on $\mathbf{R}$ if $\mathbf{R}$ has its usual metric. To see this, given $\epsilon > 0$, take $\delta = \frac{1}{3}\epsilon$. In fact, any linear function is uniformly continuous. Make sure you can show this!

Example 6.25 Every isometry is clearly uniformly continuous.

Example 6.26 Every Lipschitz-continuous function is uniformly continuous. If k is a Lipschitz constant and $\epsilon > 0$ is given, $\delta = k^{-1}\epsilon$ will do.

We originally defined continuity in terms of convergence of sequences. Can the same be done for uniform continuity? The theorem below gives an affirmative answer.

Theorem 6.27 L*et* (X, d_X) *and* (Y, d_Y) *be metric spaces,* $A \subset X$ *and* $f : X \to Y$ *be a function. Then* f *is uniformly continuous on* A *if and only if whenever* (x_n) *and* (y_n) *are sequences in* A *such that* $d_X(x_n, y_n) \to 0$*, then* $d_Y(f(x_n), f(y_n)) \to 0$*.*

Proof: We leave the proof to you as an exercise. See Exercise 14 for some hints. ∎

Example 6.28 It is not difficult to show, using this characterisation, that the function $f : (0, \infty) \to (0, \infty)$, defined by $f(x) = x^{-1}$, is not uniformly continuous if $(0, \infty)$ has the metric induced by the usual metric of $\mathbf{R}$. We ask you to check this in Exercise 14.

The characterisation of continuous functions in terms of open sets is a little disappointing. It would have been much more satisfactory if a continuous function was simply one which mapped open sets to open sets, or closed sets to closed sets. That this is too much to hope for is shown by the following examples.

Example 6.29 Let $\mathbf{R}$ have its usual metric and define $f : \mathbf{R} \to \mathbf{R}$ by $f(x) = 2$ for all $x \in \mathbf{R}$. Then f is continuous, $G = (0, 1)$ is an open subset of $\mathbf{R}$, but $f(G) = \{2\}$ is closed, but not open.

Example 6.30 Let $X = \mathbf{R}$, with the discrete metric and $Y = \mathbf{R}$, with the usual metric. The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = x$ for all $x \in \mathbf{R}$ is continuous. The set $F = (0, 1)$ is closed with respect to the discrete metric (in fact, all subsets of $\mathbf{R}$ are closed), but $f(F) = (0, 1)$ is open, and not closed, in the usual metric of $\mathbf{R}$.

These examples show the need for the terminology we introduce in the next definition. As we'll see shortly, there are other reasons as well.

Definition 6.31 *Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Then we call f*

1. **an open function** *if $f(G)$ is an open set in Y for every open set G in X;*
2. **a closed function** *if $f(F)$ is a closed set in Y for every closed set F in X.*

The function in Example 6.29 is continuous and closed, but not open. The function in Example 6.30 is continuous, but neither open nor closed (check this!). In the exercises you'll find an example of a function that is continuous and open, but not closed.

If $f : X \to Y$ is a bijection (i.e. a function that is both one-to-one and onto), then it has an inverse $f^{-1} : Y \to X$. If $A \subset X$, we can look at the inverse image of A under the function f^{-1}, i.e. at $(f^{-1})^{-1}(A)$. By definition this is just

$$(f^{-1})^{-1}(A) = \{y \in Y : f^{-1}(y) \in A\} = \{y \in Y : y = f(f^{-1}(y)) \in f(A)\} = f(A).$$

Note that in this case the inverse image of $B \subset Y$ under f (i.e the set $f^{-1}(B)$) is the same as the direct image of B under f^{-1}.

Proposition 6.32 *Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ be a continuous bijection. Then the following statements are equivalent:*

1. *f^{-1} is continuous;*
2. *f is open;*
3. *f is closed.*

Proof: The equivalence of (1) and (2) follows from

$$\begin{aligned} f^{-1} \text{ is continuous} \quad &\Leftrightarrow (f^{-1})^{-1}(G) \text{ is open for every open } G \subset X \\ &\Leftrightarrow f(G) \text{ is open for every open } G \subset X \\ &\Leftrightarrow f \text{ is open.} \end{aligned}$$

The proof of the equivalence of (1) and (3) is similar. ■

It follows from Proposition 6.32 that if f is a continuous bijection between two metric spaces such that f^{-1} is continuous, then both f and f^{-1} will be open. We could express this by saying that f "preserves open sets in both directions", i.e. f maps open sets in X to open sets in Y and f^{-1} maps open sets in Y to open sets in X. This means that we could use the function f to *identify* the topology $\mathcal{T}_X$ induced on X by d_X with the topology $\mathcal{T}_Y$ induced by d_Y on Y. There is a one-to-one correspondence (given by the function f) between the open sets of X and the open sets of Y. In this case we could think of $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ as essentially the same topological space. The function f gives a one-to-one correspondence between the elements of X and Y (since it is a bijection), and also a one-to-one correspondence between the open sets of (X, d_X) and (Y, d_Y) (since it and its inverse are both open functions).

We introduce some new terminology which is frequently used to describe this situation:

Definition 6.33 L*et* (X, d_X) *and* (Y, d_Y) *be metric spaces and* $f : X \to Y$ *a bijection.*

1. *If both* f *and* f^{-1} *are continuous, we call* f *a* **homeomorphism** *(or* **topological isomorphism***). In this case we say that* (X, d_X) *and* (Y, d_Y) *are* **homeomorphic** *(or* **topologically isomorphic***).*

2. *If* f *is an isometry, we say that* (X, d_X) *and* (Y, d_Y) *are* **isometric** *(or* **metrically isomorphic***).*

To summarise: Two metric spaces are homeomorphic if they are "the same" as topological spaces; they are isometric if they are "the same" as metric spaces. In both cases they should be linked by a bijection; in the first case the bijection should be a homeomorphism as well, in the second case an isometry.

It follows immediately from Proposition 6.32 that a continuous bijection between metric spaces is a homeomorphism if and only if it is an open map, and also if and only if it is a closed map.

Example 6.34 Let $X = (-\frac{\pi}{2}, \frac{\pi}{2})$ with the metric induced by the usual metric on $\mathbf{R}$, and $Y = \mathbf{R}$ with its usual metric. Let $f : X \to Y$ be defined by $f(x) = \tan x$. Then f is continuous, and so is $f^{-1}(x) = \arctan x$, so f is a homeomorphism, and X and Y are homeomorphic. But f is not an isometry (for example, $d(f(0), f(\frac{\pi}{4})) = 1 \neq \frac{\pi}{4} = d(0, \frac{\pi}{4})$). We can therefore say that the intervals $X = (-\frac{\pi}{2}, \frac{\pi}{2})$ and $(-\infty, \infty)$, both equipped with the metric induced by the usual metric on $\mathbf{R}$, are the same as topological spaces, but not as metric spaces.

Example 6.35 Let $\mathbf{R}^2$ and $\mathbf{C}^2$ both have their usual metrics. We have seen in Example 6.11 that the function $f : \mathbf{R}^2 \to \mathbf{C}$ defined by $f((x_1, x_2)) = x_1 + ix_2$ is an isometry, and therefore $\mathbf{R}^2$ and $\mathbf{C}$ are isometric metric spaces. We have now fulfilled our promise, made in Chapter 2, of showing that these two metric spaces are "in some sense the same".

Summary.

We defined the notion of continuity for a function between two metric spaces and characterised the concept for some well-known metric spaces. The metric on any metric space was shown to be continuous. Special classes of continuous functions (Lipschitz-continuous functions, contractions, isometries, uniformly continuous functions) were introduced. Continuity was characterised in terms of open and closed sets, and open and closed functions introduced. This led to the idea of a homeomorphism between metric spaces, and the difference between isometric and homeomorphic metric spaces.

Historical notes

Georg Friedrich Bernhard Riemann (1826 – 1866)

Riemann's father was a Lutheran minister who took responsibility for the initial schooling of his six children himself. He taught Bernhard until he was ten. At school Bernhard quickly showed an interest in mathematics. In 1846 Riemann went to the University of Göttingen, where he at first studied theology since he was encouraged to do so by his father. His interest was in mathematics, however, and he obtained permission from his father to change course. Although Gauss taught at Göttingen at the time, the mathematics department was not all that good, and Riemann moved to Berlin in 1847. There he came under the influence of Dirichlet. Riemann liked the latter's intuitive way of reasoning and adopted the same style. During this time Riemann worked on a general theory of complex functions.

In 1849 Riemann returned to Göttingen and in 1851 he finished his Ph.D. under the supervision of Gauss. In this he studied what are now called Riemann surfaces, and used topological methods in complex function theory. On Gauss's recommendation he was appointed at Göttingen. In his research work on representation of functions by trigonometric series he introduced conditions on a function to ensure its integrability (now known as Riemann integrability).

To qualify as a lecturer, Riemann had to give a public lecture. The lecture he gave *On the hypotheses that lie at the foundations of geometry* is now a classic of mathematics. In it he defines what is now known as Riemannian space. Not many people appreciated the depth of his ideas. It was only much later that his ideas turned out to be exactly what Einstein needed for his theory of relativity.

In 1857 Riemann was appointed professor and published a very important paper on the theory of abelian functions. In it he investigated the topological properties of Riemann surfaces. Dirichlet, who had been appointed to the chair in mathematics at Göttingen in 1855, died in 1859 and he was succeeded by Riemann. Shortly afterwards he was elected to the Berlin Academy of Sciences and at this occasion delivered a lecture on the distribution of prime numbers. In it he used the so-called zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

which he showed could be regarded as a function of a complex variable. Riemann conjectured that all the non-trivial roots of this function had real part equal to $\frac{1}{2}$. This is now known as the Riemann hypothesis, and remains one of the major unsolved problems of mathematics.

Riemann married in 1862. In the same year caught a bad cold which later developed into tuberculosis. He tried to improve his health by spending much time in Italy with its warmer climate. However, his health deteriorated and he died, aged 39, while on a visit to Italy in July 1866.

Henri Léon Lebesgue (1875 – 1941)

Henri Lebesgue was born in Beauvais in France in 1875. He studied at the École Normale Supérieure in Paris, and then lectured in Nancy. In 1902 he submitted his dissertation, entitled *Intégrale, longeur, aire* to the University of Nancy. In it he introduced a general theory of measure, and extended the notion of an integral from the Riemann integral to a more general integral now known as the Lebesgue integral. This was a remarkable achievement and it had a profound influence in the field of Fourier analysis. Lebesgue himself was scared of too much generalisation and did not make further contributions to this field. He published widely in other areas such as topology and Fourier analysis, however. He was appointed to the Sorbonne in Paris in 1910.

Rudolf Otto Sigismund Lipschitz (1832 – 1903)

Rudolf Lipschitz was born in Königsberg, Germany (now Kalingrad, Russia) in 1832. He made important contributions to the study of the Hamilton-Jacobi method for solving the equations of motion of dynamical systems. He is best known for his formulation of what is now known as the Lipschitz condition which ensures the existence of a unique solution to a first order differential equation. (You will see more of this in Chapter 8.) He died in Bonn, Germany in 1903.

Exercises

1. Let (X, d) be a metric space.
   (a) Let $a \in X$ be fixed. Show that the function $f : X \to X$ defined by $f(x) = a$ for every $x \in X$ is continuous.
   (b) Show that the function $id_X : (X, d) \to (X, d)$ defined by $id_X(x) = x$ for every $x \in X$ is continuous.

2. For every function $f : \mathbf{R}^2 \to \mathbf{R}$ below determine whether it is continuous when $\mathbf{R}^2$ and $\mathbf{R}$ have their usual metrics. You may use the continuity of the single-variable functions $\sin x$ and e^x without proof.
   (a) $f(x, y) = 3x - y + 2$;
   (b) $f(x, y) = \sin(x^2 + y^2)$;
   (c) $f(x, y) = e^{-3xy}$

3. For each $n \in \mathbf{N}$, let the function $f_n : [0, 1] \to \mathbf{R}$ be defined by
$$f_n(x) = nx(1 - x^2)^n.$$
   (a) Show that $\lim_{n\to\infty} f_n(x) = 0$ for every $x \in [0, 1]$.
   (b) Show that $\lim_{n\to\infty} \int_0^1 f_n(x)\, dx = \frac{1}{2}$.
   (c) Does the sequence (f_n) converge uniformly on $[0, 1]$?

4. Let $\mathbf{R}^2$ and $\mathbf{R}$ have their usual metrics and $f : \mathbf{R}^2 \to \mathbf{R}$ be defined by
$$f(x, y) = x.$$
   (a) Show that f is continuous.
   (b) Show that $f(G)$ is open for every open subset G of $\mathbf{R}^2$. [Hint: Show that every $y \in f(G)$ is the centre of an open ball contained in $f(G)$.]
   (c) Show that the set $F = \{(x, \frac{1}{x}) \in \mathbf{R}^2 : x \in (0, \infty)\}$ is closed in $\mathbf{R}^2$, but that $f(F)$ is not closed in $\mathbf{R}$.

5. Let $C([0, 1])$ have its usual supremum metric and $F : C([0, 1]) \to C([0, 1])$ be defined by $F(f) = |f|$, where $|f|(x) = |f(x)|$ for all $x \in [0, 1]$. Show that F is continuous.

6. Let $Y = \mathbf{C}$ with d_Y the usual metric, and $X = \mathbf{C}$ with d_X the metric defined by
$$d_X(z,w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w \end{cases}$$
Let $f : X \to Y$ be the identity function, i.e. $f(z) = z$. Is f is Lipschitz-continuous? If so, find a Lipschitz constant for it.

7. Let d be the lift metric on $\mathbf{R}^2$ (see Exercise 9 in Chapter 2) and let $\mathbf{R}$ have its usual metric. Let the function $f : \mathbf{R}^2 \to \mathbf{R}$ be defined by
$$f(x,y) = \begin{cases} \frac{x}{1-y} & \text{if } y \neq 1 \\ 1 & \text{if } y = 1 \end{cases}$$
Is f continuous at $(1,1)$? And at $(0,1)$? Give reasons for your answer. [Hint: Look at Exercise 7 in Chapter 3.]

8. Let $\mathbf{R}$ have its usual metric. For each of the following subsets of $\mathbf{R}$, say whether it is open, closed or neither. Use continuity where possible.
   (a) $\{x \in \mathbf{R} : \sin x \geq 0\}$;
   (b) $\{x \in \mathbf{R} : -0.1 < \cos x < 0.3\}$;
   (c) $\{x \in \mathbf{R} : 0.5 < \sin x \leq 2\}$.

9. Let $C([0,1])$ have its usual supremum metric.
   (a) Is $\{f \in C([0,1]) : 0 < \int_0^1 f(x)\,dx < 2\}$ open? Justify your answer.
   (b) Is $\{f \in C([0,1]) : \int_0^1 |f(x)|\,dx \leq 1\}$ closed? Justify your answer.

10. Let $\mathbf{R}^2$ have its usual Euclidean metric and let A be a 2×2 matrix. Define $F : \mathbf{R}^2 \to \mathbf{R}^2$ by $F(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x} \in \mathbf{R}^2$. Show that F is continuous.

11. Prove that a linear map from $\mathbf{R}^n$ to $\mathbf{R}^m$ is continuous if $\mathbf{R}^n$ and $\mathbf{R}^m$ have their usual Euclidean metrics. [Hint: Such a map can be represented by a matrix.]

12. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ be a function. Prove that f is continuous at $a \in X$ if and only if for every $\epsilon > 0$, there is a $\delta > 0$ such that $d_Y(f(x), f(a)) < \epsilon$ whenever $d_X(x,a) < \delta$. [Hint: Use the fact that $B(f(a), \epsilon)$ is an open set in Y, and therefore its inverse image must be an open set in X. Then use Proposition 5.14.]

13. Prove that every Lipschitz-continuous function is in fact uniformly continuous on its domain.

14. Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subset X$ and $f : X \to Y$ be a function.

    (a) Prove that if f is uniformly continuous on A and (x_n) and (y_n) are sequences in A such that $d_X(x_n, y_n) \to 0$, then $d_Y(f(x_n), f(y_n)) \to 0$.

    (b) Formulate and prove the converse of the statement in (a). [Hint: Try to prove the contra-positive.]

    (c) Let $X = Y = (0, \infty), d_X$ and d_Y both be the metric induced by the usual metric of $\mathbf{R}$ and $f(x) = x^{-1}$. Show that f is not uniformly continuous using the results above.

15. Prove that the metrics d_1 and d_2 on the set X are equivalent if and only if the map $f : (X, d_1) \to (X, d_2)$ defined by $f(x) = x$ for every $x \in X$ is a homeomorphism.

16. Let $a, b \in \mathbf{R}$, with $a < b$. Show that the interval (a, b) with the metric induced by the usual metric on $\mathbf{R}$ is homeomorphic to $\mathbf{R}$, with its usual metric. [Hint: Look at Example 6.34.]

17. Let $\mathbf{R}^2$ have its Euclidean metric, and let

$$S = \{(x, y) \in \mathbf{R}^2 : 0 < x < 1, 0 < y < 1\}$$

have the metric induced by the Euclidean metric of $\mathbf{R}^2$. Show that S is homeomorphic to $\mathbf{R}^2$.

Chapter 7

Spotting convergence without using limits.

In which we meet Cauchy sequences and metric spaces in which they always converge.

If one is given a sequence (x_n) in a metric space, how do you check whether it is convergent? There seems to be a pretty obvious answer to this question: Just use the definition! And what does the definition say? The sequence (x_n) converges to x if $d(x_n, x) \to 0$. But what is this x? The limit, of course. But if you already know the limit, then surely you already know that the sequence converges...

This all sounds rather confusing, especially coming at a time you may just start feeling quite comfortable with convergent sequences. It is important to realise, however, that the definition of convergence we have only allows us to check that a given sequence converges to a particular limit. To determine whether a sequence converges, we need to guess a limit first, and we can then use the definition to check whether the sequence converges to this particular point. If it does, all is well. If not, we can try another guess, and another, and so on, until we find one that satisfies the definition. But suppose we never find one that does? Does this mean that the sequence does not converge, or that we have simply not guessed the correct limit yet? You may think that it is usually quite easy to guess correctly, but a look at the sequence $((1 + \frac{1}{n})^n)$ in $\mathbf{R}$ should convince you that this is not necessarily the case.

What is it that we need then? It would be very satisfactory to have a test for convergence of a sequence that uses the terms of the sequence only, and nothing else. The test should tell us whether the sequence converges or not; what the limit is (in the case where it does converge) is not at issue. Such a test would enable us to say whether a sequence converges without having to guess a limit for it. We therefore need an intrinsic property of a sequence (one that depends on the terms of the sequence only) that will guarantee its convergence.

Where does one start looking for such a property? A good strategy is to look at properties of convergent sequences, and amongst them to try to find one that is not only *necessary*, but also *sufficient* for convergence. We can picture a convergent sequence as one for which the terms cluster around the limit, i.e. the terms eventually are all close to the limit. But if they are all close to the limit, then they must also eventually be close to one another. This is the substance of the next proposition (even though this simple idea may at first be obscured by the appearance of the dreaded epsilon).

Proposition 7.1 L*et (X, d) be a metric space and (x_n) a convergent sequence in X. Then for every $\epsilon > 0$, we can find a positive integer N_ϵ such that if $m \geq n \geq N_\epsilon$, then $d(x_m, x_n) < \epsilon$.*

Proof: Suppose $x_n \to x \in X$ as $n \to \infty$, and let $\epsilon > 0$. Then $d(x_n, x) \to 0$ as $n \to \infty$, and we can therefore find an $N_\epsilon \in \mathbf{N}$ such that $d(x_n, x) < \frac{\epsilon}{2}$ for $n \geq N_\epsilon$. If $m \geq n \geq N_\epsilon$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

■

[The notation N_ϵ is used simply to emphasise the fact that the point in the sequence beyond which the terms will be within ϵ of one another will depend on the value of ϵ; in general, the smaller ϵ, the larger N_ϵ will have to be.]

The property stated in the proposition does not look very attractive (blame it all on the ϵ!), but it has the right kind of credentials. It is a property that every convergent sequence has (i.e. it is necessary for convergence); it is a property that depends on the terms of the sequence only. This warrants giving this property a name of its own.

Definition 7.2 *Let (X, d) be a metric space. A sequence (x_n) in X is called a* **Cauchy sequence** *if for every $\epsilon > 0$, there is an $N_\epsilon \in \mathbf{N}$ such that if $m \geq n \geq N_\epsilon$, then $d(x_m, x_n) < \epsilon$.*

These sequence are named after the French mathematician Cauchy, about whom you can read more at the end of the chapter. We can now restate Proposition 7.1 using this terminology:

In a metric space, every convergent sequence is a Cauchy sequence.

It is now natural to ask whether the converse also holds: Is it true in any metric space that every Cauchy sequence is convergent?

Example 7.3 In the metric space $\mathbf{R}$, with its usual metric, every Cauchy sequence is convergent to a real number. This is not an obvious result, but in the Real Analysis course you saw how it could be deduced from some other properties of $\mathbf{R}$.

Example 7.4 Let $X = \mathbf{Q}$ have the metric induced by the usual metric on $\mathbf{R}$. For each $n \in \mathbf{N}$, let x_n be the rational number which gives the first n digits in the decimal expansion of $\sqrt{2}$, i.e. $x_1 = 1, x_2 = 1.4, x_3 = 1.41, x_4 = 1.414$ and so on. It is not too difficult to check that (x_n) is a Cauchy sequence in $\mathbf{Q}$. For any $n \in \mathbf{N}, \quad x_{n+1} - x_n \leq \frac{1}{10^{n-1}}$, and therefore if $m > n$,

$$\begin{aligned}
|x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \ldots + |x_{n+1} - x_n| \\
&\leq \frac{1}{10^{m-2}} + \frac{1}{10^{m-3}} + \cdots + \frac{1}{10^{n-1}} \\
&\leq \frac{1}{10^{n-1}} + \frac{1}{10^{n}} + \frac{1}{10^{n+1}} \ldots = \frac{1}{10^{n-1}} \cdot \frac{10}{9} \\
&\leq \frac{1}{10^{n-2}}.
\end{aligned}$$

Therefore, if we are given an $\epsilon > 0$, we can choose N_ϵ large enough to ensure that $\frac{1}{10^{N_\epsilon - 2}} < \epsilon$, and then we'll have $|x_m - x_n| < \epsilon$ if $m \geq n \geq N_\epsilon$. But the sequence (x_n) does not converge to a limit in the metric space $\mathbf{Q}$. (When considered as a sequence in $\mathbf{R}$, it does converge to $\sqrt{2}$, but $\sqrt{2} \notin \mathbf{Q}$. Certainly (x_n) cannot converge to any real number other than $\sqrt{2}$, by uniqueness of limits in a metric space.) Hence it is **not true** that every Cauchy sequence in $\mathbf{Q}$ is convergent **to an element of** $\mathbf{Q}$.

You may well feel that the last example is a bit unsatisfactory. Granted, the Cauchy sequence in $\mathbf{Q}$ does not converge to a point in $\mathbf{Q}$, but it does converge to *something*, even though this something is outside $\mathbf{Q}$. The point is that for a metric space to be complete, every Cauchy sequence in it must converge to a limit *in the metric space*; limits outside the metric space don't help. (The metric space is our *whole* world.) We'll see a little later that the situation illustrated by this example is typical of the situation in general.

What these two examples show already is that in at least one metric space the property of being a Cauchy sequence is enough to guarantee convergence **to a point in the space**, and in at least one other metric space it is not.

Example 7.5 Let $\mathbf{R}^2$ have its usual Euclidean metric, and suppose $(\mathbf{x}_n) = (x_{n1}, x_{n2})$

is a Cauchy sequence in $\mathbf{R}^2$. For $m, n \in \mathbf{N}$, we have

$$\begin{aligned} |x_{m1} - x_{n1}| &\leq \sqrt{(x_{m1} - x_{n1})^2 + (x_{m2} - x_{n2})^2} = d(\mathbf{x}_m, \mathbf{x}_n) \\ |x_{m2} - x_{n2}| &\leq \sqrt{(x_{m1} - x_{n1})^2 + (x_{m2} - x_{n2})^2} = d(\mathbf{x}_m, \mathbf{x}_n). \end{aligned}$$

Hence if $\epsilon > 0$, we can find an $N_\epsilon \in \mathbf{N}$ such that if $m \geq n \geq N_\epsilon$, then $d(x_m, x_n) < \epsilon$, and therefore also $|x_{m1} - x_{n1}| < \epsilon$. It follows that (x_{n1}) is a Cauchy sequence in $\mathbf{R}$. Since $\mathbf{R}$ with its usual metric is complete, there is an $x_1 \in \mathbf{R}$ such that $x_{n1} \to x_1$ as $n \to \infty$. In the same way we can find an $x_2 \in \mathbf{R}$ such that $x_{n2} \to x_2$ as $n \to \infty$. Since coordinatewise convergence implies convergence with respect to the Euclidean metric, it follows that $(x_{n1}, x_{n2}) = \mathbf{x}_n \to \mathbf{x} = (x_1, x_2) \in \mathbf{R}^2$. This shows that every Cauchy sequence in $\mathbf{R}^2$ converges to a limit in $\mathbf{R}^2$.

Example 7.6 In the same way it can be shown that every Cauchy sequence in $\mathbf{C}$ converges to a limit in $\mathbf{C}$ when $\mathbf{C}$ has its usual metric. (Check this!)

Example 7.7 Essentially the same proof can be used as well to show that for any $m \in \mathbf{N}$, every Cauchy sequence in $\mathbf{R}^m$ with the Euclidean metric converges to a point in $\mathbf{R}^m$.

Our hope that the property of being a Cauchy sequence in a metric space will be equivalent to being a convergent sequence in that space has been dashed by Example 7.4 (unfortunately one example is enough to do this!). The examples above show that there are many metric spaces in which Cauchy sequences are convergent. Rather than abandoning Cauchy sequences altogether because they don't always work, we adopt a strategy which may at first look very much like going for second best. The idea is to concentrate on those metric spaces for which the Cauchy property does guarantee convergence. To do this, we need a name for them.

Definition 7.8 L*et* (X, d) *be a metric space. The metric space* (X, d) *is called* **complete** *if every Cauchy sequence in* X *converges* **to an element of** X. *(A metric space for which this is not the case is sometimes called* **incomplete***.)*

It follows from the examples we've had already that $\mathbf{R}$, $\mathbf{C}$, $\mathbf{R}^2$ and more generally $\mathbf{R}^m$, for any $m \in \mathbf{N}$, are all complete metric spaces when equipped with their usual metrics. The metric space $\mathbf{Q}$, equipped with the metric induced by the usual metric on $\mathbf{R}$, is an incomplete metric space. We give another, rather more complicated, example of an incomplete metric space.

Example 7.9 Let $C([-1,1])$ have the metric defined by

$$d(f,g) = \int_{-1}^{1} |f(x) - g(x)|\, dx \qquad f,g \in C([-1,1]).$$

For every $n \in \mathbf{N}$, define the function $f_n : [-1,1] \to \mathbf{R}$ by

$$f_n(x) = \begin{cases} 0 & \text{if } \; -1 \le x \le 0 \\ nx & \text{if } \; 0 < x \le \frac{1}{n} \\ 1 & \text{if } \; \frac{1}{n} \le x \le 1 \end{cases}$$

Then $f_n \in C([-1,1])$ for each $n \in \mathbf{N}$ and if $n > m$,

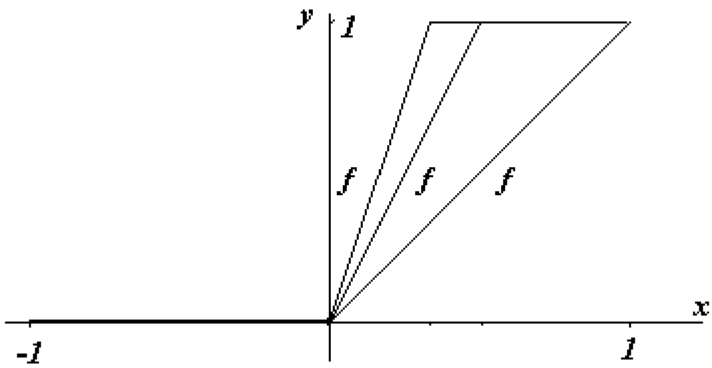


Figure 7.1: The functions f_1, f_2 and f_3.

$$\begin{aligned} d(f_n, f_m) &= \textstyle\int_0^{\frac{1}{n}} (nx - mx)\, dx + \int_{\frac{1}{n}}^{\frac{1}{m}} (1 - mx)\, dx \\ &= \tfrac{1}{2}(\tfrac{1}{m} - \tfrac{1}{n}) \\ &< \tfrac{1}{2m} \end{aligned}$$

Given $\epsilon > 0$, we can choose $N_\epsilon \in \mathbf{N}$ such that $\frac{1}{2N_\epsilon} < \epsilon$. It follows that if $n > m \ge N_\epsilon$, then $d(f_n, f_m) < \epsilon$, and hence that (f_n) is a Cauchy sequence in $C([-1,1])$. Now suppose that there is an $f \in C([-1,1])$ such that $f_n \to f$ as $n \to \infty$ with respect to the metric d. Then

$$\begin{aligned} d(f_n, f) &= \textstyle\int_{-1}^{1} |f_n(x) - f(x)|\, dx \\ &= \textstyle\int_{-1}^{0} |f(x)|\, dx + \int_0^{\frac{1}{n}} |nx - f(x)|\, dx + \int_{\frac{1}{n}}^{1} |1 - f(x)|\, dx. \end{aligned}$$

Since $d(f_n, f) \to 0$ as $n \to \infty$, and the integrands in all three integrals are non-negative and continuous, we must have $f(x) = 0$ for all $x \in [-1,0]$ and $f(x) = 1$ for all $x \in (0,1]$. But this implies that f is not continuous at $x = 0$, contradicting the fact that $f \in C([-1,1])$. It follows that (f_n) cannot converge to an $f \in C([-1,1])$, and hence that $C([-1,1])$ is not complete when equipped with this metric.

This, you may feel, is a more convincing example of an incomplete metric space. But you may still have the suspicion that if we were prepared to allow functions other than continuous functions as limits things would not have gone wrong. Here the matter is not as simple as was the case with the rational numbers, though. Which other functions should we allow? All real-valued functions on $[-1, 1]$? But how do we define a metric on this larger set of functions? We would need to be able to integrate functions other than continuous functions, and it is not at all clear how this should be done, if it can be done at all. There does seem to be a stronger case now for allowing the notion of an incomplete metric space to handle cases such as these.

So far we have defined completeness for *metric spaces*. It is possible to introduce a similar notion for *subsets* of metric spaces as well. Here is the definition:

Definition 7.10 L*et (X, d) be a metric space and A a subset of X. We say that A is a **complete subset** of X if every Cauchy sequence in A converges to a limit in A. A subset which is not complete is called incomplete.*

Example 7.11 Let $\mathbf{R}$ have its usual metric and let $A = [0, 1], B = (0, 1]$. Then every Cauchy sequence in A converges to a real number (since $\mathbf{R}$ is complete), and this limit is in $A = [0, 1]$. This shows that A is a complete subset of $\mathbf{R}$. The sequence $(\frac{1}{n})$ is a Cauchy sequence in B, but it does not converge *to a limit in* B. Hence B is an incomplete subset of $\mathbf{R}$.

The properties of completeness and closedness for subsets of a metric space are different, but have a similar ring about them. The feeling that they have a lot in common is strengthened by the example we have just given. It is therefore important to be clear about what each property means.

- A closed set contains the limit of every *convergent* sequence in it.
- A complete set contains the limit of every *Cauchy* sequence in it. Note that this already implies that every Cauchy sequence in the set converges.

It is therefore possible for a closed set not to be complete; this will happen if it contains a Cauchy sequence that does not converge. In the next two propositions we make the relationship between these two properties clearer.

Proposition 7.12 A *complete subset of a metric space is closed.*

Proof: Let A be a complete subset of the metric space (X, d), and let (x_n) be a sequence in A such that $x_n \to x \in X$. By Proposition 7.1 (x_n) is a Cauchy sequence in X, and hence in A, and since A is complete, it converges to a point y (say) in A. Then we have $x_n \to x, x_n \to y$ as $n \to \infty$. But the limit of a convergent sequence is unique, so we must have $x = y \in A$. It follows that A is closed. ■

Proposition 7.13 A *closed subset of a* complete *metric space is complete.*

Proof: Let A be a closed subset of the complete metric space (X, d), and (x_n) a Cauchy sequence in A. Then (x_n) is a Cauchy sequence in X, and since X is complete, it converges, say $x_n \to x \in X$. Since A is closed, $x \in A$, and so A is complete. ■

Corollary 7.14 *A subset of a complete metric space is complete if and only if it is closed.*

Example 7.15 Let $\mathbf{R}$ have its usual metric. Then any closed interval $[a, b]$ is a complete subset of $\mathbf{R}$, but $\mathbf{Q}$ is not.

To give a more interesting application of Proposition 7.13 we first prove the completeness of an important metric space.

Theorem 7.16 L*et X be a non-empty set and $B(X)$ the set of all bounded real-valued functions on X. For $f, g \in B(X)$, let*

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in X\}.$$

Then d is a metric on $B(X)$, and equipped with this metric, $B(X)$ is a complete metric space.

Proof: The proof that d is a metric on $B(X)$ is similar to the proof for the special case $X = [a, b]$ (see Exercise 8, Chapter 2). To prove that $B(X)$ is complete, we have to show that every Cauchy sequence in $B(X)$ converges to a limit in $B(X)$. Suppose that (f_n) is a Cauchy sequence in $B(X)$. We have to show that there is a function f in $B(X)$ to which the sequence (f_n) converges with respect to the metric of $B(X)$. To define such a function f, the first step is to find its value $f(x)$ for every $x \in X$. Since (f_n) is a Cauchy sequence, it follows that if $\epsilon > 0$ is given, we can find an $N_\epsilon \in \mathbf{N}$ such that $d(f_m, f_n) < \epsilon$ if $m \geq n \geq N_\epsilon$. Now for each $x \in X$ and for $m \geq n \geq N_\epsilon$, we have

$$|f_m(x) - f_n(x)| \leq \sup\{|f_m(x) - f_n(x)| : x \in X\} = d(f_m, f_n) < \epsilon.$$

It follows that for each fixed $x \in X$, $(f_n(x))$ is a Cauchy sequence in $\mathbf{R}$. Since $\mathbf{R}$ is complete, the Cauchy sequence $(f_n(x))$ converges to a real number which we'll denote by $\lim_{n\to\infty} f_n(x)$, as usual. We can now define a function $f : X \to \mathbf{R}$ by $f(x) = \lim_{n\to\infty} f_n(x)$. We have to show that $f \in B(X)$ and $f_n \to f$ in $B(X)$.
Let $\epsilon > 0$. Then, since (f_n) is a Cauchy sequence, we can find an N_ϵ such that $d(f_m, f_n) < \epsilon$ if $m \geq n \geq N_\epsilon$. Let $x \in X$. Then, as above we have that if $m \geq n \geq N_\epsilon$, then

$$0 \leq |f_m(x) - f_n(x)| \leq d(f_m, f_n) < \epsilon \tag{7.1}$$

Let $m \to \infty$ in (7.1), then

$$|f(x) - f_n(x)| \leq \epsilon \quad \text{if } n \geq N_\epsilon \tag{7.2}$$

But we can do this for every $x \in X$, so

$$d(f, f_n) = \sup\{|f(x) - f_n(x)| : x \in X\} \leq \epsilon \quad \text{if } n \geq N_\epsilon.$$

This shows that $d(f, f_n) \to 0$ as $n \to \infty$. To show that f is bounded, note that it follows from (7.2) that $|f(x) - f_{N_\epsilon}(x)| \leq \epsilon$ for every $x \in X$. Since f_{N_ϵ} is a bounded function, there is a $C > 0$ such that $|f_{N_\epsilon}(x)| \leq C$ for every $x \in X$. It follows that for all $x \in X$,

$$\begin{aligned} |f(x)| &\leq |f(x) - f_{N_\epsilon}(x)| + |f_{N_\epsilon}(x)| \\ &\leq \epsilon + C \end{aligned}$$

This shows that f is bounded, i.e. $f \in B(X)$, and hence $f_n \to f \in B(X)$ as $n \to \infty$. Therefore $B(X)$ is complete. ■

Corollary 7.17 *Let ℓ_∞ be the set of all bounded real sequences and for $\mathbf{x} = (x_n), \mathbf{y} = (y_n) \in \ell_\infty$, let*

$$d(\mathbf{x}, \mathbf{y}) = \sup_{n\in\mathbf{N}} |x_n - y_n|.$$

Then (ℓ_∞, d) is a complete metric space.

Proof: It is enough to note that a real sequence is a function from $\mathbf{N}$ to $\mathbf{R}$, and that we therefore have $B(\mathbf{N}) = \ell_\infty$. The result now follows by taking $X = \mathbf{N}$ in the theorem. ■

Corollary 7.18 *Let (X, d) be a metric space and let $C_b(X)$ the set of all bounded* continuous *real-valued functions on X. Then $C_b(X)$ is a complete subspace of $B(X)$.*

Proof: It follows from Proposition 7.13 and Theorem 7.16 that it will be enough to show that $C_b(X)$ is a closed subset of $B(X)$. Since convergence with respect to the supremum metric in $B(X)$ is equivalent to uniform convergence, it is enough to show that if a sequence of bounded continuous functions on X converge uniformly to a bounded function, then this limit function must be continuous as well. In the case where X is a closed bounded interval of the real line you have seen a proof of this in Real Analysis. The proof for the more general case is almost identical, and we leave it to you as an exercise to check the details. Keep in mind that if X is a closed bounded interval $[a, b]$, any continuous real-valued function on it will automatically be bounded, and so $C_b([a, b]) = C([a, b])$. ■

Although we have said quite a lot about the notion of completeness, we have not really put it to use yet. In the next chapter it will feature prominently in applications. Here we'll be content with proving one useful theorem (known as Cantor's Intersection Theorem) that uses completeness. To state it, we need a definition:

Definition 7.19 L*et (X, d) be a metric space and A a non-empty subset of X. The* **diameter** *of A, diam(A) is defined by*

$$diam(A) = \sup\{d(x, y) : x \in A, y \in A\}.$$

If $A = \emptyset$, we put diam$(A) = 0$.

Example 7.20 Let $\mathbf{R}$ have its usual metric and let
$A = (2, 5]$, $B = \mathbf{Q}$, $C = \{3\}$, $D = (-3, -1) \cup (2, 4)$.
Then $\text{diam}(A) = 3$, $\text{diam}(B) = \infty$, $\text{diam}(C) = 0$, $\text{diam}(D) = 7$.

Example 7.21 Let $\mathbf{R}^2$ have its usual Euclidean metric. If

$$A = \{(x_1, x_2) \in \mathbf{R}^2 : 0 < x_1 < 2, 1 < x_2 < 4\},$$

then $\text{diam}(A) = \sqrt{13}$.

Theorem 7.22 (*Cantor's Intersection Theorem) Let (X, d) be a complete metric space and (A_n) a decreasing sequence (i.e. $A_1 \supset A_2 \supset A_3 \supset \ldots$) of non-empty subsets of X such that diam$(A_n) \to 0$ as $n \to \infty$. Then*

$$\cap\{\overline{A_n} : n \in \mathbf{N}\} \neq \emptyset.$$

Proof: For each $n \in \mathbf{N}$, choose $x_n \in A_n$ (this can be done, since each A_n is non-empty). Then (x_n) is a Cauchy sequence in X. To see this, let $\epsilon > 0$, and choose $N_\epsilon \in \mathbf{N}$ such that $\text{diam}(A_n) < \epsilon$ for each $n \geq N_\epsilon$. If $m \geq n \geq N_\epsilon$, x_m and x_n

are both in A_{N_ϵ} (since (A_n) is decreasing), and so $d(x_m, x_n) < \epsilon$. Since (X, d) is complete, (x_n) converges to some $x \in X$. For each $n \in \mathbf{N}$, $(x_{n+k-1})_{k=1}^{\infty}$ is a sequence in A_n, and since it is a subsequence of (x_n), $x_{n+k-1} \to x$ as $k \to \infty$. This shows that $x \in \overline{A_n}$; since this is true for each $n \in \mathbf{N}$, $x \in \cap\{\overline{A_n} : n \in \mathbf{N}\}$. ■

It can easily be shown that (in the notation of the proof) $\cap\{\overline{A_n} : n \in \mathbf{N}\} = \{x\}$. Check this!

The converse of this theorem is also true; we give some hints for the proof of this fact in the exercises.

We finish this chapter with a look at the way in which metric spaces fail to be complete. Our first example of an incomplete metric space was the set $\mathbf{Q}$ of rational numbers, with the metric induced by the usual metric of $\mathbf{R}$. Although there is no arguing about the fact that $\mathbf{Q}$ is not complete according to our definition of completeness, it is also true that it is a subset of the complete metric space $\mathbf{R}$. In fact, it is now clear that we could have manufactured as many examples of incomplete spaces as we wanted in a similar way. All we need is to start with a complete metric space X, take any subset A of X that is not closed in X, and give A the metric induced by the metric on X. It follows from Proposition 7.14 that A is not complete.

Are all incomplete metric spaces of this nature? More precisely, given any incomplete metric space X, can we always find a complete metric space Y which contains it?

If we look at our second example of an incomplete metric space (Example 7.9), it is not so clear that this will be the case. As pointed out there, it is not obvious what we could use as a complete metric space which contains $C([-1, 1])$. Even our first example (the rational numbers) is not as simple an example as it initially appears. We look at it from the perspective of people who fondly believe that they know what real numbers are, and we also know that every Cauchy sequence of real numbers converges. To get a proper perspective, you should try to put yourself in the shoes of somebody who knows the rational numbers, but has never heard of real numbers. If you were in this position, how would you describe what the limits of Cauchy sequences of rational numbers are? Can you give a definition of a real number using only rational numbers (no circular definitions allowed!)?

If you are feeling far less confident about real numbers, and more generally about complete metric spaces, you are in good company. The questions we raised above are hard ones. It was only in the nineteenth century that mathematicians succeeded in giving a rigorous definition of a real number.

The good news is that the suspicion we had that incomplete metric spaces are always subsets of complete metric spaces turns out to be correct, provided that we are prepared to interpret the phrase "are subsets" liberally. We have seen in the previous chapter that we could think of two metric spaces as being "equal as metric spaces" if they are isometrically isomorphic, i.e. if there is an isometry from the one onto the other. What can be shown is that any metric space is isometrically isomorphic to a subset of a complete metric space, and it is this sense in which an incomplete metric space "is a subset of" a complete metric space. There are various ways of showing this, and none of them are particularly easy. We are going to settle for giving you the outlines of two of these proofs, and to leave the checking of the details to the enthusiastic readers! We start of with a formal statement of the result.

Theorem 7.23 L*et* (X, d_X) *be a metric space. Then there is a complete metric space* (Y, d_Y) *such that* X *is isometrically isomorphic to a subset of* Y.

Proof: *First approach:* We start with the very natural idea that we should be able to construct a complete metric space containing X by considering the set Y of all "limits" of Cauchy sequences in X. Since we do not know what these limits are, this does not really get us very far. A rather desperate idea (or so it seems at first) is to try to use the Cauchy sequences themselves as the "points" of Y, i.e think of Y as the set of all Cauchy sequences in X. Since different Cauchy sequences may have the same "limit", we have to identify all Cauchy sequences with the same "limit". The problem, as before, is that we do not know what these limits are. To get round this, we use the fact that if two sequences converge to the same limit, there terms must eventually get close to each other. We formalise this by saying that the Cauchy sequences (x_n) and (y_n) in X are *equivalent* if $d(x_n, y_n) \to 0$ as $n \to \infty$. It can be checked that this defines an equivalence relation on the set of all Cauchy sequences in X. This leads to a refinement of the original idea: we now take the points in Y to be the equivalence classes of Cauchy sequences in X with respect to this equivalence relation. It is possible to define a distance between two such equivalence classes. To do this, we define the distance between the equivalence classes $[(x_n)]$ and $[(y_n)]$ to be $\lim_{n\to\infty} d(x_n, y_n)$. It can be checked that this defines a metric on Y, and hence we can now think of Y as a metric space. But we would like X to be isometrically isomorphic to a subset of Y. To see that this is indeed the case, we associate the element $x \in X$ with the equivalence class which contains the Cauchy sequence which has every term equal to x. This defines a function from X into Y. There is then quite a bit of checking to be done to show that this function is an isometry of X into Y. The final, and hardest, part of the proof is to show that Y with the metric we have defined above is in fact a *complete* metric space.

Second approach: The second proof is much slicker than the first, but not nearly as intuitive. It uses the completeness of the metric space $C_b(X)$ (Corollary 7.18), and therefore uses the fact that **R** with its usual metric is complete. We give an outline of the proof and ask you to check the details (see Exercise 12). Fix any $a \in X$. For each $x \in X$, define the function $f_x : X \to \mathbf{R}$ by

$$f_x(z) = d_X(z, x) - d_X(z, a), \text{ where } z \in X.$$

Then it can be checked that f_x is a bounded continuous function on X, i.e.$f_x \in C_b(X)$, for each $x \in X$. We now define the function F from X to $C_b(X)$ by $F(x) = f_x$. It is not too difficult to check that if $C_b(X)$ is equipped with the supremum metric it inherits from $B(X)$, then F is an isometry of X into $C_b(X)$. Let $Y = C_b(X)$ and let d_Y be the metric of $C_b(X)$. Then X is isometrically isomorphic to the subset $F(X)$ of Y, and by Corollary 7.18 (Y, d_Y) is a complete metric space. ■

The theorem above tells us that there is a sense in which we can "enlarge" an incomplete metric space to get a complete one. Is there only one such complete space? The following very simple example shows that this is definitely not the case in general.

Example 7.24 Let **R** have its usual metric and $(0, 1]$ and $[0, 1]$ the metrics induced by the metric of **R**. Then $(0, 1]$ is an incomplete metric space, and both $[0, 1]$ and **R** are complete metric spaces containing it.

The example above suggests that it may be a good idea to look for the "smallest" complete metric space "containing" an incomplete metric space. To make this idea precise, we need the following definition.

Definition 7.25 L*et (X, d) be a metric space and A a subset of X. We say that A is* ***dense*** *in X if $\overline{A} = X$.*

Proposition 7.26 L*et (X, d_X) be a metric space, (Y, d_Y) a complete metric space and $F : X \to Y$ be an isometry of X into Y. Then $\overline{F(X)}$ is a complete subset of Y, and $F(X)$ is dense in $\overline{F(X)}$.*

Proof: This follows at once from Proposition 7.13. ■

Definition 7.27 L*et (X, d_X) and (Y, d_Y) be metric spaces. We say that (Y, d_Y) is* ***a completion*** *of (X, d_X) if (Y, d_Y) is complete and X is isometrically isomorphic to a dense subset of Y.*

It follows from Theorem 7.23 and Proposition 7.26 that every metric space has a completion. Is there only one? The fact that we have given two very different looking proofs for the theorem suggests that this would be a bit too much to expect. The next result shows that if we are prepared to relax our idea of equality, there is only one completion. As before, we have to think of metric spaces that are isometrically isomorphic as "equal" metric spaces.

Proposition 7.28 L*et* (X, d) *be a metric space and* (Y, d_Y) *and* (Z, d_Z) *be two complete metric spaces. Let* $F : X \to Y$ *and* $G : X \to Z$ *be two isometries such that* $\overline{F(X)} = Y$ *and* $\overline{G(X)} = Z$. *Then there is an isometry* $H : Y \to Z$ *such that* $H(Y) = Z$ *and* $H(F(x)) = G(x)$.

Proof: The situation described in the statement of the proposition is shown in the sketch above. What is needed is a definition of the function H. The idea is to start

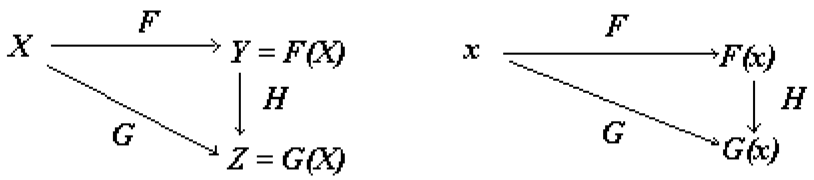


Figure 7.2: Constructing an isometry between complete metric spaces.

off by defining H on $F(X)$ only. If one thinks about it for a moment, there is only one way this can be done: If $F(x)$ is in $F(X)$, put $H(F(x)) = G(x)$. It is then straightforward to check that H is an isometry from $F(X)$ onto $G(X)$. The proof is then completed by extending the definition of H to all of $Y = \overline{F(X)}$, and to check that this gives an isometry from Y onto Z. The details are left as an exercise. Some hints are given in Exercise 13 at the end of the chapter. ■

One can interpret the proposition as saying that any two completions of a metric space are isometrically isomorphic (i.e. we may identify them as metric spaces). It is in this sense that the completion of a metric space is unique, and this allows us to talk about "**the** completion" of a metric space.

To summarise all of this rather loosely: For every metric space, there is a unique smallest complete metric space which contains it; this is known as the completion of the metric space.

Although the proof of Theorem 7.23 gives us two ways of constructing the completion of a metric space, it is not much use when it comes to trying to form an idea of what the completion looks like. To get from a metric space X to its completion, we have to "enlarge" X to obtain a complete space. Theorem 7.23 (and Proposition 7.26) tells us it can always be done, but the descriptions it gives of the points in the "enlargement" (equivalence classes of Cauchy sequences in X, or bounded continuous functions on X) are not particularly helpful if one wants to work with the elements in the completion. When working with incomplete spaces the challenge is to find useful descriptions of the elements in their completion. The example of $C([-1,1])$ equipped with the integral metric (Example 7.9) is a good case in point. It is possible to describe the elements in the completion of this space as functions, but to do this a new kind of integration (Lebesgue integration) has to be introduced. (See also the comments following Example 6.4.)

You may have wondered when we are going to try to characterise the concepts discussed in this chapter in terms of open sets. The answer is simple: we are not going to do it, because it cannot be done. The notion of a Cauchy sequence and that of completeness cannot be generalised to topological spaces. (The notion of completeness can be extended to spaces more general than metric spaces but less general than topological spaces; these spaces are known as uniform spaces.)

Summary.

The notion of a Cauchy sequence in a metric space was introduced. In any metric space, convergent sequences are Cauchy, but the converse is only true in complete metric spaces. Examples of complete and incomplete spaces were given. Complete subsets of a metric space were defined and related to closed sets. Diameters of sets were introduced and the Cantor intersection theorem was proved. The idea of a completion of an (incomplete) metric space was introduced, and the sense in which every metric space has an essentially unique completion was explored.

Historical notes

Augustin Louis Cauchy (1789–1857)

Cauchy was born in Paris at the time of the French Revolution. When he was four, conditions in Paris became so dangerous that the family moved to Arcuiel. There conditions were so hard that they soon returned to Paris. Cauchy's father took an active interest in his son's education, and sought advice from the mathematicians Laplace and Lagrange. Cauchy's formal education started with two years devoted to classical languages, after which he studied mathematics at the Ecole Polytechnique and engineering at the Ecole des Ponts et Chaussèes. After graduation he took on a job at Cherbourg, where he worked on harbour facilities for Napoleon's fleet.

Although he worked extremely hard, he made time for mathematical research and published his first paper.

In September 1812 Cauchy became ill and returned to Paris. While recuperating he continued his research and published a further paper. He did not want to return to Cherbourg, and managed to obtain a posting to the Ourcq Canal project, where he worked as a student earlier. Cauchy wanted an academic position, but failed in a number of applications. He did manage to obtain further sick leave, and a stoppage on the canal project caused by political events gave him two years in which to do research. One of the important achievements of this time was a memoir on integration which became the basis of his theory of complex functions.

In 1815 he did manage to secure a position at the Ecole Polytechnique, and the next year was awarded a prize by the French Academy of Sciences for a paper on waves. This was followed by a solution to a problem posed by Fermat, and election to the French Academy of Sciences.

In 1817 he obtained a post at the Collège de France. His textbook *Cours d'analyse* of 1821 attempted to establish a rigorous foundation for the study of the calculus. This included a rigorous definition of the notion of an integral, and a rigorous treatment of convergence of infinite series.

Cauchy's relations with his fellow scientists were not particularly good. He was a staunch Catholic and tended to bring religion into his work. He supported the Jesuits in their attack on the Academy of Sciences. His manner was often abrupt and very critical, and the young mathematicians Abel and Galois amongst others were treated badly by him.

After the revolution of 1830, Cauchy decided to leave Paris and he spent some time in Switzerland. When he returned to Paris, he was asked to swear an oath of allegiance to the new government. When he refused to do this, he lost all his positions. He went to Turin, and was appointed professor of theoretical physics there in 1832. The next year he went from there to Prague to tutor the grandson of Charles X, but Cauchy's quick temper and the prince's lack of interest meant that this was not a particularly successful venture. Cauchy returned to Paris in 1838, but could not teach because he still refused to take the oath of allegiance. This even meant that he could not take up a new position to which he was appointed. His religuous and political views also resulted in him not being appointed to a professorship at the Collège de France, even though he was the best candidate. He continued to do research in mathematical physics, astronomy and differential equations.

Political changes in France in 1848 led to Cauchy regaining his old appointments. When he applied for the chair at the Collège de France in 1850, he narrowly lost out to Liouville, and this soured relations between them. Another dispute (in which Cauchy was proved to be wrong) led to a great deal of bitterness in the last years of Cauchy's life. He died in 1857. His name lives on in many terms in mathematics, like Cauchy sequences, the Cauchy-Schwarz inequality, the Cauchy-Riemann equations and the Cauchy integral formula. He contributed to all the then-known areas of mathematics, and his contributions show an amazing creativity and insight.

Exercises

1. Show that in any metric space an eventually constant sequence is a Cauchy sequence. Are there any metric spaces in which the converse is true?

2. Prove that a finite subset of a metric space is complete.

3. Let $C([-1,1])$ have its usual supremum metric d, and let (f_n) be the sequence defined in Example 7.9. Is (f_n) a Cauchy sequence in (X,d)? Does it converge in $(C([0,1]),d)$?

4. Let $\mathbf{R}$ have its usual metric. Which of the following subsets of $\mathbf{R}$ are complete?

   (a) $[3,\infty)$;

   (b) $\{(-1)^n + \frac{1}{n} : n \in \mathbf{N}\}$;

   (c) the set of irrational numbers.

5. Let $C([-1,1])$ have its usual supremum metric.

   (a) Let p_n be the Taylor polynomial of degree n of the function $f(x) = e^x$ about $x = 0$. Does $p_n \to f$ as $n \to \infty$ in $C([-1,1])$? [Hint: Use the estimate for $|f(x) - p_n(x)|$ given by the remainder term in Taylor's Theorem.]

   (b) Is the set of all polynomial functions on $[-1,1]$ a complete subset of $C([-1,1])$?

6. Let (x_n) be a Cauchy sequence in a metric space. Show that the set $\{x_n : n \in \mathbf{N}\}$ has a finite diameter.

7. Let (x_n) be a sequence in a metric space (X, d) and for each $n \in \mathbf{N}$, put $A_n = \{x_k : k \geq n\}$. Show that (x_n) is a Cauchy sequence in X if and only if $\mathrm{diam}(A_n) \to 0$ as $n \to \infty$.

8. Prove that if A is a subset of a metric space, then $\mathrm{diam}A = \mathrm{diam}(\overline{A})$.

9. Formulate the converse of Cantor's Intersection Theorem, and then prove it. [Hint: Use the previous two questions.]

10. Let (X, d_X) be a metric space which is metrically isomorphic to the complete metric space (Y, d_Y). Show that (X, d_X) is complete.

11. Let (X, d) be a metric space and C the set of all Cauchy sequences in X. For $(x_n) \in C, (y_n) \in C$, define $(x_n) \sim (y_n)$ iff $d(x_n, y_n) \to 0$ as $n \to \infty$.

    (a) Show that $\sim$ is an equivalence relation on C.

    (b) Let Y be the set of all equivalence classes of elements of C. For $[(x_n)]$, $[(y_n)] \in Y$, put $d_Y([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n)$. Show that d_Y is a metric on Y.(Remember to check that d_Y is well defined, i.e. that $d_Y([(x_n)], [(y_n)])$ does not depend on the choice of representatives (x_n) and (y_n) from the equivalence classes $[(x_n)]$ and $[(y_n)]$.)

    (c) Show that the function $F : X \to Y$ defined by $F(x) = [(x_n)]$, with $x_n = x$ for all n, is an isometry.

12. Let (X, d_X) be a metric space and $C_b(X)$ be the set of all bounded continuous real-valued functions on X. Define a metric on $C_b(X)$ by

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in X\}.$$

    (a) Check that d is a metric on $C_b(X)$.

    (b) Fix a point $a \in X$. For each $x \in X$, define the function $f_x : X \to \mathbf{R}$ by

$$f_x(z) = d_X(z, x) - d_X(z, a).$$

    Show that $f_x \in C_b(X)$ for every $x \in X$.

    (c) Prove that for all $x, y \in X$, $\quad |f_x(z) - f_y(z)| \leq d_X(y, x)$ for all $z \in X$, and that $f_x(y) - f_y(y) = d_X(y, x)$ for $x, y \in X$.

(d) Deduce that the function $F : X \to C_b(X)$ defined by $F(x) = f_x$ is an isometry.

13. Prove Proposition 7.28. Note that the aim of the proof is to define an isometry $H : Y \to Z$, and that for $y = F(x) \in Y$, we must have $H(y) = H(F(x)) = G(x)$.

    (a) Explain why for any $y \in Y$, we have $y = \lim_{n\to\infty} F(x_n)$, for some sequence (x_n) in X.

    (b) Put $H(y) = \lim_{n\to\infty} G(x_n)$, with (x_n) as in (a). Explain why this limit exists, and also why $H(y)$ does not depend on the choice of the sequence (x_n), as long as $F(x_n) \to y$.

    (c) Prove that H maps Y onto Z.

    (d) Prove that H is an isometry.

Chapter 8

Doing it over and over again gets us there.

In which we encounter the contraction mapping theorem, and manage to solve some of the problems raised in Chapter 1.

It is time to return to the problems we have encountered in the first chapter. We now know enough about metric spaces to solve some of them. In fact, we'll see that there is quite a general problem that we can formulate in the setting of metric spaces, and that a solution to this general problem will also give us a solution to the first three problems raised in the first chapter.

The problems we want to consider are all of the following nature: Given a set X and a function $f : X \to X$, we need to find an $x \in X$ such that $f(x) = x$. It will be convenient to introduce some terminology for such a point:

Definition 8.1 L*et X be a set and $f : X \to X$ a function. A point $x_0 \in X$ is called a* **fixed point** *of f if $f(x_0) = x_0$.*

Example 8.2 Let $X = \mathbf{R}$ and $f(x) = x^3$. A fixed point of f is a real number x such that $x^3 = x$. We can write this equation as $x(x-1)(x+1) = 0$. It follows that $-1, 0$ and 1 are fixed points of f, and these are the only fixed points of f. The function $g(x) = e^x$ has no fixed points, since the fact that $e^x > x$ for all real numbers x shows that the equation $e^x = x$ has no solution. It is easy to see that $x = 0$ is a fixed point of $h(x) = \sin x$, and the fact that $x > \sin x$ for $x > 0$ shows that this is the only fixed point.

Example 8.3 Let $X = C([-\frac{1}{2}, \frac{1}{2}])$ and define $F : C([-\frac{1}{2}, \frac{1}{2}]) \to C([-\frac{1}{2}, \frac{1}{2}])$ by

$$F(f) = g, \text{ where } g(x) = 1 + \int_0^x f(t)\,dt \quad \text{for every } x \in [-\frac{1}{2}, \frac{1}{2}].$$

Then it can be checked (by substitution) that $f(x) = e^x$ is a fixed point of F. (Do this!)

In this chapter we want to look at three questions:

1. When does a function have a fixed point? (This is an example of an *existence* question.)
2. If there is a fixed point, is there only one? (This is called a *uniqueness* question.)
3. If there is a fixed point, how can it be found? (This is a question about an *algorithm* for finding fixed points.)

The method we tried in Chapter 1 could be described as a "guess and check" method. We start off by guessing a fixed point, say x_1. To check the guess, we calculate $f(x_1)$. If $x_1 = f(x_1)$, x_1 is a fixed point. If not, we use $x_2 = f(x_1)$ as our second guess. To check whether it is a fixed point, we calculate $f(x_2)$; if $x_2 = f(x_2)$, x_2 is a fixed point. If not, we use $x_3 = f(x_2)$ as our third guess. We can continue in this way until we find a fixed point. In practice, however, it is only very rarely that we are fortunate enough to find a fixed point after a finite number of steps. This procedure is more likely to produce a sequence (x_n) (where for each $n \geq 1$, $x_{n+1} = f(x_n)$) of what we hope will be approximations to a fixed point. The important practical question is: How close do these approximations get to the fixed point? Does the sequence (x_n) converge to a fixed point? To answer these questions, we need a notion of distance on X; it makes sense therefore to assume that X is a metric space.

What should the function f be like to guarantee that the sequence (x_n) converges? If (x_n) does converge, it will be a Cauchy sequence, and so the terms in the sequence will eventually bunch together. Since each term in the sequence (after the first one) is obtained by applying f to the previous term, we would like f to have the effect of reducing the distance between points. In particular, we would like x_3 and x_2 to be closer together than x_2 and x_1. If d is the metric on X, this means that we would like to have

$$d(x_3, x_2) < d(x_2, x_1) \quad \text{or} \quad d(f(x_2), f(x_1)) < d(x_1, x_2).$$

or more generally,

$$d(f(x_{n+1}), f(x_n)) < d(x_{n+1}, x_n) \quad \text{for all } n \in \mathbf{N}.$$

To play safe we could require the function f to be *distance-reducing* in the sense that

$$d(f(x), f(y)) < d(x, y) \quad \text{for all } x, y \in X, x \neq y.$$

Does a function like this always have a fixed point?

Example 8.4 Let $X = [1, \infty)$ and d be the usual metric of $\mathbf{R}$ restricted to X. Let $f(x) = x + \frac{1}{x}$. Then f is just the function f_1 considered in Example 6.14, and we saw there that f is distance reducing. But the equation $x = x + \frac{1}{x}$ has no solution (since we cannot have $\frac{1}{x} = 0$) , so f has no fixed point.

Example 8.5 Let us modify f slightly: put $g(x) = \frac{1}{2}(x + \frac{1}{x})$ (i.e. g is the function f_k of Example 6.13, with $k = \frac{1}{2}$). We showed in that example that this function is a contraction (see Definition 6.10). In this case the equation $x = g(x) = \frac{1}{2}(x + \frac{1}{x})$ has the solution $x = 1$ in X (check this!), so g has a fixed point. It follows from the same example that we can replace the $\frac{1}{2}$ in the definition of g by any $k \in (0, 1)$, and still have a contraction. It is also easy to check that for each such k we get one fixed point in $X = [1, \infty)$.

These examples suggest that it may be more profitable to concentrate on contractions, rather than distance-reducing functions. If we do this, we are able to prove one theorem which answers all three of the questions we raised at the beginning. We'll return to distance-reducing functions again later, and see that all is not lost even for such functions.

The next theorem is known as the **Contraction Mapping Theorem** or **Banach Fixed Point theorem**. The only surprise in the conditions stated in the theorem may be the fact that we insist on the metric space being complete. The reason for this is that it is not too difficult to check that the sequence (x_n) of approximations is a Cauchy sequence; completeness then guarantees that it converges.

Theorem 8.6 L*et (X, d) be a complete metric space and $f : X \to X$ a contraction. Then f has a unique fixed point in X, and for any $x_1 \in X$, the sequence (x_n) defined inductively by*

$$x_{n+1} = f(x_n), \quad n \geq 1$$

converges to this fixed point.

Proof: Since f is a contraction, we can find a $k \in (0, 1)$ such that

$$d(f(x), f(y)) \leq kd(x, y) \quad \text{for all } x, y \in X.$$

Let $x_1 \in X$. The first step of the proof is to show that the sequence (x_n) defined by $x_{n+1} = f(x_n)$ is a Cauchy sequence. To do this, we first estimate the distance between successive terms of this sequence. Let $n \in \mathbf{N}$, then

$$\begin{aligned} d(x_{n+1}, x_n) &= d(f(x_n), f(x_{n-1})) \\ &\leq kd(x_n, x_{n-1}) \\ &= kd(f(x_{n-1}), f(x_{n-2})) \\ &\leq k^2 d(x_{n-1}, x_{n-2}) \\ &\vdots \\ &\leq k^{n-1} d(x_2, x_1). \end{aligned}$$

Next we estimate the distance between any two terms of the sequence. If $m > n$,

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{n+1}) + d(x_{n+1}, x_n) \\ &\leq d(x_m, x_{n+2}) + d(x_{n+2}, x_{n+1}) + d(x_{n+1}, x_n) \\ &\vdots \\ &\leq d(x_m, x_{m-1}) + ... + d(x_{n+1}, x_n) \\ &\leq k^{m-2} d(x_2, x_1) + ... + k^n d(x_2, x_1) + k^{n-1} d(x_2, x_1) \\ &= k^{n-1}(k^{m-n-1} + ... + 1) d(x_2, x_1) \\ &\leq k^{n-1}(1 + k + k^2 + ...) d(x_2, x_1) \\ &= \tfrac{k^{n-1}}{1-k} d(x_1, x_2) \qquad (1) \end{aligned}$$

Since $0 < k < 1$, $\frac{k^{n-1}}{1-k} d(x_1, x_2) \to 0$ as $n \to \infty$. It follows that if $\epsilon > 0$ is given, we can find an $N_\epsilon \in \mathbf{N}$ such that $\frac{k^{n-1}}{1-k} d(x_1, x_2) < \epsilon$ for $n \geq N_\epsilon$. Then we have $d(x_m, x_n) < \epsilon$ for $m > n \geq N_\epsilon$. Hence (x_n) is a Cauchy sequence in X.

Since X is complete, (x_n) converges, say to $x \in X$. Now f is a contraction and hence continuous (Proposition 6.15), so $f(x_n) \to f(x)$ as $n \to \infty$. Since $x_{n+1} = f(x_n)$, we have $x_{n+1} \to f(x)$, and also $x_{n+1} \to x$ (since (x_{n+1}) is a subsequence of (x_n)). Since limits are unique, we must have $x = f(x)$, i.e. x is a fixed point of f.

To show that x is the only fixed point, assume that y is another fixed point. Then

$$d(x, y) = d(f(x), f(y)) \leq kd(x, y).$$

If $d(x, y) \neq 0$, this gives $k \geq 1$, a contradiction. Hence we must have $d(x, y) = 0$, and so $x = y$. ∎

Warnings:

1. It is important to realise that **three** conditions must be checked before applying this theorem:
    - (X, d) must be a complete metric space;

- f must map X into X;
- f must be a contraction on X.

The second condition is easily overlooked, and if this is done, it could lead to an incorrect conclusion about the existence of a unique fixed point.

2. The theorem gives **sufficient** conditions for a unique fixed point to exist (i.e. if the conditions are satisfied, the function has a unique fixed point). These conditions are not **necessary**, i.e. fixed points may exist even if the conditions are not satisfied. The theorem can therefore be used to show that a function has a unique fixed point, but it **cannot** be used to show that a function does not have a fixed point.

In practice we will have to be satisfied with an approximation in the form of one of the x_n's; we can after all, even with a computer, only calculate a finite number of terms. It is therefore important to know how close any particular x_n is to the fixed point x. At first one may think this is impossible without knowing x, but the proof of the theorem gives us a way of estimating how close x_n is to x.

Corollary 8.7 *With the notation of Theorem 8.6, we have for all $n \geq 3$,*

$$d(x, x_n) \leq \frac{k^{n-1}}{1-k} d(x_1, x_2) \qquad \textbf{\textit{(Prior estimate)}}$$

and

$$d(x, x_n) \leq \frac{k}{1-k} d(x_{n-1}, x_n) \qquad \textbf{\textit{(Posterior estimate)}}$$

Proof: The inequality (1) in the proof of the theorem gives

$$d(x_m, x_n) \leq \frac{k^{n-1}}{1-k} d(x_2, x_1) \quad \text{for all } m > n.$$

Since $x_m \to x$ as $m \to \infty$ and d is continuous, this gives

$$d(x, x_n) = \lim_{m \to \infty} d(x_m, x_n) \leq \frac{k^{n-1}}{1-k} d(x_2, x_1) \qquad (2)$$

To obtain the second estimate, fix n and let $y_m = x_{n-2+m}$. Then $y_{m+1} = f(y_m)$ for each $m \geq 1$ and $y_m \to x$ as $m \to \infty$, so we can apply the estimate (2) to the sequence (y_m) to get

$$d(x, y_2) \leq \frac{k}{1-k} d(y_1, y_2)$$

that is

$$d(x, x_n) \leq \frac{k}{1-k} d(x_{n-1}, x_n).$$

∎

Now that we have the basic result which enables us to find fixed points, we can start applying it to specific cases. As a first application, we look at the problem of finding fixed points for functions $f : I \to I$, where I is an interval on the real line. To do this, we'll need to know what a contraction looks like in this case. It will perhaps not come as a surprise that whether a function is a contraction or not in this situation depends on the rate of growth of the function.

Proposition 8.8 *Let I be an interval in $\mathbf{R}$ and $f : I \to I$ a differentiable function. Then f is Lipschitz continuous (with respect to the metric induced by the usual metric of $\mathbf{R}$) if and only if there is a constant $k > 0$ such that $|f'(x)| \leq k$ for all $x \in I$. In this case k is a Lipschitz constant for f.*

Proof: Suppose f is Lipschitz continuous, with Lipschitz constant $k > 0$. Then if $x \in I$ and h is such that $x + h \in I$, $|f(x+h) - f(x)| \leq k|x + h - x| = k|h|$. Hence $|\frac{1}{h}(f(x+h) - f(x))| \leq k$, and so

$$\begin{aligned} |f'(x)| &= |\lim_{h\to 0} \tfrac{f(x+h)-f(x)}{h}| \\ &= \lim_{h\to 0} |\tfrac{f(x+h)-f(x)}{h}| \\ &\leq k, \end{aligned}$$

and this holds for all $x \in I$.
Conversely, suppose there is a $k > 0$ such that $|f'(z)| \leq k$ for all $z \in I$. If $x, y \in I$ and $x > y$, it follows from the Mean Value Theorem that there is a $c \in (y, x)$ such that

$$f(x) - f(y) = f'(c)(x - y)$$

and so

$$|f(x) - f(y)| = |f'(c)||x - y| \leq k|x - y|,$$

and this shows that f is Lipschitz continuous with Lipschitz constant k. ∎

Corollary 8.9 *A differentiable function $f : I \to I$ is a contraction if and only if there is a constant $k \in (0, 1)$ such that $|f'(x)| \leq k$ for all $x \in I$.*

Example 8.10 Let $\mathbf{R}$ have its usual metric and $f(x) = e^x - 2$. Then $f'(x) = e^x$, and therefore f is a contraction on, for example, the interval $(-\infty, -\frac{1}{2}]$ (since $e^x \leq e^{-\frac{1}{2}} < 1$ for all $x \leq -\frac{1}{2}$). But f is not a contraction on $[-1, 1]$, for $e^x \geq 1$ for $x \geq 0$.

If $g(x) = \cos x, g'(x) = -\sin x$, so $|g'(x)| = |\sin x|$. Hence g is a contraction on $[0, \frac{\pi}{3}]$ (since $|\sin x| \le \sin(\frac{\pi}{3}) < 1$ for $x \in [0, \frac{\pi}{3}]$), but not on $[0, \frac{\pi}{2}]$ (since $\sin(\frac{\pi}{2}) = 1$). It follows in the same way that $h(x) = \sin x$ is not a contraction on $\mathbf{R}$.

*Note that to show that a differentiable function $f : I \to I$ is **not** a contraction on I, it is enough to find one $x \in I$ such that $|f'(x)| \ge 1$.*

We are now ready for our first application of the contraction mapping theorem.

Example 8.11 (See Example 1.1) Let $X = [0, \frac{\pi}{3}]$ have the metric induced by the usual metric of $\mathbf{R}$ and $g(x) = \cos x$. Then g maps X into X (since $0 \le \cos x \le 1 < \frac{\pi}{3}$ for $0 \le x \le \frac{\pi}{3}$), and it follows from the previous example that g is a contraction on X. Since X is a closed subset of the complete metric space $\mathbf{R}$, it is complete. It therefore follows from the contraction mapping theorem that g has a unique fixed point in $x_0 \in X$, and that for any $x_1 \in [0, \frac{\pi}{3}]$, the sequence (x_n) defined inductively by $x_{n+1} = \cos x_n$ converges to this fixed point.

We can use Corollary 8.7 to get an estimate for the error we make when we use x_n to approximate the fixed point. If we start with $x_1 = 0$, we get $x_2 = \cos x_1 = \cos 0 = 1$. In this case we can use $k = \sup\{\sin x : x \in [0, \frac{\pi}{3}]\} = \frac{\sqrt{3}}{2}$. The prior estimate then gives

$$|x_0 - x_n| \le \frac{(\frac{\sqrt{3}}{2})^{n-1}}{1 - \frac{\sqrt{3}}{2}}|x_2 - x_1| = 7.4641(0.8660)^{n-1}$$

This allows us to estimate the error we'll make if we use x_n to approximate x_0. We can use this estimate as soon as we know x_1 and x_2. For $n = 20$, this gives $|x_0 - x_{20}| \le 0.48508$. This kind of estimate is useful when one wants to know how many steps will be needed to achieve a certain accuracy. These are usually quite conservative estimates (i.e. the error is over-estimated). For the posterior estimate one needs to know the last two approximations in order to estimate how close the last one is to the fixed point. If we already know that $x_{19} = 0.7388$ and $x_{20} = 0.7393$, then the posterior estimate gives

$$|x_0 - x_{20}| \le \frac{\frac{\sqrt{3}}{2}}{1 - \frac{\sqrt{3}}{2}}|x_{20} - x_{19}| = 0.0032.$$

Both estimates are over-estimates of the error, but the prior estimate is clearly far more conservative.

Example 8.12 Let $X = (-\infty, -\frac{1}{2}]$, with the usual metric, and $f(x) = e^x - 2$. Then X is complete, f maps X into X and is a contraction on X (see Example 8.10).

Hence it follows from the contraction mapping theorem that f has a unique fixed point x_0 in X. If we start with $x_1 = -1$, and use $k = e^{-\frac{1}{2}}$, the prior error estimate gives

$$|x_0 - x_7| \leq \frac{k^6}{1-k}|x_2 - x_1| = 0.08,$$

and the posterior error estimate gives

$$|x_0 - x_7| \leq \frac{k}{1-k}|x_7 - x_6| = 0.000196.$$

As in the previous example the latter is much smaller than the former.

Example 8.13 We have seen that $h(x) = \sin x$ is not a contraction on $\mathbf{R}$. However, $x = 0$ is a fixed point of h on $\mathbf{R}$ (since $\sin 0 = 0$). This shows that a function may have a fixed point even if the conditions of the contraction mapping theorem are not all satisfied.

In Example 1.2 in Chapter 1 we saw that it is possible to rewrite some differential equations as integral equations, and that trying a "guess and check" method in this case produced a sequence of functions. We are now able to check whether this sequence converges, and if the limit is a solution to the differential equation. This involves applying the contraction mapping theorem to metric spaces of continuous functions. We start with an example for which we have already found a fixed point (Example 8.3) and try to reconcile this with the result obtained by using the contraction mapping theorem.

Example 8.14 Let $X = C([-\frac{1}{2}, \frac{1}{2}])$ have its usual supremum metric d and define $F : C([-\frac{1}{2}, \frac{1}{2}]) \to C([-\frac{1}{2}, \frac{1}{2}])$ by

$$F(f) = g, \text{ where } g(x) = 1 + \int_0^x f(t)\,dt \quad \text{for every } x \in [-\frac{1}{2}, \frac{1}{2}].$$

Then X is a complete metric space and since for every continuous f, g is also continuous (why?), F maps X into X. To see that F is a contraction, let $F(f_1) = g_1, F(f_2) = g_2$. Then for every $x \in [0, \frac{1}{2}]$,

$$\begin{aligned} |g_1(x) - g_2(x)| &= |\textstyle\int_0^x f_1(t)\,dt - \int_0^x f_2(t)\,dt| \\ &\leq \textstyle\int_0^x |f_1(t) - f_2(t)|\,dt \\ &\leq \sup\{|f_1(t) - f_2(t)| : t \in [0, \frac{1}{2}]\} \textstyle\int_0^x \,dt \\ &\leq d(f_1, f_2)x. \end{aligned}$$

Similarly, if $x \in [-\frac{1}{2}, 0]$ it follows that

$$|g_1(x) - g_2(x)| \leq \sup\{|f_1(t) - f_2(t)| : t \in [-\frac{1}{2}, 0]\} \int_{-x}^{0} dt \leq d(f_1, f_2)|x|.$$

Putting these two inequalities together we get

$$\begin{aligned} d(F(f_1), F(f_2)) &= \sup\{|g_1(x) - g_2(x)| : x \in [-\tfrac{1}{2}, \tfrac{1}{2}]\} \\ &\leq d(f_1, f_2) \sup\{|x| : x \in [-\tfrac{1}{2}, \tfrac{1}{2}]\} \\ &= \tfrac{1}{2} d(f_1, f_2). \end{aligned}$$

It follows from the contraction mapping theorem that F has a unique fixed point f in $C([-\frac{1}{2}, \frac{1}{2}])$. If we start with the function f_1 defined by $f_1(x) = 1$ for all $x \in [-\frac{1}{2}, \frac{1}{2}]$, the sequence (f_n), with $f_{n+1} = F(f_n)$, converges to this fixed point f. If we do the calculations, we find that

$$f_2(x) = 1 + x, \quad f_3(x) = 1 + x + \frac{x^2}{2}, \quad f_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

This is not too surprising, since we have already seen that $f(x) = e^x$ is a fixed point for F (Example 8.3), and the functions f_n obtained are just the Taylor polynomials of $f(x) = e^x$ about $x = 0$. It can also be checked that the a function y of x will satisfy the integral equation

$$y(x) = 1 + \int_0^x y(t)\, dt$$

if and only if it satisfies the initial value problem

$$\frac{dy}{dx} = y, \qquad y = 1 \text{ when } x = 0.$$

It is easy to check that the solution of the initial value problem is $y = e^x$.

You can check that with $k = \frac{1}{2}$ the prior and posterior error estimates are respectively

$$|f(x) - f_n(x)| \leq (\frac{1}{2})^{n-1} \text{ for all } |x| \leq \frac{1}{2} \text{ and } |f(x) - f_n(x)| \leq \frac{(\frac{1}{2})^{n-1}}{(n-1)!} \text{ for all } |x| \leq \frac{1}{2}.$$

Example 8.15 We can use the same method to find a solution in $C([0, 1])$ for the initial value problem

$$\frac{dy}{dx} = xy + 1, \qquad y = 2 \text{ when } x = 0$$

(Example 1.2). We saw there that it is equivalent to solving the integral equation

$$y(x) = 2 + \int_0^x (ty(t) + 1)\,dt,$$

which in its turn is equivalent to finding a fixed point in $C([0,1])$ for the function $F : C([0,1]) \to C([0,1])$ defined by

$$F(f) = g, \text{ where } g(x) = 2 + \int_0^x (tf(t) + 1)\,dt \text{ for all } x \in [0,1].$$

If $C([0,1])$ has its usual supremum metric, it is a complete metric space, and we can check in the same way as in the previous example that F is a contraction on $C([0,1])$ with Lipschitz constant $k = \frac{1}{2}$. It follows from the contraction mapping theorem that F has a unique fixed point in $C([0,1])$, which is also the unique solution to the initial value problem. Note that the differential equation we started with is linear, but that if we try to solve it using the usual method, we end up with an integral that cannot be found in terms of elementary functions. The method we use here effectively gives a series solution.

In Example 8.5 we found a unique fixed point for the function $f(x) = \cos x$ in the interval $[0, \frac{\pi}{3}]$. A rough sketch of the graph of f on $\mathbf{R}$ should be enough to convince you that this looks like the only fixed point of f on $\mathbf{R}$. But we have also seen that f is not a contraction on $\mathbf{R}$. How do we reconcile all of this? As we warned before, the contraction mapping theorem gives *sufficient* conditions for the existence of a unique fixed point, but certainly does not claim that these are the only conditions under which we get fixed points (i.e the conditions are not *necessary*).

A simple observation that gives us a clue to improving the theorem itself is the following. If x is a fixed point of a function f, then it is also a fixed point of the composition $f \circ f$ of f with itself. For if $x = f(x)$, then $f(f(x)) = f(x) = x$. It is easy to see that the same will be true if we take the composition of f with itself any finite number of times. Let's introduce some useful terminology and notation which will help us to explore this fact further.

Definition 8.16 *Let X be a set and $f : X \to X$ be a function. The* **second iterate** *of f is the function $f^2 : X \to X$ defined by*

$$f^2(x) = f(f(x)), \quad x \in X.$$

More generally, for $n \in \mathbf{N}$, the **n-th iterate** *of f is defined inductively by*

$$f^1(x) = f(x), \qquad \textit{and for } n \geq 1, \quad f^{n+1}(x) = f(f^n(x)), \quad x \in X$$

Example 8.17 If $X = \mathbf{R}$ and $f(x) = x^2 + 1$, then

$$f^2(x) = (x^2+1)^2 + 1 = x^4 + 2x^2 + 2,\ f^3(x) = f(f^2(x)) = (x^4 + 2x^2 + 2)^2 + 1.$$

If $f(x) = \cos x$, $f^2(x) = \cos(\cos x)$. Note that this is **not** the same as $(f(x))^2 = (\cos x)^2 = \cos^2 x$.

We have seen above that if x is a fixed point of a function f, it is also a fixed point of every iterate f^n of f. Is the converse true? We'll see below that an even better result is true: If some iterate of f is a contraction (and hence has a fixed point), then f has a fixed point.

Theorem 8.18 L*et* (X, d) *be a complete metric space and* $f : X \to X$ *be a function such that* f^n *is a contraction for some* $n \in \mathbf{N}$. *Then* f *has a unique fixed point in* X, *and for any* $x_1 \in X$, *the sequence defined by* $x_{n+1} = f(x_n)$, $n \geq 1$, *converges to this fixed point.*

Proof: Since f^n is a contraction, it follows from Theorem 8.6 that f^n has a unique fixed point $x \in X$, i.e. $f^n(x) = x$. Put $y = f(x)$, then $f^n(y) = f^n(f(x)) = f(f^n(x)) = f(x) = y$, so y is also a fixed point of f^n. By uniqueness of the fixed point of f^n, we have $y = x$, i.e. $f(x) = x$, so x is in fact a fixed point of f as well. Suppose $z \in X$ is another fixed point of f, i.e. $f(z) = z$. Then

$$f^n(z) = f^{n-1}(f(z)) = f^{n-1}(z) = f^{n-2}(f(z)) = ... = f(z) = z,$$

so z is a fixed point of f^n. By uniqueness again, $z = x$.

It also follows from Theorem 8.6 that for any $u \in X$, the sequence

$$u, f^n(u), f^n(f^n(u)), ...$$

will converge to x. If we take $u = x_1$, this shows that the sequence

$$x_1, x_{n+1}, x_{2n+1}, \ldots$$

converges to x. Starting with $u = f(x_1) = x_2$, we find that the sequence

$$x_2, x_{n+2}, x_{2n+2}, \ldots$$

converges to x. Continuing in this way, we find that for $1 \leq k \leq n$, the sequence

$$x_k, x_{n+k}, x_{2n+k}, x_{3n+k}, \ldots$$

converges to x. From this we can deduce that the sequence (x_n) itself must converge to x. For if $\epsilon > 0$, then we may choose for $1 \leq k \leq n$ an $N_k \in \mathbf{N}$ such that if $m \geq N_k$, then $d(x, x_{mn+k}) < \epsilon$. Let $N = \max\{N_k n + k : 1 \leq k \leq n\}$. If $p \in \mathbf{N}$, then $p = mn + k$ for some $m \in \mathbf{N}$ and some k with $1 \leq k \leq n$; if $p \geq N$, we must have $m \geq N_k$. Hence $d(x, x_p) = d(x, x_{mn+k}) < \epsilon$. ■

Example 8.19 We are now in a position to explain why $f(x) = \cos x$ has a unique fixed point in $\mathbf{R}$, even though it is not a contraction on $\mathbf{R}$. Let $g(x) = f^2(x) = \cos(\cos x)$. Then for all $x \in \mathbf{R}$,

$$|g'(x)| = |\sin x||\sin(\cos x)| \leq |\sin x||\cos x| = \frac{1}{2}|2\sin x \cos x| = \frac{1}{2}|\sin(2x)| \leq \frac{1}{2},$$

since for any $y \in \mathbf{R}, |\sin y| \leq |y|$. This shows that f^2 is a contraction on $\mathbf{R}$, and hence $f(x) = \cos x$ has a unique fixed point in $\mathbf{R}$, by Theorem 8.18.

Theorem 8.18 also enables us to prove quite a general theorem about the existence of first order differential equations. This is done by first converting an initial value problem into an integral equation, and then showing that the associated map satisfies the conditions of Theorem 8.18.

Theorem 8.20 *Let $a < b, a, b \in \mathbf{R}$ and $S = \{(x, y) \in \mathbf{R}^2 : a \leq x \leq b, y \in \mathbf{R}\}$. Let $f : S \to \mathbf{R}$ be a continuous function and suppose there is an $L > 0$ such that*

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2| \text{ for all } x \in [a, b], y_1, y_2 \in \mathbf{R}.$$

Then for any $\alpha \in \mathbf{R}$, the initial value problem

$$\frac{dy}{dx} = f(x, y), \qquad y(a) = \alpha$$

has a unique solution.

Proof: It is important to realise that in the differential equation y is a function of x, and hence the right-hand side of the equation, $f(x, y) = f(x, y(x))$ is a function of x as well. It is easy to check (see Example 1.2) that a function y of x is a solution to the given initial value problem if and only if it satisfies the integral equation

$$y(x) = \alpha + \int_a^x f(t, y(t))\, dt.$$

If y is a continuous function of x on $[a, b]$ (i.e. $y \in C([a, b])$, let $F(y)$ be the continuous function on $[a, b]$ defined by

$$F(y)(x) = \alpha + \int_a^x f(t, y(t))\, dt \text{ for } x \in [a, b].$$

Let $y_1, y_2 \in C([a, b])$, and $x \in [a, b]$. Then

$$\begin{aligned} |F(y_1)(x) - F(y_2)(x)| &= |\textstyle\int_a^x [f(t, y_1(t)) - f(t, y_2(t))]\, dt| \\ &\leq \textstyle\int_a^x |f(t, y_1(t)) - f(t, y_2(t))|\, dt \\ &\leq L \textstyle\int_a^x |y_1(t) - y_2(t)|\, dt \\ &\leq L \sup\{|y_1(t) - y_2(t)| : t \in [a, b]\} \textstyle\int_a^x\, dt \\ &= L(x - a) d(y_1, y_2), \end{aligned}$$

where d is the usual supremum metric on $C([a,b])$. It follows that

$$d(F(y_1),F(y_2)) = \sup\{|F(y_1)(x) - F(y_2)(x)| : x \in [a,b]\} \le L(b-a)d(y_1,y_2).$$

Since we do not necessarily have $L(b-a) < 1$, F need not be a contraction. If we repeat the above with $F(y_1)$ and $F(y_2)$ in the place of y_1 and y_2, we get

$$\begin{aligned} |F^2(y_1)(x) - F^2(y_2)(x)| &= |F(F(y_1)) - F(F(y_2))| \\ &\le L \textstyle\int_a^x |F(y_1)(t) - F(y_2)(t)|\,dt \\ &\le L^2 d(y_1,y_2) \textstyle\int_a^x (t-a)\,dt \\ &= L^2 \tfrac{(x-a)^2}{2} d(y_1,y_2). \end{aligned}$$

This process can clearly be repeated. After n repetitions we get the inequality

$$|F^n(y_1)(x) - F^n(y_2)(x)| \le L^n \frac{(x-a)^n}{n!} d(y_1,y_2)$$

and so

$$d(F^n(y_1),F^n(y_2)) = \sup_{x\in[a,b]} |F^n(y_1)(x) - F^n(y_2)(x)| \le \frac{L^n(b-a)^n}{n!} d(y_1,y_2).$$

Since $\lim_{n\to\infty} \frac{[L(b-a)]^n}{n!} = 0$, there is an $m \in \mathbf{N}$ such that $\frac{[L(b-a)]^m}{m!} < 1$. It follows from this that F^m will then be a contraction, and hence we may use Theorem 8.18 to deduce that F has a unique fixed point $y \in C([a,b])$. This fixed point is also the unique solution to the initial value problem.

■

Corollary 8.21 *Let S be as in the Theorem and $f : S \to \mathbf{R}$ be a continuous function such that $\frac{\partial f}{\partial y} = f_y$ is a bounded function on S. Then the initial value problem*

$$\frac{dy}{dx} = f(x,y), \quad y(a) = \alpha$$

has a unique solution in $C([a,b])$.

Proof: Suppose, for some $L > 0$, $|f_y(x,y)| \le L$ for all $(x,y) \in S$. If we fix $x \in [a,b]$, and put $g(y) = f(x,y)$, then g is differentiable, and if $y_1, y_2 \in \mathbf{R}$, with $y_2 > y_1$, then it follows from the Mean Value Theorem that there is a $c \in (y_1,y_2)$ such that $g(y_2) - g(y_1) = g'(c)(y_2 - y_1)$. Hence

$$|f(x,y_1) - f(x,y_2)| = |g'(c)||y_1 - y_2| \le L|y_1 - y_2|.$$

The conditions of Theorem 8.20 are therefore satisfied and the result follows. ■

This corollary (or some variant of it) is generally known as **Picard's Theorem**.

There are many more applications of the Contraction Mapping Theorem; we've only looked at two of the more important ones in this chapter. You will find some more in the next chapter.

Summary.

The contraction mapping theorem for contractions on complete metric spaces was proved, and error estimates for approximations to fixed points obtained in this way given. Contractions on intervals in $\mathbf{R}$ were characterised in the case of differentiable functions. Iterates of functions were defined and the idea used to prove a generalisation of the contraction mapping theorem. This was used in turn to prove an existence theorem for solutions for first order ordinary differential equations.

Historical notes

Stefan Banach (1892–1945)

Banach was born in the city of Kraków in what is now Poland in 1892, and went to school there. The mathematician Steinhaus "discovered " him on a walk through a park. He overheard Banach discussing mathematics with a friend, immediately realised his potential, and made arrangements for him to go to university. Banach obtained his doctorate from the University of Lvov in 1920. In it he defines what are now known as a normed spaces axiomatically for the first time. Normed spaces which are complete when considered as metric spaces are now known as *Banach spaces* in his honour; this name was first used by Frechet.

Banach taught at the Institute of Technology in Lvov, and then at the University of Lvov, where he became a professor in 1927. He was a brilliant, eccentric and colourful mathematician who liked spending long times in cafés discussing mathematics. During the Second World War he lived under very difficult conditions in Nazi-occupied Lvov. He was planning to take up a professorship in Mathematics at the Jagiellonian University in Kraków after the war, but died of lung cancer in 1945.

Banach is regarded as one of the founders of modern functional analysis. He also made contributions to measure and integration theory and the study of orthogonal series. In 1932 he published a book on normed spaces, *Théorie des opérations linéares*, a classic still regarded today as one of the best works on the topic. His name lives on in many important theorems, such as the Banach fixed point theorem, the Hahn-Banach theorem and the Banach-Steinhaus theorem.

Charles Émile Picard (1856–1941)

Picard was born in Paris in 1856 and appointed lecturer at the University of Paris in 1878. In 1879 he took up a professorship at Tolouse, and in 1898 moved back to Paris as professor at the Sorbonne. He made important contributions in the fields of analysis and algebraic geometry. The basic existence theorem for solutions of first order ordinary differential equations was proved by him using the method of successive approximations. He used analysis in applications to the theories of elasticity, electricity and heat.
Picard was married to the daughter of the French mathematician Hermite. Their daughter and two sons were killed in the First World War, and his grandsons wounded in the Second World War. He died in Paris in 1941.

Exercises.

1. For each of the functions $f : \mathbf{C} \to \mathbf{C}$ given below, find its fixed points (if any).

   (a) $f(z) = \overline{z} + i$;

   (b) $f(z) = -z^3$.

2. In each case below, a function $f : \mathbf{R}^2 \to \mathbf{R}^2$ is defined. Find the fixed points of f (if any). Also determine whether f is a contraction on $\mathbf{R}^2$ with its usual Euclidean metric.

   (a) $f(\mathbf{x})$ is the projection of $\mathbf{x} = (x_1, x_2)$ onto the x_2-axis;

   (b) $f(\mathbf{x})$ is the reflection of $\mathbf{x}$ in the x_1-axis;

   (c) $f(\mathbf{x})$ is the point obtained by rotating $\mathbf{x}$ anti-clockwise about the origin through $\frac{\pi}{4}$ radians.

3. Describe all the contractions on a discrete metric space X. Do the same for the Lipschitz-continuous functions from X to X.

4. In each case below, an interval I in $\mathbf{R}$, with its usual metric, and a function $f : I \to I$ are given. Determine whether f is a contraction on I. [Hint: Use Corollary 8.9.]

   (a) $f(x) = \arctan(\frac{1}{2}x), I = \mathbf{R}$;

   (b) $f(x) = \arctan x, I = [\frac{1}{2}, \infty)$;

   (c) $f(x) = e^{-x^2}, I = [0, \infty)$;

   (d) $f(x) = \ln(1 + x^2), I = [0, \infty)$.

5. Let $f(x) = e^{\frac{1}{2}x} - 1$, and let $\mathbf{R}$ and intervals in $\mathbf{R}$ have the usual metric.

   (a) Is f a contraction on $\mathbf{R}$? And on $[-1, 1]$?

   (b) Does f map $\mathbf{R}$ into itself? And $[-1, 1]$?

   (c) How many fixed points does f have in $\mathbf{R}$? [Hint: Look at the graph in Example 1.1 of Chapter 1, and use some calculus to prove what this makes you suspect.] Can they all be found using the contraction mapping theorem?

6. Let $f : \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = \frac{1}{7}(x^3 + x^2 + 1)$ and $\mathbf{R}$ have its usual metric.

   (a) Is f a contraction on $\mathbf{R}$?

   (b) Show that f maps $[0, 1]$ into $[0, 1]$.

   (c) Is f a contraction on $[0, 1]$?

   (d) How many fixed points does f have in $[0, 1]$?

   (e) If you use $x_1 = 0$ as a starting point and use iteration to try and find a fixed point for f in $[0, 1]$, estimate the error in your approximation to the fixed point after n steps. [Do **not** try to find x_n!]

(f) How many fixed points does f have in $\mathbf{R}$? Is it possible to find them all using the contraction mapping theorem? Give reasons for your answers.

7. Can the contraction mapping theorem be used to find all the fixed points of $f(x) = 4x^2e^{-x^2}$ in $\mathbf{R}$? Look at Example 1.1 in Chapter 1 and try to explain what we found there.

8. For each of the following functions $f : \mathbf{R} \to \mathbf{R}$ find the second iterate f^2 and determine whether it is a contraction; $\mathbf{R}$ has its usual metric.

   (a) $f(x) = \sin x$;

   (b) $f(x) = e^{-x}$;

   (c) $f(x) = \arctan(\frac{1}{2}x)$.

9. Show that no iterate of the function $f(x) = \sin x$ is a contraction.

10. Consider the initial value problem

$$\frac{dy}{dx} = 2e^{-x} - y, \qquad y(0) = 2.$$

   (a) Show that the contraction mapping theorem can be used to show that this problem has a unique solution in $C([0, 1])$, and find the first three approximations to this solution starting with $y_1(x) = 1$ for all $x \in [0, 1]$.

   (b) Solve the equation exactly. [Hint: It is a linear differential equation.]

   (c) Compare your answers and explain what you find.

11. Use

   (a) Theorem 8.20

   (b) Corollary 8.21

   to show that the initial value problem

$$\frac{dy}{dx} = (y + x^2)e^{x-1}, \qquad y(1) = 4$$

   has a unique solution in $C([1, 3])$.

Chapter 9

Sets that are small enough for good things to happen on them.

In which we meet compact sets and discover some of their useful properties.

In the first chapter we saw there that one way of thinking about fractal sets is to regard them as being the limit, in some sense that we did not specify, of a sequence of sets. The sense in which we take the limit depends on the way we measure the distance between sets. This suggests that it may be useful to have a notion of the distance between two subsets of $\mathbf{R}^2$, at least. So far we have not attempted to introduce anything like this. In this chapter we return to the (more general) problem of defining the distance between two subsets of a metric space, and in the process we introduce a very useful class of sets.

In Chapter 4 we have looked at something more special: the distance $d(a, A)$ between a *point* a and a set A in a metric space (X, d):

$$d(a, A) = \inf\{d(a, x) : x \in A\}.$$

There are simple examples to show that this infimum need not be attained: there need not be a point $x \in A$ such that $d(a, A) = d(a, x)$. Are there sets for which the infimum will be attained? Again, it is easy enough to find examples where this does happen. Is it perhaps possible to find a property of a set that will ensure that it will happen? Something that may help us here is the fact that the function $f_a : X \to \mathbf{R}$ defined by $f_a(x) = d(a, x)$ is continuous (Proposition 6.8). So we may ask: What property must a set have to ensure that a continuous function attains its infimum on such a set? This question may well ring a bell: an important theorem in Real Analysis says that a continuous function from $\mathbf{R}$ to $\mathbf{R}$ attains its supremum and infimum on every closed and bounded interval in $\mathbf{R}$. We have already introduced the notion of a closed set in a metric space. Can we do the same for boundedness?

A subset of $\mathbf{R}$ is bounded if and only if it is bounded below and above. It is easy to see that this is equivalent to saying that the set is contained in an interval with midpoint halfway between the lower and upper bound. This gives us a way of generalising the concept to arbitrary metric spaces.

Definition 9.1 A *subset of a metric space is* ***bounded*** *if it is contained in a closed ball in the metric space.*

It follows from what we said above that a subset of $\mathbf{R}$, equipped with its usual metric, will be bounded in the sense of this definition if and only if it is bounded in the usual sense. It also follows easily that we could just as well have used "open ball" instead of "closed ball" in the definition.

Example 9.2 A subset of $(\mathbf{R}^2, d_E)$ is bounded if and only if it is contained in a square. This is the case because every ball in this metric space is contained in a square, and every square is contained in a ball.

The notion of the diameter of a set introduced in a previous chapter enables us to give another characterisation of boundedness.

Proposition 9.3 A *subset of a metric space is bounded if and only if its diameter is finite.*

Proof: If a set is bounded, it is contained in a closed ball, and the diameter of the set is therefore less than or equal to the diameter of the ball, which is less than or equal to twice its radius. Conversely, if a set has a finite diameter k, say, choose any element a in the set. Then it can easily be checked that the set is contained in the ball $B[a; k]$. ∎

Now that we can talk about closed and bounded sets in a general metric space, it is tempting to try and prove that every continuous real valued function on a metric space attains its infimum and supremum on every closed and bounded subset of the metric space. A closer look at the proof of the fact that a continuous real-valued function on $\mathbf{R}$ attains its supremum and infimum on every closed and bounded interval in $\mathbf{R}$ shows that it uses the Bolzano-Weierstrass theorem: Every bounded sequence of real numbers has a convergent subsequence. It turns out that it is this kind of property that makes the same proof work in general metric spaces as well. This motivates the following definition:

Definition 9.4 A *subset A of a metric space (X, d) is called* **sequentially compact** *if every sequence in A has a subsequence which converges* to a point of A.

The word "compact" conveys the sense that in some sense we are talking about "small" sets. Although these sets are not necessarily small in any conventional sense, they are still "small" enough to have many good properties.

There are in fact a number of different types of compactness that can be defined in a metric space. The one we've chosen is *sequential* compactness; this seems appropriate in the light of our policy to first define notions in terms of convergence of sequences. In metric spaces this notion coincides with another notion, referred to as **compactness** (without any adjective), which turns out to be the appropriate one to generalise to topological spaces. We give the precise definition of compactness later in this chapter (Definition 9.21). Since the notions of "compactness" and "sequential compactness" coincide in metric spaces, we'll drop the adjective "sequentially" in what follows, and simply talk about "compact sets".

Example 9.5 Every closed bounded subset of $\mathbf{R}$, with its usual metric, is compact. This follows from the Bolzano-Weierstrass theorem: a sequence in such a set has a convergent subsequence, and since the set is closed, the limit of this subsequence must be in the set. The set $A = \{\frac{1}{n} : n \in \mathbf{N}\}$ is not compact, because it contains the sequence $(\frac{1}{n})$, and every subsequence of this sequence converges to 0, and 0 is not in the set.

Example 9.6 In any metric space finite sets are compact, since every sequence in such a set contains an eventually constant subsequence.

Compact sets are not that far from closed and bounded sets. In fact, in any metric space a compact set must be closed and bounded.

Proposition 9.7 L*et (X, d) be a metric space and A a compact subset of X. Then A is closed and bounded.*

Proof: Let (x_n) be a convergent sequence in A, say $x_n \to x \in X$ as $n \to \infty$. Since A is compact, (x_n) has a subsequence which converges to a point in A. But this subsequence also converges to x. Since limits are unique, x must be in A, and so A is closed.

Let us assume that A is not bounded, and try to arrive at a contradiction. Choose any $b \in A$. Since we assume A is not bounded, it will not be contained in any ball centred at b. Therefore, for each $n \in \mathbf{N}$, A will not be contained in $B(b, n) = \{x \in X : d(b, x) < n\}$. Hence we can find, for each $n \in \mathbf{N}$, an $a_n \in A$ such that $d(b, a_n) \geq n$. Since (a_n) is a sequence in the compact set A, it has a convergent subsequence (a_{n_k}), say $a_{n_k} \to a \in A$ as $k \to \infty$. Then

$$n_k \leq d(b, a_{n_k}) \leq d(b, a) + d(a, a_{n_k}).$$

Now let $k \to \infty$ in this inequality; we get

$$\infty = \lim_{k \to \infty} n_k \leq d(b, a),$$

and this is clearly a contradiction. Hence our assumption that A is not bounded was false. ■

We've already seen that in $\mathbf{R}$ the converse of this result is also true. Hence a subset of $\mathbf{R}$ is compact if and only if it is closed and bounded. It will probably not come as a surprise that this is in fact true in $\mathbf{R}^m$, for any $m \in \mathbf{N}$. Before we can prove this, we first need to find out which subsets of compact sets are themselves compact. It follows from what we have just proved that a compact subset will necessarily have to be closed; this turns out to be sufficient as well.

Proposition 9.8 L*et (X, d) be a metric space and A a compact subset of X. Then a subset B of A is compact if and only if it is closed in X.*

Proof: It follows from Proposition 9.7 that if B is compact, it is closed in X. For the converse, suppose that B is closed in X and a subset of A, and (x_n) is a sequence in B. Then it is also a sequence in A, and since A is compact, it contains a subsequence which converges to a point of A. Since this convergent subsequence is in the closed set B, its limit must also be in B. Hence B is compact. ■

Theorem 9.9 L*et $m \in \mathbf{N}$ and let $\mathbf{R}^m$ have its usual Euclidean metric. Then a subset A of $\mathbf{R}^m$ is compact if and only if it is closed and bounded.*

Proof: We give the proof for the case $m = 2$; the general case needs some obvious modifications. The one implication is Proposition 9.7. For the converse, suppose A is closed and bounded in $\mathbf{R}^2$. Then A is contained in a rectangle, say $A \subset \{(x_1, x_2) \in \mathbf{R}^2 : a \leq x_1 \leq b, c \leq x_2 \leq d\} = B$ (see Example 9.2). It follows from Proposition 9.8 that it will suffice to prove that B is compact. Let $(\mathbf{x}_n)$ be a sequence in B, with $\mathbf{x}_n = (x_{n1}, x_{n2})$. Then $a \leq x_{n1} \leq b$, i.e. (x_{n1}) is a bounded sequence of real numbers, and so by the Bolzano-Weierstrass theorem has a convergent subsequence, say $x_{n_k 1} \to x_1 \in [a, b]$ as $k \to \infty$. Now $(x_{n_k 2})$ is likewise a bounded sequence of real numbers and therefore has a convergent subsequence, say $x_{n_{k_j}2} \to x_2 \in [c, d]$ as $j \to \infty$. Note that $(x_{n_{k_j}1})$ is a subsequence of (x_{n_k1}) and hence also converges to x_1. Then $(\mathbf{x}_{n_{k_j}})$ is a subsequence of $(\mathbf{x}_n)$ and it converges coordinatewise, and hence also in the metric, to $\mathbf{x} = (x_1, x_2) \in B$. Hence B is compact. ∎

It is now very tempting to conjecture that this theorem (usually referred to as the *Heine-Borel Theorem*) holds in metric spaces generally. The next example shows that this is **not** the case.

Example 9.10 Let $C([0, 1])$ have its usual supremum metric d and put

$$A = \{f \in C([0, 1]) : 0 \le f(x) \le 1 \text{ for every } x \in [0, 1]\}.$$

Then A is bounded, for it is contained in the ball with centre at the constant 0 function and radius 1. It also follows easily from the fact that convergence with respect to d in $C([0, 1])$ implies pointwise convergence that A is closed (check this!). Now consider the sequence (f_n) in A defined by

$$f_n(x) = \begin{cases} 2^n x & \text{for } 0 \le x \le 2^{-n} \\ 1 & \text{for all other } x \in [0, 1] \end{cases}$$

Then

$$d(f_n, f_m) = 1 - 2^m 2^{-n} \ge \frac{1}{2} \text{ for } n > m.$$

Hence (f_n) cannot have any Cauchy subsequence, and therefore also no convergent subsequence, and so A is not compact.

The next result has, on the face of it, nothing to do with compactness, but we'll use it to establish a link between compactness and completeness. You may have come across it for real sequences.

Proposition 9.11 I*f a Cauchy sequence* (x_n) *in a metric space* (X, d) *has a convergent subsequence, then it is itself convergent.*

Proof: Suppose (x_{n_k}) is a subsequence of (x_n) and $x_{n_k} \to x \in X$ as $k \to \infty$. We show that $x_n \to x$ as $n \to \infty$. Let $\epsilon > 0$. We can find an $N_1 \in \mathbf{N}$ such that $d(x, x_{n_k}) < \frac{1}{2}\epsilon$ for $k \ge N_1$ (since $x_{n_k} \to x$). Since (x_n) is a Cauchy sequence, we can find an $N_2 \in \mathbf{N}$ such that $d(x_m, x_n) < \frac{1}{2}\epsilon$ if $m \ge n \ge N_2$. Let $N_\epsilon = \max\{N_1, N_2\}$; if $k \ge N_\epsilon$,

$$d(x, x_k) \le d(x, x_{n_k}) + d(x_{n_k}, x_k) < \frac{1}{2}\epsilon + \frac{1}{2}\epsilon = \epsilon,$$

since we have $n_k \ge k \ge N_\epsilon \ge N_2$. This shows that $x_k \to x$ as $k \to \infty$. ∎

Theorem 9.12 A *compact subset of a metric space is complete.*

Proof: Let A be a compact subset of the metric space (X,d) and (x_n) a Cauchy sequence in A. Since A is compact, (x_n) has a convergent subsequence. By the previous proposition (x_n) itself is convergent, and since A is closed (Proposition 9.7), the limit of (x_n) is in A. Hence A is complete. ∎

We have seen that continuous functions do not necessarily map closed sets to closed sets. The situation is far more satisfactory when it comes to compact sets.

Theorem 9.13 L*et X and Y be metric spaces and $f : X \to Y$ a continuous function. If A is a compact subset of X, $f(A)$ is a compact subset of Y.*

Proof: Let (y_n) be a sequence in $f(A)$. For each n, we can find an $x_n \in A$ such that $y_n = f(x_n)$. Since (x_n) is a sequence in the compact set A, it has a convergent subsequence (x_{n_k}), say $x_{n_k} \to x \in A$. Since f is continuous, $f(x_{n_k}) \to f(x) \in f(A)$ as $k \to \infty$, and therefore $(f(x_{n_k}))$ is a convergent subsequence of (y_n), with its limit in $f(A)$. This shows that $f(A)$ is compact. ∎

We are now in a position to generalise the result about continuous functions mentioned at the start of this chapter. We need a useful property of compact subsets of $\mathbf{R}$.

Lemma 9.14 A *non-empty compact subset A of $\mathbf{R}$, with its usual metric, contains a smallest and a largest element.*

Proof: Since A is bounded, $x = \sup A$ exists. For each $n \in \mathbf{N}$, $x - \frac{1}{n}$ is not an upper bound of A, so there is an $x_n \in A$ such that $x - \frac{1}{n} < x_n \leq x$. Then (x_n) is a sequence in A and $x_n \to x$. Since A is closed, $x \in A$.

The proof for $\inf A$ is similar. ∎

Theorem 9.15 I*f (X,d) is a compact metric space, $\mathbf{R}$ has its usual metric and $f : X \to \mathbf{R}$ is a continuous function, then f attains its maximum value and its minimum value on X.*

Proof: It follows from Theorem 9.13 that $f(X)$ is a compact subset of $\mathbf{R}$, and hence by the lemma it contains a smallest and largest element. These are the minimum and maximum values of f on X. ∎

Example 9.16 If A is a compact subset of the metric space (X, d) and $a \in X$, then there is an $y \in A$ such that $d(a, y) = \inf\{d(a, x) : x \in A\} = d(a, A)$, since the function $f_a : X \to \mathbf{R}$ defined by $f_a(x) = d(a, x)$ is continuous.

The next result shows that on compact metric spaces continuity of an inverse function comes " for free".

Theorem 9.17 I*f X is a compact metric space, Y a metric space and $f : X \to Y$ a continuous bijection, then f is a homeomorphism.*

Proof: Since f is a bijection, $f^{-1} : Y \to X$ exists, and it is only necessary to show that it is continuous. By Proposition 6.32 it is enough to show that f is a closed function. Let F be a closed subset of X. Since X is compact, F is compact (Proposition 9.8). Hence $f(F)$ is compact (by Theorem 9.13), and hence closed (by Proposition 9.7). ■

Example 9.18 Let $\mathbf{R}$ have its usual metric and $f : [a, b] \to \mathbf{R}$ be a continuous injective (one-to-one) function. Let Y be the range of f. Then the inverse $f^{-1} : Y \to [a, b]$ exists and is continuous.

In Chapter 8 we saw that a "distance-reducing" function on a complete metric space need not have a fixed point. If the metric space is compact, we can salvage the contraction mapping theorem for functions that are only distance-reducing, rather than contractions.

Theorem 9.19 L*et (X, d) be a compact metric space and $f : X \to X$ a function such that $d(f(x), f(y)) < d(x, y)$ for all $x, y \in X, x \neq y$. Then f has a unique fixed point in X. For any $x_1 \in X$, the sequence (x_n) defined inductively by*

$$x_{n+1} = f(x_n), \quad n \in \mathbf{N}$$

converges to this fixed point.

Proof: We begin by noting that x is a fixed point of f if and only if $d(x, f(x)) = 0$. The function $F : X \to \mathbf{R}$ defined by $F(x) = d(x, f(x))$ has only non-negative values. It follows from our first remark that it will suffice to prove that there is an $x \in X$ such that $0 = F(x) = d(x, f(x))$. If we can show that F is continuous on X, it will follow from the fact that X is compact and Theorem 9.15 that F attains its smallest value on X. To show that F is continuous, let $x, y \in X$. Then

$$\begin{aligned} d(x, f(x)) &\leq d(x, y) + d(y, f(y)) + d(f(y), f(x)) \\ &\leq d(x, y) + d(y, f(y)) + d(x, y) \end{aligned}$$

and so

$$|F(x) - F(y)| = |d(x, f(x)) - d(y, f(y))| \leq 2d(x, y).$$

This inequality shows that F is Lipschitz-continuous, and therefore continuous. Therefore F attains its least value on X, i.e. there is an $x_0 \in X$ such that $F(x_0) \leq F(x)$ for all $x \in X$. We show that $F(x_0) = 0$. If this is not the case, we have $0 < F(x_0) \leq F(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = F(x_0)$, a contradiction. Hence $F(x_0) = 0$ and x_0 is a fixed point for f. It is also unique, for if we also have $z = f(z), z \neq x_0$, then $d(x_0, z) = d(f(x_0), f(z)) < d(x_0, z)$, a contradiction again.

If $x_1 \in X$ and $x_{n+1} = f(x_n)$ for $n \geq 1$, we need to show that $x_n \to x_0$. We have

$$d(x_0, x_n) = d(f(x_0), f(x_{n-1})) < d(x_0, x_{n-1}).$$

It follows from this that the real sequence $(d(x_0, x_n))$ is decreasing and bounded below by 0. Hence it converges; it will suffice to show that the limit is 0. Since X is compact, the sequence (x_n) has a convergent subsequence (x_{n_k}), say $x_{n_k} \to x$ as $k \to \infty$. Then the continuity of the distance function implies that $d(x_0, x_{n_k}) \to d(x_0, x)$. Since $(d(x_0, x_n))$ is a decreasing sequence, it follows that $d(x_0, x_n) \to d(x_0, x)$. Hence it suffices to show that $d(x, x_0) = 0$. Since $x_{n_k} \to x$, $x_{n_k+1} = f(x_{n_k}) \to f(x)$ (f is continuous). As before, $d(x_0, x_{n_k+1}) \to d(x_0, f(x))$ and hence $d(x_0, f(x)) = d(x_0, x)$. If $x_0 \neq x$, we have $d(x_0, x) = d(x_0, f(x)) = d(f(x_0), f(x)) < d(x_0, x)$, a contradiction. So we must have $x_0 = x$, and hence $d(x_0, x) = 0$. ■

We are now ready to consider the problem mentioned at the start of the chapter: How should the distance between two subsets of a metric space be defined? There are several ways of doing this; one that has turned out to be appropriate especially in the study of fractals leads to a metric known as the **Hausdorff metric**. This metric can be used to measure the distance between sets *in a metric space.* We start with a metric space, and then construct a second metric space in which the "points" are subsets of the first metric space, and the distance between them is given by the Hausdorff metric. As we'll see below, we need to restrict the type of subset allowed as points, or elements, of the second metric space. This is where compactness turns out to be useful.

We have already defined what we mean by the distance between a point x and a set A in a metric space (X, d). For two subsets A and B of X, we now put

$$d(A, B) = \sup\{d(x, B) : x \in A\}.$$

Note that as long as B is not empty, it follows then that $d(x, B) < \infty$ for every $x \in A$. If A is also compact, $\{d(x, B) : x \in A\}$ is a bounded set of real numbers, and hence $d(A, B) < \infty$. For this reason we'll only consider $d(A, B)$ in the case where both A and B are non-empty compact sets. We denote the family of all non-empty compact subsets of X by $\mathcal{K}(X)$.

It is tempting to think that with $d(A,B)$ as defined above we now have a metric. A little experimentation shows that in general we do not have $d(A,B) = d(B,A)$. To get around this we define the Hausdorff metric d_H on $\mathcal{K}(X)$ by

$$d_H(A,B) = \max\{d(A,B), d(B,A)\}, \qquad A, B \in \mathcal{K}(X).$$

It can be checked that this does indeed give a metric on $\mathcal{K}(X)$; we ask you to check the details in the exercises.

The definition of the Hausdorff metric above does not really give us a geometrical feeling for what it means. It is helpful to introduce what is sometimes known as the r-neighbourhood of a set. If $r > 0$ and $A \in \mathcal{K}$, the r-neighbourhood of A is the set

$$N_r(A) = \{x \in X : d(x,a) < r \text{ for some } a \in A\}.$$

It can be shown (see Exercise 8d) that for $A, B \in \mathcal{K}$,

$$d_H(A,B) = \inf\{r > 0 : A \subset N_r(B) \text{ and } B \subset N_r(A)\}.$$

Once we have this result, it is easy to see that the Hausdorff distance between A and B is defined in such a way that A and B are within a distance r of each other if every point in A is within a distance r of some point in B, and every point in B is within a distance r of some point in A. The sketch below shows two sets in $\mathbf{R}^2$ for which the Hausdorff distance between them is less than r.

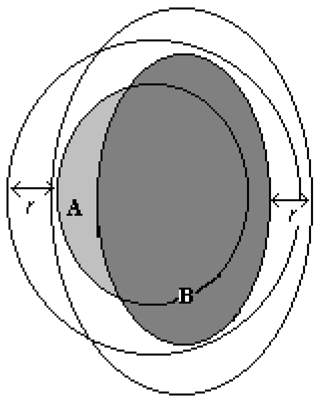


Figure 9.1: The Hausdorff distance between A and B is less than r.

It can also be shown that $(\mathcal{K}(X), d_H)$ is a complete metric space (this is much harder, so we'll be content to state this without proof). This implies, amongst other things, that the contraction mapping theorem holds in this metric space. This is the idea behind many of the algorithms used to produce fractals.

Compactness can also help us to understand the relationship between continuity and uniform continuity better. We have seen that uniformly continuous functions are continuous. There are continuous functions that are not uniformly continuous. On compact sets, however, continuous functions are always uniformly continuous.

Theorem 9.20 *Let (X, d_X) and (Y, d_Y) be metric spaces, A a compact subset of X and $f : X \to Y$ a continuous function. Then f is uniformly continuous on A.*

Proof: See Exercise 9. ∎

We have not said much about topologies lately. Completeness is not a topological property, and it is also clear that the notion of a contraction, and hence the contraction mapping theorem, only make sense in the setting of a metric space. The definition we have used for compactness (or, to be more precise, sequential compactness) is phrased in terms of sequences. Is it possible to give an equivalent definition of compactness which is phrased in terms of open sets only? The answer is yes, but the proof of the equivalence of the two definitions in a metric space setting is not straightforward. We give the definition in terms of open sets below, and simply state the equivalence. We'll explore these ideas in a little more depth in the exercises and tutorials. In what follows we'll use the terminology *sequentially compact* for the concept defined in Definition 9.4, and reserve the term *compact* for the concept defined below.

Definition 9.21 *Let (X, d) be a metric space and $A \subset X$.*

1. *We say a family of subsets $\{A_i : i \in I\}$ of X covers A, or is a* **covering** *of A, if $A \subset \cup\{A_i : i \in I\}$. (Here I is an index set, and could be finite or infinite.)*
2. *We say A is* **compact** *if every family of open subsets of X which covers A has a finite subfamily which still covers A (i.e. whenever $\{G_i : i \in I\}$ is a family of open sets such that $A \subset \cup\{G_i : i \in I\}$ then there is a* finite *set $J \subset I$ such that $A \subset \cup\{G_i : i \in J\}$.)*
3. *We say A is* **totally bounded** *if for every $\epsilon > 0$, there is a finite set $\{x_1, x_2, \ldots, x_n\} \subset X$ such that*
$$A \subset \cup_{i=1}^{n} B(x_i; \epsilon),$$
i.e. for every $\epsilon > 0$, A can be covered by a finite number of open balls of radius ϵ.

The following theorem links all these concepts in the setting of a metric space. We give it without proof; in the exercises you will find hints to help you to prove some of the implications.

Theorem 9.22 L*et* (X, d) *be a metric space and* $A \subset X$*. Then the following statements are equivalent:*

1. *A is sequentially compact;*
2. *A is totally bounded and complete;*
3. *A is compact.*

This theorem explains why in the setting of a metric space we could call a sequentially compact set compact. The definition of compactness above is phrased in terms of open sets only; it will therefore make sense in topological spaces as well. This is in fact the way compact sets are defined in topological spaces. Since the notion of convergence of sequences makes sense in topological spaces as well (see Proposition 5.26), the definition of sequential compactness also makes sense in topological spaces. In general topological spaces the two notions are not equivalent, however.

It is clear that the definition of total boundedness only makes sense in a metric space.

Summary.

Bounded and sequentially compact sets in metric spaces were defined. Compact sets are always closed and bounded; the converse is true in $\mathbf{R}^m$, but not generally. Compact sets are complete. Continuous functions preserve compactness; this leads to a number of useful results. The Hausdorff metric enables us to measure the distance between non-empty compact subsets of a metric space. Compactness (defined in terms of open sets) and total boundedness were defined, and the relationship of these concepts to sequential compactness and completeness given.

Historical notes

Bernard Placidus Johann Nepomuk Bolzano (1781 – 1848)

Bolzano was born in Prague (now in the Czech Republic) and went to the University of Prague in 1796 to study philosophy and mathematics. From the start he was interested in the philosophical and foundational aspects of mathematics; as he put it "those parts of mathematics which was at the same time philosophy". In 1800 he started three years of theological study and at the same time began on a doctoral thesis on geometry. In it he investigated the nature of proof in mathematics. In 1804 he was awarded the doctorate, and ordained as a priest in the Roman Catholic church. He was also appointed as a professor in philosophy and religion at the University of Prague. He was a pacifist and had strong views on economic justice which did not make him popular with the authorities. This lead to him losing his position in 1819, being placed under house arrest and forbidden to publish.

Bolzano embarked on a series of books on the foundations of mathematics. The first volume was published in 1810, the second was written but never published. Instead he began work on an attempt to free calculus from the concept of an infinitesimal. He wanted to develop rigorous definitions of the notions of limit, convergence and derivative that did not depend on geometry. He used his new approach to give a proof of the intermediate value theorem, and also gave a definition of a what is now called a Cauchy sequence before Cauchy did!
After 1817, Bolzano did not publish any mathematical work for many years, but worked on a theory of science and knowledge. He did return to the foundations of mathematics, and worked on an attempt to put the whole of mathematics on a logical foundation. In the process he discovered a number of paradoxes concerned with infinite sets. He was the first to use the term "set", and to give an example of an infinite set whose elements is in one-to-one correspondence with the elements of a proper subset of itself. He anticipated Cantor's theory of infinite sets in many ways, but much of his work was only published after his death in 1848.

Félix Édouard Justin Émile Borel (1871 – 1956)

Borel was born in Saint Affrique in France in 1871. His first appointment was at the École Normale Supérieure in Paris, in 1896. He was the first to develop a theory of the measure ("size") of a set of points. This theory and the work of Baire and Lebesgue led to the modern theory of functions of a real variable. Borel also developed a systematic theory of divergent series. His work was so influential that a chair in the theory of functions was created for him at the Sorbonne in Paris in 1909.

He was awarded the Croix de Guerre for his contribution to the French war effort in 1918.

Borel was the first to define the notion of a game of strategy, and wrote a series of papers on game theory in the 1920's. At the same time he entered politics and served as Minister of the Navy in the period 1925 – 1940. During the Second World War he was arrested by the Vichy regime in France and briefly imprisoned. After his release he worked for the French Resistance, and was awarded the Resistance Medal (1945) and the Grand Croix Légion d'Honneur (1950). He died in Paris in 1956.

Heinrich Eduard Heine (1821 – 1881)

Heine was born in Berlin in 1821. He was taught by Gauss and Dirichlet. His most important contributions were to analysis, and he is perhaps best known for the so-called Heine-Borel theorem characterising compact sunsets of $\mathbf{R}^n$. The concept of a uniformly continuous function was first formulated by him, and his interest in compactness was stimulated by the proof of the fact that a continuous function on a compact set is uniformly continuous. Heine also did work on Legendre polynomials and Bessel functions. He died in Halle, Germany in 1881.

Karl Theodor Wilhelm Weierstrass (1815 – 1897)

Weierstrass was born in Ostenfelde, Germany, the eldest of four children. His father, a well-educated man, changed jobs often and this resulted in Karl having to change schools frequently. His mother died when he was 12, and his father remarried two years later. In 1829 his father took on a job in Paderborn, and Karl went to school in there and took on a part-time bookkeeping job to help the family. He did very well at school, especially in mathematics. His father wanted him to follow a career in finance, and therefore enrolled him for a courses in law, finance and economics at the University of Bonn in 1834.

Karl wished to study mathematics, but was afraid to disobey his father's wishes. His way of resolving the conflict was not to attend lectures at all, but to spend the next four years trying to enjoy student life.

Karl did study mathematics on his own, and became interested in elliptic functions. He managed to solve a problem posed by Abel, and this made him decide to study mathematics, even if it went against his father's wishes. He gave up his studies and left the university without taking any examinations. His father was greatly upset, but arranged for Karl to enroll at the Academy in Münster in 1839 in order to qualify as a teacher. There he was encouraged by Gudermann, who also had an interest in elliptic functions, to continue his mathematical career. During this time he laid the foundations for his theory of a function of a complex variable. He finished his training as teacher in 1842, and started teaching at secondary schools. He found this a time of "unending dreariness and boredom". In 1850 his health began to suffer, probably as a result of the mental and physical strain brought on by trying to do mathematics and satisfy the demands of his teaching job.

In 1854 Weierstrass published a paper on Abelian functions which immediately brought him to the attention of the mathematical world. Although he did not succeed in getting a university appointment at the time, his next paper in 1856 led to a number of offers from universities. He accepted an offer from the Industry Institute in Berlin, and was subsequently offered a chair at the University of Berlin, a position he was only able to take up a few years later. Weierstrass was a very popular lecturer and attracted students from all over the world. He gave a course on the foundations of analysis in 1860; his emphasis was on rigour and led to his discovery of a continuous but nowhere differentiable function. His health deteriorated further and this led to a complete collapse in 1861. When he returned to work a year later he had to lecture sitting down, with a student writing on the blackboard for him.

In 1863 Weierstrass began to formulate his theory of real numbers. His lectures on this and other topics were eventually published and exerted an enormous influence on the teaching of analysis, so much so that the term "Weierstrassian rigour" is still in use today. A large number of famous mathematicians benefited from his teaching. One of the most famous was Sofia Kovalevskaya, who was taught privately by him, because women were not allowed admission to the university at the time. He was instrumental in securing her a position in Stockholm and in getting the University of Göttingen to award her an honorary doctorate.

Weierstrass' perfectionism resulted in him publishing little, but he did supervise the initial work on publication of his complete works. Much of this only appeared after his death, from pneumonia, in 1897.

Exercises

1. Write out in detail the proof that a subset of a metric space is bounded if and only if has finite diameter (Proposition 9.3).

2. Let $C([0,1])$ have the supremum metric. Which of the following subsets of $C([0,1])$ are bounded? Give reasons for your answers.

   (a) $\{f \in C([0,1]) : f(0) = 0\}$;

   (b) $\{f \in C([0,1]) : -1 \leq f(x) \leq 1 \text{ for all } x \in [0,1]\}$;

   (c) $\{f_n : n \in \mathbf{N}\}$, where $f_n(x) = \sin n\pi x$ for $x \in [0,1]$.

3. Which of the following subsets of $\mathbf{R}^2$ with its usual Euclidean metric are compact? Give reasons for your answers.

   (a) $\{(x_1, x_2) \in \mathbf{R}^2 : |x_1| \leq 2, |x_2| < 1\}$;

   (b) $\{(x_1, x_2) \in \mathbf{R}^2 : 2x_1 + x_2 = 3\}$;

   (c) $\{(x, \cos x) : x \in [-\pi, 2\pi]\}$.

4. Characterise the compact subsets of $\mathbf{C}$, with its usual metric. [Hint: $\mathbf{C}$ is isometric to $\mathbf{R}^2$ when both have their usual metrics.]

5. Say, with reasons, which of the following subsets of $\mathbf{C}$, with its usual metric, are compact.

   (a) $\{z \in \mathbf{C} : z^7 = 2i\}$;

   (b) $\{z \in \mathbf{C} : 0 < |z - i| \leq 2\}$;

   (c) $\{z \in \mathbf{C} : z$ has an argument in $[0, \frac{\pi}{4}]\}$.

6. Let $C([0,1])$ have the supremum metric. For each of the following subsets of $C([0,1])$ say whether it is compact. Give reasons for your answers.

   (a) $\{f \in C([0,1]) : f(0) = 0\}$;

   (b) $\{f \in C([0,1]) : 0 < f(x) \leq 1$ for every $x \in [0,1]\}$.

7. Let A be a subset of a metric space (X, d) with the property that there is a sequence (x_n) in A such that for some $r > 0$, $d(x_n, x_m) \geq r$ for all $n, m \in \mathbf{N}$. Show that A is not compact.

8. Let (X, d) be a metric space.

   (a) Give an example to show that we may have $d(A, B) \neq d(B, A)$ for subsets A and B of X.

   (b) Prove that the Hausdorff metric is indeed a metric on $\mathcal{K}(X)$.

   (c) Give examples to show that the Hausdorff metric will no longer be a metric if it is applied to the family of all subsets of X (i.e. $\mathcal{P}(X)$) rather than to the family $\mathcal{K}(X)$.

   (d) Let $r > 0$ and $A \in \mathcal{K}$. Put

   $$N_r(A) = \{x \in X : d(x, a) < r \text{ for some } a \in A\}.$$

   Prove that for $A, B \in \mathcal{K}$,

   $$d_H(A, B) = \inf\{r > 0 : A \subset N_r(B) \text{ and } B \subset N_r(A)\}.$$

9. Prove Theorem 9.20. [Hint: Use Theorem 6.27. Start of by assuming f is not uniformly continuous, write out what this means in terms of the characterization in Theorem 6.27 and derive a contradiction.]

10. Prove that every sequentially compact set in a metric space is totally bounded. [Hint: Suppose that A is sequentially compact but not totally bounded. Show that there is an $\epsilon > 0$ and a sequence (x_n) in A such that $d(x_n, x_m) > \epsilon$ for all $m \neq n$, and show that this contradicts the fact that A is sequentially compact (see Exercise 7).]

11. Prove that every compact subset (in the sense of Definition 9.21) of a metric space is totally bounded.

12. Prove that every totally bounded and complete subset of a metric space is sequentially compact. [Hint: If (x_n) is a sequence in such a set, the total boundedness can be used to show that it has a subsequence which is a Cauchy sequence.]

Chapter 10

And now all comes together.

In which we meet sets that cannot be taken apart, or connected sets.

We often talk about the graph of a continuous function as being "in one piece", and by this we usually mean that there is no natural way of thinking of the graph as being made up of two or more separate pieces. This is one of many important examples of sets which cannot be so separated. In this chapter we explore the possibility of making this idea precise in a more general setting than that of $\mathbf{R}$ or $\mathbf{R}^2$.

It is in fact easier to define first of all what we mean by saying what it means for a set **not** to have this property, i.e. to say what it means for a set to be in "two pieces". A first attempt may be to say that a set A is "in two pieces" if there are disjoint non-empty sets A_1 and A_2 such that $A = A_1 \cup A_2$. But

$$\mathbf{R} = (-\infty, 0) \cup [0, \infty), \text{ with } (-\infty, 0) \cap [0, \infty) = \emptyset.$$

So $\mathbf{R}$ is the union of two disjoint non-empty sets, but we certainly do not think of $\mathbf{R}$ as being in two pieces. To have a "natural separation" we have to place more restrictions on A_1 and A_2.

Definition 10.1 A *metric space X is* **disconnected** *if there are two non-empty, disjoint,* closed *sets F_1 and F_2 such that $X = F_1 \cup F_2$. A pair (F_1, F_2) like this is called a* **disconnection** *of X.*

Before we look at examples, we note that we may as well have used open sets in the definition.

Proposition 10.2 T*he following statements are equivalent for a metric space* (X, d)*:*

1. X *is disconnected;*
2. *there are non-empty, disjoint open sets* G_1 *and* G_2 *such that* $X = G_1 \cup G_2$*;*
3. X *contains a proper subset which is both open and closed. (Note that a subset is called "proper" here if it is neither* $\emptyset$ *nor* X*.)*

Proof: (1) $\Rightarrow$ (2): If X is disconnected, we have $X = F_1 \cup F_2$, with F_1 and F_2 non-empty, disjoint and closed. Since $F_1' = F_2$, F_1 is open as well; similarly F_2 is open. Hence we may take $G_1 = F_1, G_2 = F_2$.
(2) $\Rightarrow$ (3): If $X = G_1 \cup G_2$ with G_1 and G_2 non-empty, disjoint and open, then G_1 is also closed ($G_1' = G_2$), and $\emptyset \neq G_1 \neq X$.
(3) $\Rightarrow$ (1): If A is a proper subset of X which is both open and closed, then put $F_1 = A, F_2 = A'$. Then F_1 and F_2 are closed, disjoint and non-empty (since A is a proper subset of X), and $X = F_1 \cup F_2$. ■

Example 10.3 Every discrete metric space with more than one point is disconnected, since every singleton is a proper subset which is both open and closed.

You may always have felt a little uncomfortable with the fact that a subset of a metric space could possibly be open and closed at the same time. The fact that the whole metric space and the empty set both have this property seems unavoidable, but somehow it feels unnatural for other sets to share it with them! The result we have just proved gives some justification for this feeling, in the sense that it shows that this kind of pathological behaviour can only occur in disconnected spaces.

So far we have just defined what we mean by the whole metric space being disconnected. What about subsets? If A is a subset of a metric space (X, d), the metric d on X induces a metric d_A on A, and with this metric A becomes a metric space in its own right. The natural thing to do is to say that A is disconnected if and only if the metric space (A, d_A) is disconnected. If we do this it will be helpful to know what the open sets in the metric space (A, d_A) look like. Fortunately we already know the answer to this: A subset B of A is open in (A, d_A) if and only if there is an open set G in X such that $B = A \cap G$ (Proposition 5.18).

All of this motivates the following definition:

Definition 10.4 A *subset* A *of a metric space* X *is disconnected if* (A, d_A) *is a disconnected metric space. This means that* A *is disconnected if there are open subsets* G_1 *and* G_2 *of* X *such that* $A \cap G_1$ *and* $A \cap G_2$ *are non-empty disjoint sets such that* $A = (A \cap G_1) \cup (A \cap G_2)$*.*

Example 10.5 Let $\mathbf{R}$ have its usual metric. The set $A = [0, 1) \cup (1, 2]$ is disconnected. (Take $G_1 = (-1, 1), G_2 = (1, 3)$.) Likewise, the set $\mathbf{Q}$ is disconnected. (Take $G_1 = (-\infty, \sqrt{2}), G_2 = (\sqrt{2}, \infty)$.)

Example 10.6 Let $\mathbf{C}$ have its usual metric. Check that the set

$$\{z \in \mathbf{C} : |\mathrm{Re} z| \geq 1\}$$

is disconnected.

We can now define a set to be "in one piece" if it is not "in two pieces".

Definition 10.7 A *metric space, or a subset of a metric space, is* **connected** *if it is not disconnected.*

The form of this definition makes it very difficult to prove that a set is connected, since it requires us to show that **no** disconnection of the set can possibly be found. To get around this problem, we try to build up a stock of connected sets by starting in familiar territory. We start with probably the most familiar of metric spaces, $\mathbf{R}$ with its usual metric, and try to identify the connected sets in it. A natural guess would be that these are the intervals in $\mathbf{R}$. To show that this is indeed the case, we need to say precisely what we regard as an interval:

Definition 10.8 A *subset A of* $\mathbf{R}$ *is called an* **interval** *if whenever $x, y \in A$ and $x < z < y$ then $z \in A$.*

Theorem 10.9 A *subset of* $\mathbf{R}$*, with its usual metric, is connected if and only if it is an interval.*

Proof: We first prove that a connected set is an interval by showing that if $A \subset \mathbf{R}$ is not an interval, then it is disconnected. If A is not an interval, then we can find $x, y \in A, x < y$ and a $z \notin A$ with $x < z < y$. If we take $G_1 = (-\infty, z), G_2 = (z, \infty)$, then both are open sets in $\mathbf{R}$ and it follows that $A \cap G_1$ and $A \cap G_2$ are non-empty and disjoint and have union A. Therefore A is disconnected.

To prove that an interval is a connected set, we suppose A is both an interval and disconnected and try to find a contradiction. Since A is disconnected, we can find open sets G and H in $\mathbf{R}$ such that $A \cap G$ and $A \cap H$ are non-empty, disjoint and $A = (G \cap A) \cup (A \cap H)$. Choose $x \in G \cap A, y \in H \cap A$; without loss of generality we may assume $x < y$. Since A is an interval and $x, y \in A$, we have $[x, y] \subset A$. We put $z = \sup\{t \in A : t \in [x, y] \cap G\}$; the supremum exists because the set is bounded and non-empty. Then we have $z \in [x, y] \subset A$. To get a contradiction we show that $z \notin G \cup H$.

We first show that $z \notin G$. Suppose $z \in G$, then we must have $z < y$ (we cannot have $z = y$, since $y \notin G$). Since G is open in $\mathbf{R}$, it is possible to find an $\epsilon > 0$ such that $B(z; \epsilon) \subset G$ and also $[z, z + \epsilon) \subset [x, y]$. From this we get $[z, z + \epsilon) \subset [x, y] \cap G$, and in particular $z + \frac{1}{2}\epsilon \in [x, y] \cap G \cap A$, which contradicts the definition of z. So $z \notin G$.

Next we show $z \notin H$. Suppose $z \in H$; then $z > x$ (we cannot have $z = x$, since $x \notin H$). Again using the fact that H is open in X, we can find a $\delta > 0$ such that $(z - \delta, z] \subset [x, y] \cap H$. This implies that $z - \delta$ is an upper bound for $[x, y] \cap G$, and this contradicts the definition of z. So $z \notin H$.

Hence $z \in A$, but $z \notin G \cup H$, a contradiction. This proves that an interval is connected. ∎

Corollary 10.10 $\mathbf{R}$ *is connected, and hence the only subsets of* $\mathbf{R}$ *which are both open and closed are* $\mathbf{R}$ *and* $\emptyset$.

We next prove a characterisation of connected spaces which at first appears very artificial, but turns out to be a great help when trying to prove that sets are connected. To formulate this result, we introduce some notation. We'll use the abbreviation $\mathbf{2}$ for the metric space $(\{0, 1\}, d_D)$, where d_D is the discrete metric on the set $\{0, 1\}$.

Proposition 10.11 A *metric space* X *is connected if and only if every continuous function* $f : X \to \mathbf{2}$ *is a constant function.*

Proof: Suppose X is disconnected. Then by Proposition 10.2, X has a proper subset A which is both open and closed. Define the function $f : X \to \mathbf{2}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \in A \\ 1 & \text{if } x \notin A \end{cases}$$

There are only four open sets in $\mathbf{2}$, and it is easy to check that the inverse image of each one of them under the function f is an open set in X. Hence f is continuous, and clearly not constant.

Conversely, suppose there is a continuous function $f : X \to \mathbf{2}$ which is not constant. Put $G_1 = f^{-1}(\{0\}), G_2 = f^{-1}(\{1\})$. Since f is continuous, G_1 and G_2 are both open, and $X = G_1 \cup G_2$. Also G_1 and G_2 are clearly disjoint and non-empty (the latter because f is not constant). This shows that X is disconnected. ∎

Proposition 10.12 *Let A be a connected subset of a metric space X. Then $\overline{A}$ is connected as well.*

Proof: Let A be a connected subset of X and $f : \overline{A} \to \mathbf{2}$ be continuous. Then $f(\overline{A}) \subset \overline{f(A)}$ by Theorem 6.18. Since A is connected and f restricted to A is continuous, it follows from Proposition 10.11 that f must be constant on A. Therefore $f(A)$ is a singleton and also closed, so $f(\overline{A}) \subset f(A)$ implies that $f(\overline{A})$ is a singleton as well. Hence f is a constant function on $\overline{A}$ and therefore $\overline{A}$ is connected, again by Proposition 10.11. ■

Theorem 10.13 *Let X and Y be metric spaces, X connected and $f : X \to Y$ a continuous function. Then $f(X)$ is a connected subset of Y.*

Proof: Let $g : f(X) \to \mathbf{2}$ be continuous. We have to show that g is constant. Then $g \circ f : X \to \mathbf{2}$ is continuous (Proposition 6.6) and constant (since X is connected). Then g must be constant on $f(X)$. (If $f(x_1), f(x_2) \in f(X)$, then $g(f(x_1)) = g(f(x_2))$.) Hence $f(X)$ is connected. ■

Corollary 10.14 *(The Intermediate Value Theorem) Let $f : [a, b] \to \mathbf{R}$ be continuous and y a number between $f(a)$ and $f(b)$. Then there is an $x \in [a, b]$ such that $y = f(x)$.*

Proof: The interval $[a, b]$ is connected, and therefore $f([a, b])$ is connected, hence an interval. ■

Corollary 10.15 *Let I be an interval in $\mathbf{R}$ and $f : I \to \mathbf{R}$ a continuous function. Then the graph of f, i.e. the set*

$$\{(x, f(x)) \in \mathbf{R}^2 : x \in I\}$$

is a connected subset of $\mathbf{R}^2$.

Proof: Define the function $F : I \to \mathbf{R}^2$ by $F(x) = (x, f(x))$. Then F is continuous (if $x_n \to x$ in I, then $f(x_n) \to f(x)$, so $F(x_n) = (x_n, f(x_n)) \to (x, f(x)) = F(x)$). Therefore $F(I)$ is connected. But $F(I)$ is just the graph of f. ■

In general a union of a family of connected sets need not be connected. (Find an example to convince yourself of this!) The next theorem gives a positive result in this direction.

Theorem 10.16 *Let $\{A_i : i \in I\}$ be a family of connected subsets of a metric space X such that $\cap\{A_i : i \in I\} \neq \emptyset$. Then $A = \cup\{A_i : i \in I\}$ is connected.*

Proof: Let $f : A \to \mathbf{2}$ be continuous. For each $i \in I$, f restricted to A_i is continuous. Since A_i is connected, f restricted to A_i is constant. Let $x \in \cap\{A_i : i \in I\}$. Then if $y \in A$, $y \in A_i$ for some $i \in I$. Since $x, y \in A_i$, $f(y) = f(x)$. Since $y \in A$ was arbitrary, this shows that $f(y) = f(x)$ for all $y \in A$, i.e. f is constant on A, and so A is connected. ■

Example 10.17 The family of all lines through the origin in $\mathbf{R}^2$ is a family of connected sets (why?) with non-empty intersection (since all of them contain the origin). Hence its union is connected. But the union is $\mathbf{R}^2$.

We now show that it is possible to write every metric space as a disjoint union of rather special connected sets – ones that are not contained in any larger connected sets. Not surprisingly, such sets are known as components.

Definition 10.18 L*et X be a metric space and $x \in X$. The largest connected subset of X containing x is called the* **component** *of X containing x.*

It follows from Proposition 10.12 that every component must be closed (if not, its closure would be a larger connected set).

Example 10.19 If a metric space X is connected, it has only one component, namely X itself. In a discrete metric space, on the other hand, the components are the singletons.

Proposition 10.20 L*et X be a metric space. Then every $x \in X$ is contained in a unique component of X; X is therefore a union of disjoint components.*

Proof: Since for every $x \in X$, $\{x\}$ is a connected set containing x, the family of all connected sets containing x is non-empty, and has non-empty intersection. By Theorem 10.16 the union of this family is connected, and it is clearly the largest connected set containing x, that is the component containing x. Any two components must be disjoint, because if they were not, their union would be a larger connected set containing both, by Theorem 10.16. ■

This result can be used to characterise the open subsets of $\mathbf{R}$ with its usual metric: A set in $\mathbf{R}$ is open if and only if it is a countable union of disjoint open intervals. This improves the characterisation given in Corollary 5.17. We give some hints about the proof of this fact in the exercises.

You may have noticed that we defined connectedness (or rather disconnectedness) in terms of open sets from the start. It is a good example of a property that does not lend itself to a definition in terms of sequences. It is also clear that our definition will make good sense in any topological space, not only in metric spaces.

Summary.

Disconnected metric spaces, and disconnected subsets of metric spaces were defined. This led to the definition of connected metric spaces and sets. The connected subsets of $\mathbf{R}$ were characterised as the intervals. Closures and continuous images of connected sets are connected, and so are graphs of continuous functions from $\mathbf{R}$ to $\mathbf{R}$. Unions of connected sets were investigated, and the notion of a component of a metric space introduced.

Exercises

1. For each of the following subsets of $\mathbf{R}^2$, with its usual metric, say whether it is connected or not. If not, give a disconnection.
   (a) $\{(x, y) \in \mathbf{R}^2 : xy > 0\}$;
   (b) $\{(x, y) \in \mathbf{R}^2 : 1 < x^2 + y^2 < 3\}$;
   (c) $\{(x, \sin x) \in \mathbf{R}^2 : x \in (-\pi, 2\pi]\}$;
   (d) $\{(x, y) \in \mathbf{R}^2 : |x| > 2\}$.
2. Let A and B be connected subsets of a metric space X. Which of the following statements are true? Give a proof or counterexample to substantiate your answer.
   (a) $A \cup B$ is connected.
   (b) $A \cap B$ is connected.
   (c) A' is connected.
3. Use the method of Example 10.17 to show that the following subsets of $\mathbf{R}^2$ are connected:
   (a) $\{(x, y) \in \mathbf{R}^2 : x^2 + y^2 \leq 2\}$;
   (b) $\{(x, y) \in \mathbf{R}^2 : |x| \leq 2, |y| < 1\}$.
4. Let (A_n) be a sequence of connected subsets of a metric space X such that $A_n \cap A_{n+1} \neq \emptyset$ for every n. Show that $\cup_{n=1}^{\infty} A_n$ is connected.
5. Let X be a metric space and $A \subset B \subset X$. Prove that if A is connected as a subset of B if and only if it is connected as a subset of X.

6. Show that if A is a connected subset of a metric space X, then there is a component C of X such that $A \subset C$.

7. Prove that a clopen connected subset of a metric space X is a component of X.

8. Let $\mathbf{R}$ have its usual metric and let G be an open subset of $\mathbf{R}$.

   (a) Show that if C is a component of G, then it is open in $\mathbf{R}$. [Hint: If $x \in C$, show that there is an open interval containing x and contained in C.]

   (b) Show that C is an interval. (Hint: Use Exercise 5).

   (c) Deduce that G is a union of disjoint open intervals.

   (d) Show that there are at most countably many intervals in this union. [Hint: There is a rational number in each open interval.]

Appendix A

Some useful facts from the past.

In this appendix we list some definitions, results and inequalities that you should have encountered in earlier courses, but may have forgotten. If you need more information about the topics mentioned here, you should consult your notes or textbook for the courses in which you were first introduced to them.

A.1 The absolute value of a real number.

The *absolute value* of a real number x is defined by

$$|x| = \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0. \end{cases}$$

The absolute value of x may be interpreted geometrically as the distance between x and 0 on the real line.

The absolute value is also given by

$$|x| = \sqrt{x^2}.$$

The following are useful properties of the absolute value:

1. $|x| \geq 0$ for all $x \in \mathbf{R}$
2. $|x| = 0$ if and only if $x = 0$
3. $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbf{R}$
4. $|xy| = |x||y|$ for all $x, y \in \mathbf{R}$
5. $||x| - |y|| \leq |x - y|$ for all $x, y \in \mathbf{R}$

A.2 Complex numbers.

A complex number z can be represented in *Cartesian form*: $z = a + ib$, where $a, b \in \mathbf{R}$ and $i^2 = -1$. The real number a is called the *real part* of z, written $\mathrm{Re} z$, and b is called the *imaginary part* of z, written $\mathrm{Im} z$. The set of all complex numbers is denoted by $\mathbf{C}$. Geometrically the complex number $z = a + ib$ can be represented by the point with coordinates (a, b) in the plane, and in this context the plane is referred to as the *complex plane*.

The *complex conjugate* of the complex number $z = a + ib$ is the complex number $\overline{z} = a - ib$. Geometrically, $\overline{z}$ is the reflection of z in the *real axis* (the line $\mathrm{Im} z = 0$). Here are some useful properties of the conjugate; $z, w \in \mathbf{C}$:

1. $\overline{\overline{z}} = z$
2. $\overline{z + w} = \overline{z} + \overline{w}$
3. $\overline{z - w} = \overline{z} - \overline{w}$
4. $\overline{zw} = \overline{z}\ \overline{w}$
5. $\overline{\frac{z}{w}} = \frac{\overline{z}}{\overline{w}}$, where $w \neq 0$.

The *modulus* of a complex number $z = a + ib$ is given by

$$|z| = |a + ib| = \sqrt{a^2 + b^2}.$$

The modulus of z can be thought of geometrically as the distance between 0 and z in the complex plane. The modulus of a complex number has the same properties as the absolute value of a real number:

1. $|z| \geq 0$ for all $z \in \mathbf{C}$
2. $|z| = 0$ if and only if $z = 0$
3. $|z + w| \leq |z| + |w|$ for all $z, w \in \mathbf{C}$
4. $|zw| = |z||w|$ for all $z, w \in \mathbf{C}$
5. $||z| - |w|| \leq |z - w|$ for all $z, w \in \mathbf{C}$
6. $|\overline{z}| = |z|$ for all $z \in \mathbf{C}$.

Complex numbers can also be written in *modulus-argument*, or *polar*, form. If $z = a + ib$ and θ is the angle (in radians) between the positive real axis and the line segment from 0 to z in the complex plane, then $a = |z|\cos\theta, \quad b = |z|\sin\theta$ and so we can write $z = |z|(\cos\theta + i\sin\theta)$. We call θ an *argument* of z. Note that if θ is an argument of z, then so is $\theta + 2\pi k$, for any $k \in \mathbf{Z}$. The *principal argument* of z is the argument θ for which $-\pi < \theta \leq \pi$.

If we put $e^{i\theta} = \cos\theta + i\sin\theta$, then we can write

$$z = |z|(\cos\theta + i\sin\theta) = |z|e^{i\theta};$$

this is the modulus-argument form for z.

A.3 Inner products and norms for vectors.

Let n be a positive integer. An *real n-vector* $\mathbf{x}$ is an n-tuple of real numbers: $\mathbf{x} = (x_1, x_2, \ldots x_n)$, where $x_1, x_2, \ldots, x_n \in \mathbf{R}$. The set of all real n-vectors is denoted by $\mathbf{R}^n$. Complex n-vectors are defined in the same way, with complex numbers replacing real numbers. The set of all complex n-vectors is denoted by $\mathbf{C}^n$.

For $\mathbf{x} = (x_1, x_2, \ldots, x_n), \mathbf{y} = (y_1, y_2, \ldots, y_n) \in \mathbf{R}^n$, the *inner product* or *dot product* of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

For $\mathbf{x} = (x_1, x_2, \ldots, x_n), \mathbf{y} = (y_1, y_2, \ldots, y_n) \in \mathbf{C}^n$, the *inner product* or *dot product* of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = x_1\overline{y_1} + x_2\overline{y_2} + \cdots + x_n\overline{y_n}.$$

The norm of $\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbf{R}^n$ or $\mathbf{C}^n$ is defined by

$$||\mathbf{x}|| = \sqrt{|x_1|^2 + |x_2|^2 + \cdots + |x_n|^2}.$$

The norm and the inner product are related by the *Cauchy-Schwarz inequality*:

$$|\mathbf{x} \cdot \mathbf{y}| \leq ||\mathbf{x}||\,||\mathbf{y}|| \quad \text{for } \mathbf{x}, \mathbf{y} \in \mathbf{R}^n \text{ or } \mathbf{C}^n.$$

A.4 Bounds for sets of real numbers.

We say that the real number c is a *lower bound* for the set A of real numbers if $c \leq x$ for all $x \in A$; in the same way d is an *upper bound* for A if $x \leq d$ for all $x \in A$.

A set which is both bounded above and below is called *bounded*. Every non-empty set A of real numbers that is bounded below has a *greatest lower bound* or *infimum*, denoted by $\inf A$. A non-empty set A that is bounded above has a *least upper bound* or *supremum*, denoted by $\sup A$. It is useful to keep in mind that if $c = \inf A$ and $a > c$, then a cannot be a lower bound of A and hence there must be an $x \in A$ such that $x < a$. A similar result holds for the supremum.

A.5 Convergent sequences of real numbers.

A sequence (x_n) of real numbers *converges* to the real number x if for every $\epsilon > 0$ there is an $n_\epsilon \in \mathbf{N}$ such that for all $n \geq n_\epsilon$, $|x - x_n| < \epsilon$.

The following are useful facts about convergent sequences:

1. The limit of a convergent sequence is unique.
2. Every convergent sequence is bounded.
3. Every subsequence of a convergent sequence is itself convergent, and has the same limit as the original sequence.
4. If (x_n) is a convergent sequence and $a \leq x_n \leq b$ for every $n \in \mathbf{N}$, then $a \leq \lim_{n\to\infty} x_n \leq b$.
5. If (x_n) and (y_n) are convergent sequences, then the sequence
   (a) $(x_n + y_n)$ is convergent, and $\lim_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$;
   (b) $(x_n - y_n)$ is convergent, and $\lim_{n\to\infty}(x_n - y_n) = \lim_{n\to\infty} x_n - \lim_{n\to\infty} y_n$;
   (c) $(x_n y_n)$ is convergent, and $\lim_{n\to\infty}(x_n y_n) = (\lim_{n\to\infty} x_n)(\lim_{n\to\infty} y_n)$;
   (d) $(\frac{x_n}{y_n})$ is convergent, and
   $$\lim_{n\to\infty}\left(\frac{x_n}{y_n}\right) = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}$$
   provided that $y_n \neq 0$ for all n and $\lim_{n\to\infty} y_n \neq 0$;
   (e) $(\sqrt{x_n})$ is convergent and $\lim_{n\to\infty}\sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}$, provided $x_n \geq 0$ for all n.
6. If (x_n) and (y_n) are convergent sequences and $x_n \leq y_n$ for every n, then $\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n$.

A.6 Continuous real functions.

Let I be an interval on the real line and $f : I \to \mathbf{R}$ be a function. Then we say that f is continuous at $a \in I$ if for every $\epsilon > 0$, there is a $\delta > 0$ such that if $x \in I, |x - a| < \delta$, then $|f(x) - f(a)| < \epsilon$. We say that f is continuous on I if it is continuous at every point in I.

If I is a closed and bounded interval (i.e. $I = [a, b]$) and f is continuous on I, then f is bounded on I (i.e. there is a $C > 0$ such that $|f(x)| \le C$ for all $x \in I$), and f attains its maximum and minimum value on I (i.e. there are $c, d \in I$ such that $f(c) \le f(x) \le f(d)$ for all $x \in I$).

The link between convergence and continuity is given by the following result: If $f : I \to \mathbf{R}$ is continuous at $x \in I$ and $x_n \to x$ in I, then $f(x_n) \to f(x)$ in $\mathbf{R}$.

If the functions $f : I \to \mathbf{R}$ and $g : I \to \mathbf{R}$ are both continuous at $a \in I$, then each of the following functions is continuous at a:

1. the function $f + g$ which maps x to $f(x) + g(x)$;
2. the function $f - g$ which maps x to $f(x) - g(x)$;
3. the function fg which maps x to $f(x)g(x)$;
4. the function $\frac{f}{g}$ which maps x to $\frac{f(x)}{g(x)}$ provided $g(a) \ne 0$.

If I and J are intervals, $f : I \to \mathbf{R}$ is continuous at $a \in I$, $f(a) \in J$ and $g : J \to \mathbf{R}$ is continuous at $f(a)$, then the composition $g \circ f$ is continuous at a.

The absolute value function which maps $x \in \mathbf{R}$ to $|x|$ and the square root function which maps $x \in [0, \infty)$ to $\sqrt{x}$ are continuous.

If a real-valued function f is differentiable at $a \in \mathbf{R}$, then it is continuous at a. Hence all polynomials, rational functions, exponential functions, logarithmic functions and trigonometric functions are continuous where they are defined.

A.7 Integrals

If a function $f : [a, b] \to \mathbf{R}$ is continuous, it is integrable on $[a, b]$.

If $f : [a, b] \to \mathbf{R}$ and $g : [a, b] \to \mathbf{R}$ are both continuous and $c \in \mathbf{R}$, then

1. $\int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$;

2. $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$;

3. if $f(x) \le g(x)$ for all $x \in [a,b]$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$;

4. $|\int_a^b f(x)\,dx| \le \int_a^b |f(x)|\,dx$;

5. if $|f(x)| \le M$ for all $x \in [a,b]$, then $|\int_a^b f(x)\,dx| \le M(b-a)$.

If $f : [a,b] \to \mathbf{R}$ is continuous, then for every $x \in [a,b]$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

A.8 Sets and direct and inverse images of sets under functions.

The empty set is the set which contains no elements, and is denoted by $\emptyset$.

A set A is a subset of a set B, written $A \subset B$ (or in some texts $A \subseteq B$) if every element of A is also an element of B.

Two sets A and B are equal ($A = B$) if and only if $A \subset B$ and $B \subset A$.

The complement of a set A in a set X is denoted by $X \setminus A$ and consists of all elements in X that are not in A. If all sets considered are subsets of some "universal" set X, then we write A' for $X \setminus A$, and call it the complement of A.

The intersection of two sets A and B is the set, denoted by $A \cap B$, consisting of all elements that are in both A and B. Likewise, the intersection of the family $\{A_i : i \in I\}$ of sets is the set denoted by $\cap\{A_i : i \in I\}$ consisting of all elements that are in every set A_i of the family. (Here I is a so-called index set used to label the members of the family of sets.)
More technically, $\cap\{A_i : i \in I\} = \{x \,:\, (\forall i \in I)(x \in A_i)\}$.

The union of two sets A and B is the set, denoted by $A \cup B$, consisting of all elements that are in either A or B, or in both. Likewise, the union of the family $\{A_i : i \in I\}$ of sets is the set denoted by $\cup\{A_i : i \in I\}$ consisting of all elements that are in at least one of the sets A_i of the family.
More technically, $\cup\{A_i : i \in I\} = \{x \,:\, (\exists i \in I)(x \in A_i)\}$.

Unions, intersections and complements are related by De Morgan's Laws:

1. $(\cap\{A_i : i \in I\})' = \cup\{A_i' : i \in I\}$;

2. $(\cup\{A_i : i \in I\})' = \cap\{A_i' : i \in I\}$.

The distributive laws hold for unions and intersections:

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Similar rules hold for unions and intersections of arbitrary families.

If X and Y are sets, f is a function from X to Y ($f : X \to Y$) and A and B are subsets of X and Y respectively, then

1. the direct image of A under f is the subset $f(A)$ of Y defined by
$$f(A) = \{f(x) : x \in A\}$$
2. the inverse image of B under f is the subset $f^{-1}(B)$ of X defined by
$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

Note that the inverse image is defined even if the function f does not have an inverse.

A.9 Partial orders

A partial order $\leq$ on a non-empty set X is a relation on X such that, for all $x, y, z \in X$,

1. $x \leq x$ (reflexivity);
2. if $x \leq y$ and $y \leq x$, then $x = y$ (anti-symmetry);
3. if $x \leq y$ and $y \leq z$, then $x \leq z$ (transitivity).

If $\leq$ is a partial order on the set X, then we call $(X, \leq)$ a partially ordered set; if it is clear what the partial order is, we abbreviate this to "the partially ordered set X."

Appendix B

Answers and hints to the Exercises.

B.1 Chapter 1

1. Starting with $x = 0$ the approximation $x = -0.13781$ is obtained. For $x = 2$ the sequence diverges to infinity.

2. For the one direction, integrate the differential equation and use the initial condition; for the other direction, use the integral equation to find the value of y at 0 and differentiate the integral equation to get the differential equation.

3. $y_3(x) = 2 + x + x^2 + \frac{1}{3}x^3 + \frac{1}{8}x^4, \quad y_4(x) = 2 + x + x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{15}x^5 + \frac{1}{48}x^6.$
   Starting with $y_1(x) = x$, we get $y_2(x) = 2 + x + \frac{1}{3}x^3$,
   $y_3(x) = 2 + x + x^2 + \frac{1}{3}x^3 + \frac{1}{15}x^5$,
   $y_4(x) = 2 + x + x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{15}x^5 + \frac{1}{105}x^7$.

5. No. Every Fourier polynomial of f has the value $\frac{1}{2}$ at $x = 0$, while $f(0) = 0$.

B.2 Chapter 2

1. (b) For the triangle inequality, consider the different cases that can arise separately.

2. Straightforward from definition.

3. (a) For (M1), use the fact that $\arctan x = \arctan y$ implies $x = y$.

(b) Use the fact that $|\arctan x| < \frac{\pi}{2}$.

(c) For d_f to be a metric, f must be one-to-one. For it to be a bounded metric, f must be a bounded function.

4. (a) Note that $|r_1 - s_i| = 1$ if $r_i \neq s_i$ and $|r_i - s_i| = 0$ if $r_i = s_i$.

   (b) Use the facts that $|x_i - y_i| \leq 1$ for all n, that $\sum_{n=1}^{\infty} s^{-n}$ is a convergent series and the comparison test for series.

5. Similar to the proof for the discrete metric.

6. Note that d' can only have values in the set $\{2^{-n} : n = 0, 1, 2, \ldots\} \cup \{0\}$.

7. (b) $d((x_n), (y_n)) = ||(x_n) - (y_n)|| = \sum_{n=1}^{\infty} |x_n - y_n|$.

8. (a) For $f, g \in B([a, b])$, the set $\{|f(x) - g(x)| : x \in [a, b]\}$ is a non-empty bounded set of real numbers, and hence has a supremum. A bounded, discontinuous function need not attain its maximum value on $[a, b]$.

   (b) Since a continuous function attains its maximum on $[a, b]$, the supremum is also the maximum in this case.

   (c) 2.

9. (a) Consider the cases $x_2 = y_2$ and $x_2 \neq y_2$ separately.
   (e) (ii)

$$\begin{aligned}
\mathbf{x} \in B[(1,0);2] &\Leftrightarrow d((x_1, x_2), (1, 0)) \leq 2 \\
&\Leftrightarrow |x_1 - 1| \leq 2 \text{ and } x_2 = 0 \text{ or } |x_1| + |x_2| + 1 \leq 2 \text{ and } x_2 \neq 0 \\
&\Leftrightarrow -1 \leq x_1 \leq 3 \text{ and } x_2 = 0 \text{ or } |x_1| + |x_2| \leq 1 \text{ and } x_2 \neq 0
\end{aligned}$$

10. (b) (i) $\{z \in \mathbf{C} : d(0, z) < 1\} = \{0\} \cup \{z \in \mathbf{C} : |z| + |0| < 1\} = \{z \in \mathbf{C} : |z| < 1\}$.
    (ii) $\{z \in \mathbf{C} : d(i, z) < 1\} = \{i\} \cup \{z \in \mathbf{C} : |z| + |i| < 1\} = \{i\} \cup \{z \in \mathbf{C} : |z| + 1 < 1\} = \{i\} \cup \{z \in \mathbf{C} : |z| < 0\} = \{i\} \cup \emptyset = \{i\}$.

11. (b) The taxi metric on $\mathbf{R}^2$.

12. For the triangle inequality for d^* use the fact that the function $f : [0, \infty) \to [0, \infty)$ defined by $f(t) = \frac{t}{1+t}$ is increasing, and the triangle inequality for d. For the boundedness, show that $d^*(x, y) \leq 1$ for all $x, y \in X$.

B.3 Chapter 3

1. In a discrete metric space the only way we can have $d(x_n, x) \to 0$ is if $d(x_n, x) = 0$ eventually.

2. Use your Real Analysis notes!

3. (a) 2^n
   (b) Eventually constant sequences.
   (c) Eventually constant sequences.

4. Eventually constant sequences.

5. (a) Use the inequality
$$\int_0^1 |f(x) - f_n(x)|\, dx \leq \max\{|f_n(x) - f(x)| : x \in [0,1]\} \int_0^1 dx.$$
   (b) False. Consider (for example) the sequence (f_n), where $f_n(x) = x^n$, which converges to the constant 0 function with respect to d_1, but does not converge with respect to d_m.

6. (a) For the first implication, use the fact that $\sum_{k=n+1}^{\infty} 2^{-k} = 2^{-n}$; for the second implication, prove the contrapositive.
   (b) Consider the sequence with n-th term the string with the first n positions as in $\mathbf{x}$ and 0 everywhere else.

7. A sequence $((x_{1n}, x_{2n})) \in \mathbf{R}^2$ converges to $(x_1, x_2) \in \mathbf{R}^2$ with respect to the lift metric iff either $x_{2n} = x_2$ eventually and $x_{1n} \to x_1$ with respect to the usual metric of $\mathbf{R}$, or $x_1 = 0$ and $|x_{1n}| + |x_{2n} - x_2| \to 0$.

8. A sequence (z_n) in $\mathbf{C}$ converges to $z \in \mathbf{C}$ with respect to the given metric if and only if either $z_n = z$ eventually, or $z = 0$ and $|z_n| \to 0$ as $n \to \infty$.

B.4 Chapter 4

1. (a) Not closed, closure is $\{z \in \mathbf{C} : 1 \leq |z| \leq 2\}$, boundary is $\{z \in \mathbf{C} : |z| = 1 \text{ or } |z| = 2\}$;
   (b) Not closed, closure is $\{z \in \mathbf{C} : 0 \leq \arg z \leq \frac{\pi}{4}, |z| \leq 1\}$, boundary is $\{z \in \mathbf{C} : 0 = \arg z, |z| \leq 1\} \cup \{z \in \mathbf{C} : \frac{\pi}{4} = \arg z, |z| \leq 1\} \cup \{z \in \mathbf{C} : 0 \leq \arg z \leq \frac{\pi}{4}, |z| = 1\}$;

   (c) Closed, closure and boundary both equal $\{z \in \mathbf{C} : |z| = n, n \in \mathbf{N}\}$;
   (d) Not closed, closure and boundary both equal $\{z \in \mathbf{C} : |z| = \frac{1}{n}, n \in \mathbf{N}\} \cup \{0\}$.

2. See above.

3. (a) Closed. Boundary is empty.
   (b) Not closed. Closure is $\{z \in \mathbf{C} : 0 < \arg z < \frac{\pi}{4}, 0 < |z| < 1\} \cup \{0\}$, boundary is $\{0\}$.
   (c) Closed. Boundary is empty (if $\mathbf{N} = \{1, 2, \ldots\}$) or $\{0\}$ (if $\mathbf{N} = \{0, 1, 2, \ldots\}$).
   (d) Not closed. Closure is $\{z \in \mathbf{C} : |z| = \frac{1}{n}, n \in \mathbf{N}\} \cup \{0\}$, boundary is $\{0\}$.

4. For the metric of Question 3, $A = \{1 - i\}$ and $d(i, A) = 1 + \sqrt{2}$.

5. (a) Yes. Note that the set equals $\cap\{A_x : x \in (0, 1)\}$, where $A_x = \{f \in B([0, 1]) : f(x) = 0\}$. Since each A_x is closed (see Example 4.5), the given set is the intersection of a family of closed sets, hence closed.
   (b) Look at the proof that the limit of a uniformly convergent sequence of continuous functions is continuous in your Real Analysis notes.

6. Use the hints given for the proof of Proposition 4.26.

7. Use the definition of a point of accumulation. To obtain infinitely many distinct points, use the fact that once you have one, you can apply the definition again to a ball with radius less than the distance between x and your previous point.

B.5 Chapter 5

1. (a) Not open, interior is $\{\mathbf{x} \in \mathbf{R}^2 : 0 < x_1 < 2, -1 < x_2 < 1\}$.
   (b) Not open, interior is $\{\mathbf{x} \in \mathbf{R}^2 : -1 < x_2 < 1\}$.
   (c) Open. Interior is therefore equal to the set.

2. (a) Open. If z is in the set, any open ball with centre z and radius less than 1 will contain only z. Interior is equal to the set.
   (b) Not open. 0 is in the set, but no open ball with centre 0 is contained in the set. Interior is the set $\{z \in \mathbf{C} : z \neq 0 \text{ and } \mathrm{Re} z \geq 0, \mathrm{Im} z \geq 0\}$.

3. (a) True (the interior of any set is open).

(b) False. Try the rationals and irrationals in the reals, for a counter-example.

(c) False. Try the rationals, for a counter-example.

4. (a) Note that if an open ball contains x, there is an open ball centred at x contained in the first ball.

   (b) Use the fact that $\mathrm{int}(A) = (\overline{A'})'$ and the fact that $x \in \overline{A'}$ iff $d(x, A') = 0$.

5. Use Proposition 5.26.

6. Use Proposition 5.26 and Exercises 3.1 and 3.4.

7. (a) Similar to the proofs for the special case $X = Y = \mathbf{R}$.

   (b) Use Proposition 5.26.

8. No, for the first question, yes for the second.

9. For (T2), note that if $x \in \cap_{k=1}^{n} G_k$, then for each k there is an $r_k > 0$ such that $B(x; r_k) \subset G_k$. Take $r = \min\{r_k : k = 1, \ldots, n\}$ and show that $B(x; r) \subset \cap_{k=1}^{n} G_k$.

10. (b) Use the description of the topology induced by a quasi-metric given in Question 9.

B.6 Chapter 6

1. Straightforward, using definition of continuity.

2. All three functions are continuous. To show that the function f in (b) is continuous, let $(x, y) \in \mathbf{R}^2$ and suppose $((x_n, y_n))$ is a sequence in $\mathbf{R}^2$ converging to (x, y) with respect to the usual (Euclidean) metric on $\mathbf{R}^2$. Then (by Example 3.6) $x_n \to x$ and $y_n \to y$. Hence $x_n^2 \to x^2$ and $y_n^2 \to y^2$, and therefore $x_n^2 + y_n^2 \to x^2 + y^2$ by well-known limit theorems for convergent real sequences. Since the function $\sin x$ is continuous, it follows that $f((x_n, y_n)) = \sin(x_n^2 + y_n^2) \to \sin(x^2 + y^2) = f(x, y)$, and hence f is continuous at (x, y).

3. (a) Use the ratio test to show that the series $\sum_{n=1}^{\infty} nx(1 - x^2)^n$ converges and deduce from this that $\lim_{n\to\infty} f_n(x) = 0$.
   (b) Use the substitution $u = 1 - x^2$ to show that $\int_0^1 f_n(x)\,dx = \frac{n}{2(n+1)}$.
   (c) No. If it did, it would contradict (a) and (b).

4. (b) If $z \in f(G), z = f((x, y)) = x$ for some $(x, y) \in G$. Since G is open, there is an $r > 0$ such that $B((x, y); r) \subset G$. Show that $(z - r, z + r) \subset f(G)$.
(c) To show that F is closed, show that if $(x_n, \frac{1}{x_n}) \to (x, y)$ in $\mathbf{R}^2$, then we must have $x > 0$. Also note that $f(F) = (0, \infty)$.

5. Use the inequality $||f(x_n)| - |f(x)|| \leq |f(x_n) - f(x)|$.

6. Note that for $z \neq z_n$, $d_Y(z, z_n) = |z - z_n| \leq |z_n| + |z| = d_X(z, z_n)$, so f is Lipschitz-continuous with Lipschitz-constant 1.

7. f is continuous at $(1, 1)$ (if $(x_n, y_n) \to (1, 1)$ with respect the lift metric, we must have $x_n \to 1$ and for all n, $y_n = 1$. Hence $f(x_n, y_n) = 1$ for all n). f is not continuous at $(0, 1)$ (consider the sequence $((0, \frac{n}{n+1}))$, for example).

8. (a) Closed; equals $f^{-1}([0, \infty))$, where $f(x) = \sin x$. f is continuous and $[0, \infty)$ is closed. Not open: 0 is in the set, but there is no open ball around 0 contained in the set.
(b) Open; equals $f^{-1}((-0.1, 0.3))$, where $f(x) = \cos x$. f is continuous and $(-0, 1, 0.3)$ is open. Not closed: find a sequence in the set that converges to a point outside it.
(c) Open; equals $f^{-1}((0.5, 2))$ (since $f(x) = \sin x \leq 1$ for all x). Not closed.

9. (a) Open; set equals $G^{-1}((0, 2))$. $G : C([0, 1]) \to \mathbf{R}$ defined by $G(f) = \int_0^1 f(x)\, dx$ is continuous (see Example 6.4) and $(0, 2)$ is open.
(b) Closed; set equals $H^{-1}([0, 1])$ where $H = G \circ F$, with G the function in (a) and F the function in Exercise 5. H is continuous and $[0, 1]$ closed.

10. Use the definition of matrix multiplication and Example 3.6.

11. This can be done in the same way as the previous question, using Example 3.8.

12. Use the hint.

13. Given $\epsilon > 0$, take $\delta = \frac{\epsilon}{k}$, where k is a Lipschitz constant.

14. (b) Negate the definition of uniform continuity and for each n, use $\delta = \frac{1}{n}$ in the negation.
(c) Use $x_n = \frac{1}{n}, y_n = \frac{1}{2n}$.

15. Use Proposition 5.26(2).

16. Combine the function of Example 6.34 with a linear function that maps (a, b) onto $(-\frac{\pi}{2}, \frac{\pi}{2})$.

17. Use the same idea as in Exercise 16.

B.7 Chapter 7

1. Yes; discrete metric spaces.
2. The only Cauchy sequences in a finite set are eventually constant sequences.
3. No, no.
4. (a) Complete (b) Not complete (c) Not complete.
5. (a) Yes (b) No, by (a) the set is not closed.
6. The proof is similar to the proof that a Cauchy sequence of real numbers is bounded (see your Real Analysis notes).
7. If $\operatorname{diam}(A_n) \to 0$, then given $\epsilon > 0$, there is an $n_\epsilon \in \mathbf{N}$ such that $\operatorname{diam}(A_n) < \epsilon$ for $n \geq n_\epsilon$. Now if $n \geq m \geq n_\epsilon$, then $x_n, x_m \in A_n$, so $d(x_n, x_m) \leq \operatorname{diam}(A_n) < \epsilon$. This shows that (x_n) is a Cauchy sequence. The proof of the converse is similar.
8. Since $A \subset \overline{A}$, $\operatorname{diam}(A) \leq \operatorname{diam}(\overline{A})$. For the converse, show that if $x, y \in \overline{A}$, then $d(x, y) \leq \operatorname{diam}(A)$.
9. The converse says that if in a metric space X every decreasing sequence (A_n) of non-empty subsets of X such that $\operatorname{diam}(A_n) \to 0$ has the property that $\cap_{n=1}^{\infty} \overline{A_n} \neq \emptyset$, then X is complete.
10. Let $f : X \to Y$ be an isometry of X onto Y. If (x_n) is a Cauchy sequence in X, show that $(f(x_n))$ is a Cauchy sequence in Y. By completeness of Y, $f(x_n)) \to y \in Y$. Show that this implies that (x_n) converges in X.
11. (b) Remember to check that d_Y is well-defined, i.e. that if $(x'_n) \sim (x_n)$ and $(y'_n) \sim (y_n)$, then $d_Y([(x_n)], [(y_n)]) = d_Y([(x'_n)], [(y'_n)])$.
12. (b) Use the continuity of the metric.
    (c) Use Lemma 6.7.
    (d) Use the results in (c).
13. (a) Use the fact that $\overline{F(X)} = Y$.
    (b) Show that (x_n) is a Cauchy sequence in X, and therefore that $(G(x_n))$ is a Cauchy sequence in the complete space Z. To show that $H(y)$ is independent of the choice of the sequence (x_n), let (x'_n) be another sequence such that $y = \lim_{n\to\infty} F(x'_n)$ and use the fact that F and G are both isometries to show that $\lim_{n\to\infty} G(x'_n) = \lim_{n\to\infty} G(x_n)$.
    (c) Use the facts that $\overline{F(X)} = Y$ and $\overline{G(X)} = Z$.
    (d) Use the continuity of metrics and the fact that F and G are both isometries.

B.8 Chapter 8

1. (a) All complex numbers of the form $z = a + \frac{1}{2}i$, $a \in \mathbf{R}$ are fixed points.
   (b) There are three fixed points: $z = 0, -i, i$.

2. (a) Every point on the x_2-axis is a fixed point; not a contraction.
   (b) Every point on the x_1-axis is a fixed point; not a contraction.
   (c) The origin is the only fixed point; not a contraction.

3. Only constant functions are contractions. Every function is Lipschitz-continuous.

4. Use the fact that f will be a contraction on I iff $\sup_{x\in I} |f'(x)| < 1$.
   (a) Contraction (b) Contraction (c) Contraction (d) Not a contraction.

5. (a) No ($f'(2) > 1$, for example). Yes, since $|f'(x)| \le \frac{1}{2}e^{\frac{1}{2}} < 1$ for all $x \in [-1, 1]$.
   (b) Yes. Yes.
   (c) f has two fixed points on $\mathbf{R}$. Only one ($x = 0$) can be found by using the contraction mapping theorem.

6. (f) No, only the fixed point $x = 0$ can be found using the contraction mapping theorem. Using the derivative $f'(x)$ we can show that f is a contraction in a small interval around $x = 0$, but not near the other two fixed points.

8. (a) $f^2(x) = \sin(\sin x)$ has derivative $\cos(\sin x)\cos x$, and the derivative is 1 at $x = 0$. Hence f^2 is not a contraction.
   (c) $f^2(x) = \arctan(\frac{1}{2}\arctan(\frac{1}{2}x))$ has derivative

   $$\frac{1}{1 + (\arctan(\frac{1}{2}x)^2} \frac{\frac{1}{4}}{1 + \frac{1}{4}x^2}$$

   which has a maximum value of $\frac{1}{4}$ on $\mathbf{R}$. Hence f^2 is a contraction.

9. The derivative of every iterate of $\sin x$ is of the form $\cos(g(x))\cos x$, for some function $g(x)$ such that $g(0) = 0$. Hence this derivative equals 1 if $x = 0$, and so the iterate cannot be a contraction.

10. (a) Integrating the differential equation and using the initial condition gives $f(x) = y = 2 + \int_0^x (2e^{-t} - f(t))\,dt$.

(b) Define the function $F : C([0,1]) \to C([0,1])$ by

$$F(f)(x) = 2 + \int_0^x (2e^{-t} - f(t))\,dt.$$

A calculation similar to the one in Example 8.14 gives

$$|F(f_1)(x) - F(f_2)(x)| \le d(f_1, f_2) \int_0^x dt = d(f_1, f_2)x$$

for $x \in [0,1]$, and hence $d(F(f_1), F(f_2)) \le d(f_1, f_2)$. This means that F is not necessarily a contraction. If we do the same calculation with F^2 in the place of F, we get, for all $x \in [0,1]$,

$$\begin{aligned} |F^2(f_1)(x) - F^2(f_2)(x)| &= |F(F(f_1)) - F(F(f_2))| \\ &\le \tfrac{x^2}{2} d(f_1, f_2), \end{aligned}$$

and so

$$d(F^2(f_1), F^2(f_2)) = \sup\{|F^2(f_1)(x) - F^2(f_2)(x)| : x \in [0,1]\} \le \frac{1}{2} d(f_1, f_2).$$

This shows that F^2 is a contraction, and hence F has a unique fixed point in $C([0,1])$ by Theorem 8.18.

(c) To use Corollary 8.21, put $f(x,y) = 2e^{-x} - y$. Then $\frac{\partial f}{\partial y} = -1$, which is clearly bounded on $\{(x,y) \in \mathbf{R}^2 : 0 \le x \le 1\}$.

11. (a) Write the initial value problem as an integral equation, and then use the same method as in the previous question to show that there is a solution.

    (b) Put $f(x,y) = (y + x^2)e^{x-1}$ and show that $\frac{\partial f}{\partial y}$ is bounded for $1 \le x \le 3, y \in \mathbf{R}$.

B.9 Chapter 9

1. Follow the outline given in the proof.

2. (a) Not bounded. Note that if $f_n(x) = nx$, then f_n is in this set for $n = 0, 1, 2, \ldots$, and $d(f_0, f_n) = n$, so the diameter of the set is infinite.

   (b) Bounded (the diameter of the set is 2).

   (c) Bounded.

3. (a) Not compact (not closed).

(b) Not compact (not bounded).

(c) Compact (both closed and bounded, which in $\mathbf{R}^2$ is enough to ensure compactness).

4. A subset of $\mathbf{C}$ is compact iff it is closed and bounded.

5. (a) Compact (the set is finite, hence closed and bounded).

   (b) Not compact (not closed).

   (c) Not compact (not bounded).

6. (a) Not compact (not bounded – see Question 2(a)).

   (b) Not compact (not closed).

7. Show that no subsequence of (x_n) can be a Cauchy sequence (take $\epsilon = r$ in the definition of a Cauchy sequence), and deduce that no subsequence of (x_n) can be convergent.

8. (a) In $\mathbf{R}$ with its usual metric, take $A = [0, 2]$, $B = [1, 5]$ for example.

   (b) For (M1), make use of the fact that sets in $\mathcal{K}(X)$ are closed to show that $d(A, B) = 0$ implies that $A \subset B$, and similarly for $d(B, A) = 0$. (M2) follows from the definition. For (M3) first prove $d(A, B) \leq d(A, C) + d(C, B)$, then similarly for $d(B, C)$ and then deduce the triangle inequality for d_H.

   (c) Give an example to show that for unbounded sets distances will not be finite.

   (d) Show that if r is larger than the infimum, then $A \subset N_r(B)$ and $B \subset N_r(A)$, and this implies that $d_H(A, B) \leq r$. The converse inequality follows similarly.

9. Use the hint.

10. If A is not totally bounded, there is an $\epsilon > 0$ such that no finite set of open balls with radius ϵ will cover A. Choose any $x_1 \in A$. Since A is not contained in $B(x_1; \epsilon)$, there is an $x_2 \in A$ such that $x_2 \notin B(x_1; \epsilon$, i.e. $d(x_1, x_2) \geq \epsilon$. Now continue in this way; in the next step use the fact that A is not contained in $B(x_1; \epsilon) \cup B(x_2; \epsilon)$.

11. For every $\epsilon > 0$, $\{B(x; \epsilon) : x \in A$ is an open cover of A and therefore has a finite subcover.

12. Start with an arbitrary sequence in the set and use total boundedness to find a subsequence with every term in a ball with radius $\frac{1}{2}$; then find a subsequence of this subsequence such that every term is in a ball with radius $\frac{1}{3}$, and so on. Finally form a subsequence by picking the first term from the first subsequence, the second from the second subsequence, and so on. The resulting subsequence is Cauchy, hence convergent.

B.10 Chapter 10

1. (a) Disconnected.
   (b) Connected.
   (c) Connected.
   (d) Disconnected.

2. (a) Not necessarily connected. In **R** with its usual metric $A = [0, 1]$ and $B = [2, 3]$ are both connected, but $A \cup B$ is not.
   (b) Not necessarily connected. Consider, for example, the case of two circles in $(\mathbf{R}^2, d_E)$ which intersect in two points.
   (c) Not necessarily. Consider the interval $[0, 1]$ in **R** with its usual metric.

3. Try to write each set as the union of a family of line segments with one point in common.

4. First deduce from Theorem 10.16 that $A_1 \cup A_2$ is connected. Then prove by induction that $\cup_{k=1}^{n} A_k$ is connected for each n. Finally use Theorem 10.16 again.

5. Use Proposition 10.4, and show that A is disconnected as a subset of B iff it is disconnected as a subset of X.

6. If $x \in A$, the component of X containing x must contain C (since this component is the largest connected set containing x.

7. Let A be the clopen set, and suppose $A \subset B$. If $A \neq B$, we wold have $B = (A \cap B) \cup (A' \cap B)$, with A an A' open, and $A \cap B$ and $A' \cap B$ non-empty and disjoint, implying that B is disconnected. Hence A cannot be contained in a larger connected set.